SHADOW FUTURES

How Markets Choose Winners—and Lose Sight of the Alternatives.



MARTIN ERLIC

SHADOW FUTURES

How Markets Choose Winners—and Lose Sight of the Alternatives

MARTIN ERLIC

Copyright © 2026 Martin Erlic. All rights reserved.

First edition · October 2026

Contents

The other life of a song 4

The evidence success leaves behind 12

1 · The winner and the explanation 23

2 · The comparison budget 41

3 · Shadow futures 58

4 · The histories that became infrastructure 75

5 · Who gets to be tested 95

6 · Who deserves the winnings 115

7 · The intelligence factory 132

8 · A billion answers 150

9 · The first rung 167

10 · Competition for future evidence 183

11 · Taxing lost comparison 200

12 · An economics of possible worlds 212

The next opportunity 228

Notes 232

References 249

Acknowledgments 258

The other life of a song

In one version of a small online music market, a song called Lockdown finished first. In another, it finished fortieth out of forty-eight. The band was called 52Metro. The recording competing for attention was the same.[1]

First place invites a story. The band understood its audience. The song had something the others lacked. Listen again, knowing it won, and you might hear the qualities that explain its success. Fortieth invites a different story. The song failed to connect. Other bands understood something 52Metro did not. Listen again, knowing it lost, and its shortcomings might become obvious.

Usually, we get to tell only one of those stories. Someone succeeds, and the rest of us work backward. We examine the product, the strategy, the years of effort. The ending helps us decide which details mattered. We rarely get to check our explanation against another history in which the same work met a different fate.

For Lockdown, someone had kept the other chart.

MusicLab was a website where people could discover unfamiliar music. In research published in 2006, Matthew Salganik, Peter Dodds, and Duncan Watts studied choices made by 14,341 participants across two experiments involving forty-eight songs. Some listeners saw no download totals. Others could see what earlier listeners had downloaded. These listeners were randomly assigned to eight separate worlds, each with its own evolving popularity record. Every song began with zero downloads.[2]

A download in one world did nothing to the chart in another. Different groups of listeners encountered the same recordings, but accumulated different histories. In one experiment, popularity appeared as numbers beside songs in a grid. In the other, the socially

influenced listeners saw a list ranked by popularity. Earlier choices could help determine what the next person noticed.[1]

The stronger the social influence, the more unequal and unpredictable success became across worlds. Appeal still mattered. Songs that attracted listeners without popularity information rarely went from the very best to the very worst, or vice versa. But appeal did not dictate the finishing order.[2]

Lockdown had acquired something more revealing than a good review or a bad one: another result. Its first-place finish was real. So was its fortieth-place finish. Any convincing account of its success now had to leave room for both.

The same question turned up a few years later among projects asking for real money.

On Kickstarter, researchers led by Arnout van de Rijt selected 200 new, unfunded projects. They gave an initial donation to 100 chosen at random. The first backer, whose presence later visitors could see, had picked a project by chance. The researchers then watched who else would put money behind it.[3]

Among projects that received no experimental donation, 39 percent subsequently attracted money from other donors. Among those given the initial donation, 70 percent did. That later support excluded the researchers' own contribution. The first donation had helped bring others behind it.[3]

An observer arriving afterward would see projects with several backers and projects with none. The successful pitches might seem clearer, their ambitions more compelling. Without knowing about the random donation, the observer could give the pitch credit for an advantage assigned by an experiment.

The researchers also tested larger initial advantages. Their effects showed diminishing returns: more manufactured success did not produce proportionately more subsequent success.[3] An initial boost mattered, but increasing it did not keep buying the same additional response.

People can respond sensibly to a record of success while helping create the next success in that record. The later backers' money is real. The project may put it to excellent use. Yet counting those backers does not tell us how much support the project would have received without its experimental head start.

Return to Lockdown. Add another thousand listeners to the world where it finished first. We might learn much more about that audience's enjoyment. Those additional listeners would not recreate the world where the song finished fortieth. More observations of one history and more chances to compare histories are different things.

Markets do more than distribute money and attention. They help determine which alternatives get tried, which improve through use, and which produce a record anyone can inspect. A market can become very busy while leaving a particular question poorly answered: how much did productive work, rather than the path of earlier rewards, contribute to this success?

I call the cumulative chance of rewards going somewhere other than the current favorite the comparison budget. At each opportunity, take the chance that someone else receives the reward, then add those chances across opportunities. The favorite can change. This counts chances, rather than actual rival victories; it does not make every purchase or download an independent experiment. The question is how much opportunity the allocation process leaves for informative comparison.

MusicLab makes the distinction tangible. Forty-eight song titles can remain on a screen even while attention flows mainly toward a few. Keeping the titles available is different from getting the recordings heard. A long list of alternatives can coexist with a thin record of what they can do.

In my allocation models, the comparison budget limits how precisely a reward history can reveal the effects of work, quality, and investment. Under specified conditions, opportunities outside the favorite can dwindle fast enough that even an indefinitely growing record cannot reliably settle attribution. A small chance that persists

can still accumulate. The way those chances change over time matters, alongside their size today.

Success breeding success is an old idea. Robert Merton described how recognition could bring scientists resources for further work. Armen Alchian explained market selection under uncertainty decades before MusicLab.[4] A lead can keep growing while opportunities to check its causes disappear. The growing reward and the shrinking capacity to explain it can be consequences of the same process.

Lockdown's other chart suggests what is missing. I call these feasible alternative histories shadow futures, borrowing words from Steven Johnson's account of Stuart Kauffman's adjacent possible.[5] They share the same underlying economic conditions and rules, but early events differ. Later choices and effort can respond. The recording stays the same; different first listeners leave a different chart for everyone afterward.

Sometimes policy changes which alternatives can become real. In 1956, an American antitrust settlement opened 7,820 existing Bell System patents to royalty-free licensing. Bell Laboratories had helped create technologies including the transistor. Other businesses now had access to thousands of patented inventions without paying royalties.[6]

Decades later, Martin Watzinger and his colleagues studied the consequences. Comparing Bell patents with matched patents outside Bell, they found a lasting increase in subsequent innovation outside telecommunications. They did not find the same increase within telecommunications, where access to customers remained obstructed. Permission to use an invention was not always enough to make a business from it.[6]

The result gives the missing comparison a physical form: equipment that another company might build, offer, and improve. An invention behind a licensing barrier cannot tell us everything about what other inventors could do with it. Opening access changes the rules; it is an intervention, rather than a replay of an unchanged

economy. But it can let possibilities previously confined to speculation develop observable results.

A restriction can prevent useful discovery before it raises a price. A rival may lose its route to a first customer, leaving no finished product for buyers to compare. Preserving that route can matter before there is a price dispute. Its value depends on what another supplier could test and what people could learn, as well as the costs of making the trial possible.

Preserving that route becomes more urgent when the thing being bought can help decide who gets hired, what work gets done, and which ideas receive attention.

Consider a customer-support worker with an AI assistant suggesting replies during a conversation. The worker can accept, edit, or ignore them. This was the setting studied by Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, using data from 5,172 agents. Their published study found an average increase of 15 percent in issues resolved per hour after access to the assistant, with especially strong improvements among less experienced and lower-skilled workers.[7]

Here is a reason to welcome the technology. A newer worker can draw on assistance that helps close a gap in experience. The study also found evidence of learning. AI can make useful knowledge more available to people who have had less time to acquire it. Any account of the future of work has to make room for gains like these.

The researchers examined periods when the AI system went offline and agents continued handling their customer conversations. Workers who had used the assistant still handled conversations faster than they had before gaining access. The pattern was stronger among people with more previous exposure to its suggestions. This was evidence consistent with learning: some benefit could remain when the immediate stream of advice stopped. A worker's improved performance could therefore reflect both what the software supplied now and something the worker had acquired through using it.[7]

That interruption gives the comparison a concrete purpose. Watching a successful assisted conversation tells us what worker and tool accomplished together. Watching work continue during an outage helps investigate what the worker can do when suggestions disappear. Neither record alone describes every possible workplace. Together, they let us ask a sharper question about how a productive ability develops. The distinction matters when an employer decides whether software should guide beginners, handle their tasks itself, or change the training through which the next generation learns the job.

But a customer-support result cannot settle what happens in a software project, any more than Lockdown's winning chart can settle its fortunes everywhere. The work, tools, and opportunities to measure them differ. To see how quickly the comparison itself can change, follow what happened to researchers studying programmers.

METR, an organization that evaluates AI systems, ran an experiment in early 2025 in which experienced developers completed tasks with or without AI tools. In that setting, AI access made completion slower. When METR expanded the research later that year, it encountered a different problem: some developers did not want to participate if participation meant doing work without AI.[8]

They were being offered fifty dollars an hour to work on their own projects. Still, some declined. Others held tasks back from random assignment because they did not want those tasks landing in the condition where AI was forbidden. In surveys, between 30 and 50 percent of developers reported withholding some tasks for that reason.[8]

By February 2026, METR said the newer results gave an unreliable measure of the productivity effect. Changed pay and difficulties measuring time also complicated matters. Its researchers thought AI was likely helping developers more than before, while emphasizing how weakly their data measured the size of that improvement.[8]

Think of a task the programmer keeps out of the study. It may be exactly the task on which AI saves the most time. The developer has a reason to keep using the tool, while the researcher has a reason to

want the missing comparison. Both can be acting sensibly. The work gets done, but that opportunity to measure the difference has gone.

METR began redesigning its experiments. When developers change how they work, researchers have to change how they measure it. The question is how to preserve useful comparisons without asking people to abandon the tools they now rely on.

The same practical question runs from the programmer's desk to the data center. Training a model, running it to produce answers, and comparing it with another system are different activities. Running a trained model is called inference. A service can generate an enormous record of answers while leaving unanswered whether a different model, supplier, or organization of work would serve us better. AI can also make those comparisons cheaper. Which possibility prevails depends on how we deploy it.

The consequences reach the pay packet. Think again of the customer-support worker resolving more problems with AI assistance. The worker is doing useful work. So are the people who built the tool. The employer supplies an organization in which both can operate. Measuring the productivity improvement helps us understand what changed. It does not, on its own, tell us how to divide the additional income among them.

Now remember the Kickstarter projects. Their later support was influenced by a donation the researchers deliberately assigned. In ordinary business, we rarely have such a clearly marked intervention. We see the revenues after customers, reputation, investment, and opportunity have acted on one another. That makes the familiar claim that someone deserves precisely what the market paid them much harder to establish than the claim that they created something valuable.

Taxation enters here because governments must choose rules even when contribution cannot be divided neatly from position. Incentives matter; promises and legitimate entitlements matter. Uncertainty about a fortune's origins supplies no tax rate. It does require us to

explain our tax rules without pretending that every pound or dollar arrives with a reliable account of who caused it.

When closing a route to customers prevents other people from learning something useful, part of the cost may fall outside the business making that choice. Access rules, public trials, or a carefully justified tax could address that loss. Their justification depends on what people have lost the chance to learn, how it affects their decisions, and whether the remedy is worth its costs.

We already know how to treat some comparisons as worth maintaining. At Rothamsted in England, agricultural experiments begun in 1843 have preserved differences in how plots are treated across generations. Soil and plant samples have been kept for more than 180 years. Some plots have received particular fertilizer or manure treatments since the beginning, apart from fallow periods. Methods have also changed to keep the experiments useful.[9]

Those fields are productive in two senses. They grow crops, and their maintained differences help produce knowledge. Maximizing this year's harvest on every plot would answer a different question from the one the continuing experiments make possible. Someone has to value the comparison enough to keep it alive.

A song, a first donation, a patent, an AI assistant: each produces something people can value. Each also changes what happens next. The economy's rewards help decide which possibilities become familiar and which remain out of sight.

Lockdown finished first. Lockdown finished fortieth. The second chart survived because somebody built a place for it. The harder question is what happens to the futures for which we never make room.

The evidence success leaves behind

The advertisement that got the credit

A customer searches for eBay, clicks an advertisement for eBay, and buys something on eBay. For the advertiser, the journey looks reassuringly complete. There was an ad, then a click, then a sale. Each step can be recorded. Money went out to attract the customer; money came back when the customer bought. Here, apparently, is an unusually clean account of a contribution.

Except that the customer was searching for eBay already.

Economists Thomas Blake, Chris Nosko, and Steven Tadelis studied large advertising experiments at the company. When advertising against its own brand terms was suspended, customers largely substituted unpaid search links for paid ones. The brand advertisements had no measurable short-term benefit. Experiments with more general search terms found a different pattern: new and infrequent customers could be influenced, while frequent customers accounted for much spending without a corresponding change in purchases.[10]

One brand-keyword test stopped eBay advertising on Yahoo and Microsoft's search platform while retaining Google advertising as a comparison. Almost all the lost paid-search traffic and attributed sales reappeared through unpaid links. Customers had another route to the shop. Closing the paid route revealed how many sales the advertising ledger had mistakenly credited.[10]

The advertisements existed. Someone designed them, arranged their placement, and paid for them. Some were useful to some customers. The difficult question was the difference they made to a

specified outcome. Counting activity was an excellent way to establish activity. Establishing contribution required something more.

For the shopper who wanted eBay, the alternative was close at hand: another link on a search page. In other settings, the alternative is much harder to recover. It might be another business that never attracted a first customer, another worker who never received an assignment, or another technology that never developed a record of successful use. We can study what happened in great detail while remaining uncertain about what those alternatives would have revealed.

An early reward can affect the opportunity to earn the next reward. When that happens, the market is also helping select the evidence that later observers will use to explain its success. The economy we inhabit is partly a system for producing things and partly a system for deciding which possibilities become observable.

To measure the advertisement's short-term contribution, eBay needed customers who could arrive without it. Recording the paid clicks more accurately would not recreate that comparison.

How success changes the evidence

Economists and sociologists have long studied how success builds on itself. Armen Alchian explained how profits select surviving firms without requiring perfect foresight. Robert Merton described how recognition brings scientists resources for further work. W. Brian Arthur showed how increasing returns can let early adoption influence which technology becomes established.[11]

Network science offers another familiar mechanism. In preferential attachment, a web page that already has many links is more likely to attract another link. Albert-László Barabási and Réka Albert combined that process with network growth to explain a power-law distribution of links, with a small number of highly connected hubs.[12] Growth, lower costs, and unequal outcomes can accompany either plentiful comparisons or very few. Explaining a winner's contribution

requires evidence about the opportunities that actually remain to test alternatives.

Consider a platform that sends more tasks to workers with longer records. The feedback can widen their lead and get useful work done. It also determines what anyone can learn about workers who rarely receive a task. A thousand completed jobs for the favorite might reveal little about someone never hired. The problem concerns the production of evidence, before any judgment about whether the winner deserves the income.

The work and the difference it made

I use contribution uncertainty for uncertainty about the effect of productive inputs on rewards in a specified economic setting. The inputs might include effort, quality, investment, or risk-bearing. The question concerns how an outcome would change if an input changed, while the relevant economic rules continued to apply. A result can be valuable, and the work behind it entirely real, while the size of that effect remains difficult to recover.[13]

Increasing visits to a website differs from increasing purchases. Attracting a new customer differs from giving an existing customer a more convenient route. A short-term experiment may leave longer-term effects unresolved. Contribution has a specific outcome, comparison, and time horizon.

Lockdown's two rankings raise a related question with a different design. MusicLab preserved independently developing markets for the same songs. The song's first-place finish and its fortieth-place finish were both outcomes of listeners choosing. They tell us that one observed ranking cannot be treated as a complete account of what that recording was capable of achieving. They do not calculate the value of every rehearsal or assign a percentage of the song's success to talent.[14]

A recording can give pleasure in several possible histories and receive very different amounts of attention in each. The pleasure is

real. How much of the reward it explains remains a question about causes.

In George Akerlof's account of the used-car market, the seller may know more about the car's quality than the buyer does. A buyer unable to distinguish good cars from bad ones may offer a price that drives good cars away. The information problem changes which trades take place.[15]

My starting question allows the inputs to be observed. Suppose the buyer can inspect the work, rather than suspecting a hidden defect. That still leaves the effect of those inputs on the outcome to be learned. Knowing how much an advertising campaign cost does not identify the sales it created. Knowing what a musician recorded does not reveal every audience the recording could have found. Better disclosure and better comparison can solve different problems.

Kickstarter's experimental first donations helped some projects attract subsequent backing. In an ordinary business, an early customer can also finance better engineering, more practice, and a stronger organization. The advantage becomes productive when it improves what later customers receive. Asking whether the business is good and asking how it became good are separate inquiries.[16]

Who gets a chance

Each choice does two things: it allocates resources now and helps determine what the next observer will be able to inspect. An expanding record of success can therefore coexist with fewer opportunities to evaluate alternatives.

For each allocation, given the history so far, identify the candidate with the greatest chance of receiving the reward. Take the chance that someone else receives it instead. Accumulate those chances across the opportunities under study. This total is the comparison budget. The favorite can change. The total concerns the chances offered to alternatives, rather than the number of times an outsider actually wins.[13]

A list of awards and rejections does not, by itself, tell us those chances. The budget is not a count of winners or the number of firms on a website. To apply the idea empirically, we need evidence about the allocation process: which alternatives are eligible, what changes their prospects, and what is recorded beyond the award. The quantities cannot be read straight from market shares.

MusicLab helps us see why a catalogue is insufficient. A title remaining on a list does not mean listeners encounter it on comparable terms. Conversely, a single business can arrange useful tests of alternatives, as eBay did with advertising. Counting organizations and counting informative opportunities are different inquiries. Institutional design can matter as much as the number of logos on the screen.

Reinforcement had already been connected to estimation failure in urn models by Line Chloé Le Goff and Philippe Soulier.[17] In my allocation models, the comparison budget places an upper bound on the information the reward record supplies about productive input effects. Under further conditions, a finite lifetime budget prevents any one method from reliably recovering contribution across every explanation the model allows. Independent measurements can change the available evidence.[13]

A tiny but persistent chance of selecting an outsider can keep accumulating. A shrinking chance can also accumulate without limit if it shrinks slowly enough. Even then, comparisons must distinguish the input effects we care about. A trial that never varies the relevant engineering tells us little about its contribution, however often it runs.

Another transaction can demonstrate that an established service continues to work well while revealing little about an untested alternative. Commercial value and information about a particular comparison are separate properties of the same event.

Field experiments, independent technical evaluations, and measured outcomes can distinguish contributions that an award record leaves uncertain. Their information adds to what can be learned.

eBay could change where advertising appeared and observe what followed. Its advertising ledger did not have to remain the last word. A feasible intervention could supply the comparison that the existing record lacked.

Other possible histories

The seed of this book's name was a phrase in Steven Johnson's account of Stuart A. Kauffman's adjacent possible. Kauffman, author of *At Home in the Universe*, explored self-organization and complexity. The adjacent possible describes possibilities reachable from what already exists. Johnson called it "a kind of shadow future." Existing materials and discoveries put some next steps within reach; taking them can make further combinations possible.[18]

Think of Bell's transistor. An invention can become a component in devices its inventors never built. Who can work with it matters to which possibilities are explored. Johnson's image made me ask an economic question: what happens to possibilities that depend on opportunities going to someone who never gets them?

I chose *shadow futures* because the phrase holds together two views of the same opportunity. Before a customer places an order, more than one business may have a future that could grow from it. Afterward, we can follow the business that won: the revenue it receives, the investment it makes, the work it learns to do. The other path recedes from view. It may never leave enough of a record for us to judge what could have developed there.

I use the term for feasible alternative allocation histories with the same starting conditions, underlying productive effects, and rules for how people respond. The first customer can change; so can the later investment financed by that first sale. The rules stay fixed while events unfold differently. Kauffman's adjacent possible concerns the opening of new possibilities. My question concerns what we can learn when only one of the available paths becomes observable.

MusicLab makes the idea visible. Lockdown could finish first or fortieth without becoming a different recording. The researchers kept

the other chart. An ordinary business rarely gets an equivalent record of the opportunities it missed ten years ago.

The comparison budget connects those missing paths to the paper's impossibility result. It adds the chances of awards going outside the current favorite. Under the model's rules for how productive work affects those chances, a finite lifetime total can leave different accounts of contribution impossible to distinguish reliably from the reward record. Even watching that record grow indefinitely cannot provide a method that recovers the right contribution across every explanation the model allows.[13]

That is why I chose a phrase about unrealized futures for a result about knowledge. A path we do not observe can contain a comparison we need to explain the one we do. Another order for the winner adds to its history. It need not tell us what the rival would have learned, built, or earned if an earlier order had gone the other way.

This book is itself a product of the adjacent possible. Kauffman's work, Johnson's image, and the economic literature made a further question available to me. Its future will depend on who encounters it and what they can do with it. A professional economist may give these ideas a larger audience, sooner than I can. Credentials, an established readership, and access to the right conversations can open doors that an argument alone cannot. That is another way opportunity can shape an idea's reception.

If someone else tests the idea more rigorously, explains it better, or helps more people use it, I will be content with that outcome. Credit matters, as does the freedom to challenge a claim. What I do not want is an epistemic monopoly over an idea that could benefit humanity. I want the comparison to remain open, including when an alternative to my account proves better. I cannot ask readers to take shadow futures seriously and then assume that the history giving me the most recognition must be the best one.

Changing an early award within unchanged rules explores another history. Changing the rules is an intervention. eBay altered an arrangement and measured the response. A simulated alteration

would also need evidence that its programmed responses fit the actual market.

Predicting what comes next and explaining why the outcome has the shape it does answer different questions. In Lockdown's winning world, an observer who expects more listeners to choose an already popular song could make useful forecasts. Those forecasts would not reveal the other world's chart. Getting the next choice right does not itself account for the song's lead.

Blackwell and Dubins showed that, under specified conditions, observers' forecasts converge as evidence accumulates. Hao Zhang later demonstrated a statistical setting in which useful prediction coexists with failure to recover particular underlying parameters. Prediction and explanation can demand different things from the same record.[19]

An early research grant makes the stakes concrete. Thijs Bol, Mathijs de Vaan, and Arnout van de Rijt studied Dutch scientists whose applications fell just above or below a funding threshold. The near-winners subsequently accumulated more than twice as much further funding over eight years as the near-losers, excluding the initial grant itself. Part of the gap arose because unsuccessful applicants entered later funding competitions less often. The distinction involved who continued to present work for evaluation, as well as how evaluators treated it.[20]

The research proposal that never reaches another panel leaves a different kind of absence from the one a rejected proposal leaves. We can inspect the latter decision. The former requires asking which earlier events affected participation. When opportunity influences the decision to compete again, a record of later funding awards can conceal part of the process producing its own pool of candidates. That is a reason to follow the people outside the winner's history, even when the published results look complete.

From evidence to institutions

At the customer-support desk studied by Brynjolfsson, Li, and Raymond, AI assistance helped workers resolve more issues per hour on average. The gains were particularly strong among less experienced and lower-skilled workers. Useful capability was reaching people who could benefit from it.[21]

Training future workers raises a further question. Today's assistant might help beginners learn, or do the tasks through which they would have learned. Comparing their later abilities requires following what they attempt, the guidance they receive, and what they can do afterward.

METR's developer research exposes one difficulty. As programmers became less willing to do certain work without AI, the research team found that participation and task selection made its follow-up estimates difficult to interpret. The work being studied was changing which comparisons people would supply. The team responded by reconsidering its design. Learning about the technology required attention to the institutions producing evidence about it.[22]

The reach of a judgment changes the stakes. Smith's artisan faced a market limited by transport; another town could offer another buyer. A platform can carry one rating across towns, while a shared infrastructure provider can control the route used by many apparently separate businesses. Physical limits persist, and old guilds also obstructed entry. What changes when a common gate spans the market is how far an exclusion travels.[23]

More computing can make experiments cheaper while routing more work through the same model, service, or evaluator. Where one institution controls the comparisons others need to judge alternatives, it holds an epistemic monopoly over those comparisons.[24] Productive scale and control over discovery can grow together.

Bell's 1956 settlement provides a historical warning against treating access as a slogan. Opening existing patents to royalty-free licensing was followed by greater innovation outside

telecommunications. The study did not find the same result inside the sector, where barriers to reaching customers remained. A policy can loosen one constraint while leaving another in place. An entrant needs a usable route from permission to performance.[25]

Lost comparison can impose a cost that never appears on the customer's bill. A gatekeeper may improve its service while closing a route others need to develop or test theirs. When that closure deprives other people of valuable evidence, part of its cost falls outside the gatekeeper's decision. Many buyers can also favor the same route without one supplier controlling the whole market. Their transactions record the work completed; the missing trial leaves no failed product to count.

For such a market, waiting for a price increase can mean waiting until an alternative has lost its feasible beginning. Existing competition policy considers innovation and exclusion as well as prices.[26] Protecting a workable route to independent evaluation gives those concerns an earlier point of action. Credible threats of lasting closure should trigger a presumption for keeping that route open, with security, investment, and the costs of access included in its terms.

For taxation, explaining an income and changing future incentives are different tasks. The reward record may fail to settle how much contribution produced a fortune. Conduct that prevents useful comparison may also impose costs on others. Establishing those costs can justify a corrective charge. Financing shared research and wider access to productive capital provides a further reason for taxation, without first proving that particular recipients were morally undeserving.

Consider a capable team that cannot afford its first credible trial while existing owners use their returns to consolidate the routes to customers. Leaving both positions untouched is a choice about the next opportunity. Using part of capital income and large inheritances to finance shared research and broader access to assets is another. The second can support an economy with room for new judgments

and new owners. Its tax rates and spending choices must be assessed through investment, avoidance, administration, and the benefits people actually receive.

An independent test, access to existing records, and a change in market rules produce different evidence at different costs. A useful remedy improves a consequential decision enough to justify those costs. Sometimes the evidence will favor keeping the existing arrangement.

A different account of success

At Rothamsted, continuing agricultural experiments make the value of maintained comparison unusually tangible. Different treatments have been preserved while methods have also been adapted to keep the experiments relevant. The record matters because someone has continued the arrangements that produce it. Useful knowledge has a material history: plots, treatments, samples, labor, and decisions about which differences to keep.[27]

A first customer can finance a better product. A patent license can open a field of work. A distribution agreement can close it again. These decisions help determine whose ability gets demonstrated, how capital accumulates, and what competition policy and taxation can change.

A complete record of the eBay customer's paid click could have encouraged a confident story about the advertisement. Making another route observable changed the explanation. In more consequential settings, the missing route may be a business that never reaches a customer. Its absence from the record is precisely what makes it easy to overlook.

The winner and the explanation

An ending looking for a beginning

Put Lockdown at the top of the chart and imagine that the other chart has disappeared. The first thing we know is that listeners selected it. The next thing we may want to know is what made them do so. We could listen for a memorable passage, compare the song with its competitors, and form a view about the audience. A good explanation might begin with something that really is in the recording. The trouble starts when we decide that the ranking has told us how much that feature mattered.

Lockdown's fortieth-place finish exposes the gap between a recording's appeal and its rank. Appeal could help the song in both worlds. The different rankings also depended on the sequence of attention, choices, and displayed popularity it met in each one.[28]

Follow a MusicLab listener. The website offered unfamiliar songs by unfamiliar bands. A participant chose a recording, listened, and was asked to rate it on a scale from one star to five. Afterward came an opportunity to download it. Downloading was optional. The experiment recorded several steps between a title appearing on a screen and a song acquiring another mark of popularity.[28]

A rating and a download answered different questions. The rating expressed a response after hearing the recording; in the social-influence worlds, the download entered the popularity record shown to subsequent listeners. Imagine someone who liked a track but chose not to keep it. That person's favorable judgment would not become another download on the chart. Even this simple market

contained more than one way of being successful. The researchers had to choose which outcome to explain.

Now change the screen. In the first experiment, songs appeared in a grid, with positions assigned randomly for each participant. Some listeners also saw download counts. In the second, socially influenced listeners saw a single column ordered by popularity. An early download could now help move a title up the list encountered by the next listener. The recording had stayed fixed while its route to an audience changed.[28]

Think of a title near the bottom. To attract a listener, it has to be noticed among alternatives carrying more evidence of earlier attention. Once heard, it still has to win a download. A final download total combines those stages. A musician interested in reaching listeners would want to know where the obstacle lay: getting the title seen, getting the song played, or getting someone to keep it. The total alone cannot separate them.

Looking first at popular titles can make sense. A listener with ten spare minutes has a practical search problem. Earlier downloads may offer a useful shortcut. But that shortcut helps decide which recordings receive the next hearing. The chart both reports choices and influences the conditions under which further choices occur. It cannot be read as forty-eight recordings repeatedly taking the same unchanged test.

For Lockdown, separate worlds let the researchers investigate what an ordinary chart conceals. Each world began without a download advantage, then acquired its own record. An admirer hearing the first-place song has evidence about an encounter with that recording. The fortieth-place world adds evidence about the process through which encounters accumulate. Enjoying the song answers a real question. Explaining its finishing position requires the wider record that the experiment preserved.

Ordinary economic success comes with a similar invitation. We see the company that reached customers, the worker who developed a reputation, or the project that secured its backing. We can inspect

choices those winners made. What is harder to inspect is how another beginning might have changed the circumstances under which the same choices were available or effective. That is why evidence about the beginning deserves attention before we start reading the ending as a verdict.

Giving success away

The Kickstarter researchers did something direct about beginnings. They intervened before selected projects had raised any money. The projects already had proposals, ambitions, and creators. The researchers did not change those proposals. They changed whether an early visitor encountered a project with a first donation or a project still waiting for one. This allowed later support to be compared with a difference in initial success deliberately introduced by the research design.[29]

Improving a pitch and adding its first backer could both increase subsequent funding. Without the experimental assignment, an observer might mistake the second mechanism for the first. A creator could offer a polished explanation of why a successful pitch worked, and much of that explanation might be sensible. It would still need to account for the advantage that came from already having support.

The experiment found that projects given an initial donation were more likely to attract subsequent backing. Later support was counted separately from the researchers' donation. It was not enough to announce that the treated projects finished with more money, since the researchers had put money into them. The revealing result concerned what other people did afterward. The design separated the initial gift from the additional support that followed it.

That gives us a useful discipline for other claims about success. When someone says an investment brought a return, ask which part of the return has been counted. A first customer supplies revenue directly and may help attract the next customer. Those are different benefits. If we count the initial payment as evidence of the power to

attract later payments, we have smuggled the intervention into the outcome. The Kickstarter design avoided that elementary trap.

The team did not stop at crowdfunding. Its experiments also intervened in product-review ratings, Wikipedia awards, and petition signatures. For Wikipedia, the subjects were highly productive editors; a randomly selected subset received a community award. The researchers then observed further awards from other people. Recognition could influence subsequent recognition even among people who were already contributing substantial work.[29]

The editor is a particularly instructive figure. The work exists before the experimental award. A later observer who sees several honors can reasonably think the editor has done useful things. That may be entirely correct. What the number of honors cannot establish unaided is how much each later recognition reflects the work and how much reflects the fact that the editor has already been recognized. Genuine contribution and a feedback effect can occupy the same page.

There is also no need to suppose that every later donor or award-giver merely copies a number. Early support can invite closer inspection. Closer inspection can reveal something worth supporting. An institution may benefit from such shortcuts when nobody can investigate every candidate. The experimental result concerns the consequence of receiving the early advantage. Determining the precise channel, and whether it improves decisions, requires further evidence rather than an assumption that influence is automatically foolish.

The researchers' larger interventions supply a useful warning against a seductive sequel. They compared smaller and larger manufactured advantages and found diminishing marginal returns. In the funding treatment, moving from one experimental donor to four did not create a proportionate multiplication of later success. A system can amplify the first step and respond much less strongly to the next few. The force of feedback is a question about its rules and setting.[29]

That result prevents us from taking the favorite story about a snowball and making it the entire theory. Some early differences persist. Some run out of force. Some are overcome by other differences. If every unequal result were offered as evidence of irresistible cumulative advantage, nothing could count against the idea. The experiments are useful precisely because they can show both amplification and its limits.

For a project creator, the practical difference might be between improving the proposal and finding a way to secure credible initial backing. Both could deserve effort. An observer trying to learn which bottleneck matters should not treat the final funding total as if it had already performed that diagnosis. The first donor has changed the situation in which the next donor decides. Understanding the record means following that change through time.

The extra vote

A similar problem can begin with something smaller than a donation. In an experiment published in 2013, Lev Muchnik, Sinan Aral, and Sean Taylor randomly gave online comments an initial positive vote, a negative vote, or no vote. Over five months the study covered 101,281 comments on a social news website. The manipulation changed a visible score attached to words that had already been written.[30]

An artificial positive vote increased the likelihood that the next viewer would vote positively by 32 percent relative to the control group. Final mean scores were 25 percent higher. The percentages describe relative differences in those outcomes; they do not mean that a third of readers had been converted from disagreement to agreement. One extra mark on the screen had changed the subsequent path of evaluation.[30]

Here the productive object is unusually easy to hold in view. The comment did not become better written when the researchers clicked the positive button. A later reader could still find a good argument in it. But the experimental advantage was not an improvement in the argument. If a final score were used as a precise measure of the

quality of the words, some of the effect of the initial vote would be assigned to the writing.

The study's site did not order comments by their popularity. That is a useful detail because it stops us from explaining the whole result as the familiar consequence of being moved higher on a ranked page. The displayed score was tied to the comment when people evaluated it. Social information could affect the subsequent rating even without that particular route through a popularity ranking.[30]

A comment score can help a reader find something useful while also reflecting the votes earlier readers happened to see. Its convenience and its reliability as a measure of quality depend on different features of the rating process.

Negative votes behaved differently. Readers tended to correct the artificial negative treatment, leaving final mean ratings without a statistically detectable difference from controls. The upward and downward nudges were not symmetrical.[30]

The topic mattered too. The researchers detected positive herding in politics, culture and society, and business. They did not detect it in economics, information technology, fun, or general news. Those categories had comparable activity levels. The same button and the same visible score did not produce a uniform response across the site.[30]

Compare the comment with Lockdown. In both settings, someone encounters an object together with information about earlier responses. The object may be appealing. The earlier response may also matter. The observer who sees only the accumulated result faces a mixture of those influences. The two experiments make the mixture easier to study by introducing or separating the relevant histories.

What they establish is specific enough to be useful. An early signal can affect subsequent judgment. It need not determine every outcome, work equally in both directions, or leave the underlying work unchanged in a longer economic setting. We have evidence for a mechanism and reasons to investigate its strength. We do not yet have

a general permission to read all success as accidental or all rankings as worthless.

Recognition becomes an opportunity

In his account of scientific recognition, Robert Merton used the image of a forty-first chair. The French Academy had forty places, but the history of French thought contained distinguished people who never occupied one. Merton's examples included Descartes, Molière, and Proust. An institution with a fixed number of honors cannot turn the boundary between its members and everyone else into a complete ranking of achievement.[31]

There are forty available chairs. An outstanding person can remain outside without anyone needing to conclude that the work was poor. A prize can be prestigious because its recipients are excellent, while failing to include everyone of comparable excellence. The quality of the winners and the completeness of the distinction are separate properties.

Merton went further. Recognition could become a resource for future research. Eminent scientists could receive disproportionate credit for contributions, and credit could help bring facilities for further work. His argument drew in part on Harriet Zuckerman's interviews with Nobel laureates. The honor was both an account of earlier achievement and a possible influence on what the scientist could do next.[31]

That second role changes the economic question. An extra chair would alter an honor roll. Access to a laboratory can alter the work that later appears on the roll. If we inspect achievement only after the opportunities have diverged, we may find a real difference in productive capacity. The difference can be an outcome of the earlier allocation as well as a reason for the later one.

The Dutch grant study introduced earlier makes this process less dependent on a striking phrase. Bol, de Vaan, and van de Rijt examined applicants close to a funding threshold. Those just above it later accumulated more than twice as much additional grant money

over eight years as those just below. The original award was excluded from that total. A narrow early distinction was followed by a much wider difference in further funding.[32]

An applicant just above the funding threshold had won; one just below it had lost. Comparing those nearby applicants reduced the differences that would separate the entire pool of winners from the entire pool of rejected scientists. Their talent could still vary. The study's design and assumptions made it possible to investigate the award's effects without assigning the whole later gap to initial superiority.

The competitions had names: Veni for the early-career award and Vidi for the midcareer grant. Among applicants closest to the Veni threshold, about 59 percent of winners later applied for Vidi, compared with 40 percent of nonwinners. The researchers checked whether the application difference simply reflected people leaving academia or the country; it did not. An earlier decision had changed which proposals reached a later panel, before that panel judged anything.[32]

A scientist can stop applying, a supplier can stop bidding, and a musician can stop trying to reach an audience. The alternatives do not wait patiently for evaluation. An apparently thin field may reflect earlier decisions that drove people away. The missing proposal is harder to inspect than one that received a low score.

There are several possible practical responses, and they need different evidence. Helping promising unsuccessful applicants reapply would address participation. Giving a reviewer a clearer account of how close a prior decision was might affect the interpretation of a failure. Funding everyone would involve a much larger commitment with different costs. The study makes a mechanism visible; choosing a response requires asking which change would improve the institution's work.

Subsequent funding gives a scientist resources for further work. The discoveries made with those resources are another outcome. More money might permit important research, but the size of the

grant cannot tell us how much scientific value it produced. An institution can create a gap in resources and a gap in recognition without those gaps carrying identical information about achievement.

The comparison with the online comment is instructive. An artificial vote changes the evaluation of fixed words. A grant can help change the future words, experiments, and projects themselves. Both involve feedback, but the productive consequences need not be the same. Understanding cumulative advantage requires following what the reward does after it is received, rather than treating attention, money, equipment, and achievement as interchangeable quantities.

What failure selects

A story about early advantage should be able to survive an encounter with a successful loser. In a 2019 study, Yang Wang, Benjamin Jones, and Dashun Wang compared junior scientists whose applications narrowly received or missed US National Institutes of Health funding. Early rejection was associated with greater departure from the NIH system. Among those who remained active, however, near-misses subsequently produced work with greater citation impact than narrow winners.[33]

Among those who stayed, some did distinguished work after an early rejection. Others left the funding system. Judging the setback only by the survivors would omit those departures; judging it only by departures would omit the later achievements. Both groups were part of the original applicant pool.

For a scientist still submitting proposals, the ability to continue may be part of what distinguishes that person from someone who stops. Later success among survivors does not by itself show that giving everyone the same early setback would improve scientific work. The researchers investigated whether selection alone could explain their findings and argued that it did not fully do so.[33]

Imagine reading a shelf of papers by researchers who overcame rejection. The papers are real achievements. The biographies might offer useful accounts of persistence. But the shelf contains no place for

papers that were never written by people who lacked the resources, inclination, or opportunity to continue. Admiring the work on the shelf and evaluating the institution's treatment of the original applicants are different tasks. One requires attention to achievements; the other also requires attention to absences.

This is the same discipline we needed for Lockdown's missing chart. An outcome can be impressive without revealing every route by which the observed group came to exist. A scientist's later work may genuinely be better. The additional question is what an observer can infer about a process that also affects who remains visible. The existence of a successful survivor cannot, unaided, make the cost of the missing careers disappear.

Nor should we turn the funding result into a convenient reversal of the Matthew effect. Further funding and citation impact measure different outcomes. A person can receive less money and produce influential work. Another can receive more money for research that is valuable in ways the chosen citation measure does not capture. The question of which reward predicts which achievement must remain specific enough for evidence to answer it.

Subsequent researchers tested both patterns across a wider set of funding programs. A 2025 reviewed preprint by Vincent Traag and colleagues examined fourteen programs from six funders in Europe and North America. It found the relationship between early and later funding more robust than the apparent advantage in later citation impact following rejection. The latter varied with setting and method; selection into reapplication offered an important explanation.[34]

That is useful news for anyone tempted by a universal story. A striking finding about one group can identify a mechanism worth studying without becoming a rule for all careers. Replication can change our confidence about how far it travels. If a comparison is valuable enough to influence policy, it is valuable enough to examine under arrangements that might reveal its limits.

For a funding agency, the practical choice is not whether to believe that winners are talented or that losers are resilient. It is how to

allocate resources while learning what those allocations do. Does an early award change later applications? Does it help people do better work? Does a rejection remove promising candidates from future consideration? Different records are needed for the different answers. A single later honor roll cannot contain them all.

An early rejection need not settle a scientist's career. Another funder, laboratory or employer can provide a route back. People respond to setbacks, and institutions can revise earlier judgments. Which opportunities remain available, to whom, and for how long helps determine whether an early disadvantage persists.

The road with the fuel

Long before online charts, Armen Alchian offered an economical way to separate a successful outcome from a wise decision. His 1950 paper imagined thousands of travelers leaving Chicago along randomly chosen roads. Gasoline stations existed on only one road. Travelers who happened to choose it could continue; the others would run out of fuel. The continuing travelers might look unusually far-sighted to someone who saw only their progress.[35]

Alchian's environment selected surviving behavior without requiring every individual to understand the selection in advance. If gasoline supplies moved to another road, the pattern of continued travel would change even without travelers changing their paths.[35]

The travelers made real progress along the fueled road. Someone who saw only their continued movement might mistake survival for foresight. Knowing how they chose their roads changes that explanation without changing the distance they traveled.

In the grant studies, the scientist who keeps applying remains eligible for later awards. The one who stops no longer appears in the same competition. Their continued presence or absence says little on its own about the earlier experiences that changed their choices and opportunities.

If the surviving travelers also learned along the way, their lucky route could support real improvements in skill. Selection and learning could then operate together. An early advantage can make productive work possible without measuring how much that work caused the eventual reward.

For a Wikipedia editor, recognition can bring another person to a useful contribution. For a business, revenue can finance improvements. These are different mechanisms. Alchian's example exposes an inference from survival; explaining a particular reward system still requires investigating how it selects, rewards, and develops its participants.

Markets can select viable activities under uncertainty. People can imitate, adapt, and learn without possessing a perfect account of the economy. Choosing a business that works is one task; measuring how much each input contributed to its rewards is another. The first can be useful while the second remains unresolved.

The response should be to examine what would discriminate between explanations. In the travelers' example, discovering where the fuel was located changes the interpretation of the journey. In the online experiments, the researcher's record of the assigned advantage changes the interpretation of the later score. In a real procurement process, a useful independent trial might do similar work. Each additional observation earns its value by answering a question the winner's progress could not settle alone.

We can therefore learn from a successful traveler without treating every mile as a certificate of foresight. The same discipline lets us learn from a successful company without turning every sale into a complete causal audit. It asks us to understand the selection process as well as the selected outcome, and to notice which comparisons that process leaves available for the next observer.

The car the buyer can inspect

Akerlof's used-car model makes another kind of missing evidence easy to picture. A seller learns whether the car is good through ownership. A prospective buyer cannot distinguish its quality as well. The offered price reflects that uncertainty, and a good car's owner may refuse to sell at it. The information problem can remove good cars from the set the buyer gets to choose among. Quality and selection interact.[36]

The buyer's uncertainty helps determine which cars actually appear in this market. Inspecting only cars offered at the resulting price would not show the full population of cars their owners might have sold under different information arrangements. The missing good car matters even though no transaction involving it appears in the sales record. Akerlof's example illustrates how a market's rules of information can help select its observable contents.

The contribution problem I want to isolate can remain even after a different uncertainty has been resolved. Suppose, in an explicitly imagined extension, an inspection tells the buyer the condition of each car accurately. The hidden-defect problem in the example has changed. But that knowledge would not tell a dealership how much its advertising added to sales, how much a salesperson's work affected a customer's choice, or how a prior reputation brought that customer through the door.

This does not make inspections unhelpful. An inspection answers a valuable question about the car. It may enable a trade that otherwise would not occur. The mistake would be asking the inspection to provide an account of every input's effect on the dealership's income. A certificate of condition and an estimate of contribution are different products, even if both are called information.

eBay's paid-search study provides a real counterpart. There was no need to doubt that an advertisement had appeared or that a customer had clicked. The records could establish the event. The experimental suspension showed how much apparent advertising success could be supplied through another route instead. The question was about an

effect, and the added evidence came from changing the conditions under which that effect could be observed.[37]

It helps to picture the two records side by side. In one, the customer follows the paid link. In the other, the customer can reach the same destination without that advertisement. We are interested in the difference in purchasing, rather than the mere existence of the first route. A detailed account of the first route is useful for running a website. The second arrangement is useful for investigating what the advertisement contributes.

Broader searches required another test. Someone typing “cell phone” need not have eBay in mind. Using Google's geographic bidding system, the researchers stopped nonbrand advertising for sixty days in 68 US market areas, selected using a random draw followed by matching on past sales. Sales could then be compared with places where advertising continued. New and infrequent customers responded more than frequent customers. An advertisement could be useful to a shopper without creating the purchase that its click would later appear to explain.[37]

This is where a familiar response to disputes about success reaches its limit. Show the work. Publish the records. Demonstrate that the product functions. Those are useful demands, and sometimes they settle the question actually being asked. But they establish contribution only to the extent that the observations distinguish the relevant alternatives. When the dispute concerns how much a factor changed an outcome, evidence that the factor existed can leave a substantial part unresolved.

In MusicLab, everyone could in principle inspect the same recording. The result did not require one world to receive a secretly defective version. The experimental setting separates the song from some of the history through which listeners encounter it. A recording can therefore be known and its possible market outcomes still varied. This is the observed-input question in a form that does not depend on concealing the product.

Making a useful product establishes an achievement. Measuring how much of its maker's income that achievement caused requires a further comparison. Respect for the work can coexist with uncertainty about the size of the reward attributable to it.

What success makes possible

An early advantage can finance productive capabilities that become valuable in their own right. Experience with customers can improve a product; revenue can pay for research. Later success may then have better immediate reasons than the original allocation did.

The improvement does not identify its own origin. Opportunity and the response to it have acted together. Another recipient might have developed an equally useful capability, a worse one, or something different. Observing the actual improvement cannot choose among those possibilities.

The work performed along the successful path remains real. A useful explanation must account for both its productive effect and the circumstances that made it feasible. Labeling the first advantage luck cannot measure the later work away; documenting that work cannot establish what a different initial allocation would have produced.

Obtaining the missing comparison can be expensive. Switching systems, training staff, or running an experiment consumes resources and may interrupt reliable service. The economic question is which uncertainty matters enough to justify investigating, and whether a smaller test can resolve it.

Bell's patent opening makes the distinction between knowledge and opportunity concrete. The historical study found that access to inventions and access to customers were different constraints. Outsiders could receive permission to use knowledge while remaining unable to turn it into a market record in the obstructed sector.[38]

Recognition can invite attention to existing work; revenue can finance new capability. Both change the record the next evaluator encounters. Treating every later award as a fresh judgment on

unchanged starting abilities misses how earlier awards shaped the conditions of judgment.

The grant studies reveal another response: participation itself can change after an early result. When preparing an application is costly, repeated rejection can make another attempt unattractive. The observed field of candidates is partly an outcome of the reward process, rather than a fixed list on which that process merely acts.

By then a directory of remaining suppliers may conceal as much history as it reveals. It tells the buyer who is available now. It does not show all the firms that could have developed under different earlier allocations. This is one reason concentration deserves careful description. A smaller field could reflect superior performance, productive scale, restricted access, discouraged entry, or several influences together. Counting the survivors cannot settle the account of how they became the survivors.

Turning the chart upside down

Salganik and Watts devised a particularly revealing challenge to the idea that a displayed record simply reports what an audience wants. In another MusicLab experiment, published in 2008, they allowed a popularity order to develop and then reversed the displayed download counts in two new worlds. A song with many downloads was presented as having few; one with few was presented as having many. After this intervention, new downloads were recorded honestly.[39]

The swap could be remarkably concrete. Before inversion, She Said by Parker Theory had 128 downloads and Florence by Post Break Tragedy had nine. In the altered displays, their apparent counts were exchanged. No band needed to improve a recording for the screen to give later listeners a different account of its success. The researchers had changed the history people believed they were entering.[39]

The first MusicLab experiment followed independently developing histories under unchanged arrangements. The later experiment deliberately altered the popularity record to test how people

responded. It changed the chart that listeners saw, even though the recordings stayed the same.

Some apparent popularity became self-fulfilling. Yet the inversion did not permanently reverse the market as a whole. The songs with the greatest independent appeal tended to recover their popularity. The manipulation also reduced overall downloads. The result supports influence while showing that influence encounters preferences and can affect how much people take from the market, as well as which song receives their attention.[39]

That recovery is an important part of the ending. The best songs were not helpless objects waiting for a number to tell them what they were worth. Listeners still heard recordings and responded to them. A useful argument about history needs space for those responses to overcome some disadvantages. Otherwise, it would have no way to explain why the inversion did less than a fully self-fulfilling account might predict.

The lower download total raises a separate question about the whole market. A band could benefit from a favorable display while listeners collectively found less music they wanted to keep. An individual's rank and an audience's total enjoyment need not move together. The experiment measures downloads rather than every aspect of welfare, but it makes clear why explaining who wins is different from evaluating the outcome for everyone affected.

For Lockdown's first chart, we now have several warnings against explaining too much from the ending. Initial support can attract further support. A positive vote can influence later evaluation. A grant can change who applies again. A manipulated chart can divert attention, and strong preferences can also resist it. The evidence gives us several mechanisms, rather than a single principle under which every winner is either inevitable or arbitrary.

The next step is to ask what the record itself can teach us once those mechanisms are operating. Suppose another thousand choices arrive. They might test something new, or repeat a route already strongly favored by the existing history. We need an account of the

opportunities for informative comparison inside that sequence. Counting transactions is an appealing place to start, but it cannot be the end of the inquiry.

The comparison budget

A movie behind several doors

The film is Good Will Hunting. One image emphasizes Matt Damon and Minnie Driver; another emphasizes Robin Williams. In a 2017 account of artwork personalization, Netflix engineers used these possibilities to illustrate a choice their system had to make. A member drawn to romance might respond to one invitation, a member drawn to comedy to another. The same film could be introduced through different parts of its appeal.[40]

There is a small economic experiment hidden in that image. Showing an invitation supplies a chance for somebody to respond. Showing a different invitation supplies a chance to learn something else. An artwork that remains in a designer's folder can be beautiful, but the folder does not contain the audience's response to seeing it. An additional play following the established image does not automatically answer the question about the image still in the folder.

Netflix's engineers described a practical response: deliberately vary some selections, record how they were selected, and use responses to improve later choices. They explained that learning could carry an immediate cost when a member received an image other than the one currently predicted to work best. Their account concerned how to combine that experimentation with personalization, rather than simply displaying the same apparent winner indefinitely.[40]

A company with a large audience can create comparisons inside its own service. The same scale that helps an established choice dominate can spread the cost of testing another. What matters is how the service assigns those opportunities.

The image also keeps us honest about the question. Comparing two invitations to watch Good Will Hunting would not measure the

contribution of every actor, the director, the screenwriter, or the company carrying the film. A useful experiment can answer a narrow question very well. Its strength depends partly on resisting the urge to attach every neighboring question to the same result.

The engineers also asked what happens when an image changes between visits. A member might remember a film by its picture and then fail to recognize the new invitation. Or the replacement might finally persuade the member to try it. If several pictures preceded a play, deciding which deserved credit became harder. The team treated the frequency of image changes as part of the learning problem.[40]

Getting somebody to press play was not enough, either. A striking picture could attract an audience that soon abandoned the film. The engineers described using the quality of engagement when labeling their observations, to avoid rewarding that kind of clickbait. Now the record needed to connect the member, title, image, and subsequent response. A million presentations would be less revealing if the file lost those connections.[40]

The comparison budget names one feature of the process producing such records: accumulated allocation chance outside the current favorite. Brief probability examples will make that accounting concrete before the documented systems show what an application additionally requires.

What we add up

Start with two alternatives that each have a one-half chance of receiving an allocation. Either can be designated the current favorite when they tie. The chance outside it is one half, which becomes the first addition to the comparison budget. This is a probability before the result is known, rather than a prediction of the realized winner.

At the next allocation, suppose the leading alternative has a three-in-four chance. The chance outside it contributes one quarter. If the other alternative becomes the favorite at a third allocation with the same margin, the addition is again one quarter. Across these three decisions, the additions total one.

That total does not promise exactly one outsider victory. The awards could repeatedly go to the favorite or sometimes go elsewhere. A quarter added before an allocation stays a quarter in the accounting, whichever alternative wins.

The favorite is redefined at every step. The quantity concerns the concentration of the next opportunity, conditional on the history, rather than loyalty to a permanent incumbent.

The favorite is defined by the chance of receiving the allocation, not by an independent verdict that it is the best product. A buyer can favor a familiar supplier because switching is costly. An experimenter can deliberately give a plausible challenger substantial attention. The reason for the choice and the quality of the option both matter economically, but neither should be smuggled into the word "favorite." The budget first describes how opportunities are distributed.

Calling this a budget does not mean that somebody deposited a fixed fund of experimental tokens at the beginning. The quantity accumulates as opportunities arrive. We can ask how much has accumulated so far, or what the remaining sequence could add under a specified process. The economic issue is whether continuing the market continues to supply meaningful possibilities outside its current favorite.

With three alternatives, an eight-in-ten chance for the favorite leaves a combined two-in-ten chance outside it. Adding another name changes nothing if it receives no chance and the existing probabilities remain the same. A longer directory is not itself a larger comparison budget.

A company can preserve a variety of informative trials while remaining a single company, as Netflix's artwork experiments show. For image selection, the relevant alternatives are the pictures users could see. Counting corporate registrations would count something else.

The comparison budget describes opportunities through which evidence can emerge. It leaves service quality, price, workers'

incomes, and consumers' enjoyment to be assessed separately. Its economic value depends on what those opportunities reveal and which decisions the evidence improves.

The allocation needs a consistent unit. For a study of image selection, candidate images could be the alternatives and a presentation the allocated opportunity. Whether somebody subsequently plays the film is an additional outcome. For a procurement study, suppliers and contracts could take those roles instead. We cannot count presentations for one question, switch to completed plays for another, and call the resulting total the same budget. The definition becomes useful by fixing the economic question and the process before counting.

When verified productive inputs enter the allocation rule in my model, and their differences remain within the stated bounds, the comparison budget places an upper limit on the information the allocation history provides about their effects. Scarce outside chances can therefore limit how precisely contribution can be recovered from the record. A large budget can still leave attribution unresolved if the trials never distinguish the relevant inputs.

The news story that was not shown

Yahoo's front page gives us a documented example of an organization designing a record for later comparison. In research presented in 2010, Lihong Li, Wei Chu, John Langford, and Robert Schapire studied personalized news recommendation. Part of the underlying traffic came from a randomized bucket: candidate articles were selected at random for the highlighted position, with user and article information and the resulting click or nonclick recorded.[41]

The unshown article creates the difficulty. A person encounters one highlighted story, and the system sees whether that person clicks it. The same encounter does not reveal whether the person would have clicked every other candidate. The researchers' procedure started from that limited observation, rather than treating a visitor's arrival as a completed test of every story in the pool.

Random selection made a different use of the record possible. A proposed recommendation rule could be evaluated against logged encounters in which its choice matched the article actually selected. Under the paper's assumptions, this replay procedure supplied a reliable way to evaluate the rule using existing experimental traffic. The record was useful because of how the articles had been chosen and which responses had been retained.[41]

It is a beautiful answer to one version of the data problem. The observer does not invent a visitor's reaction to an unseen story. The evaluation uses reactions to stories that were genuinely shown. Its credibility comes from the logging design and the assumptions connecting those encounters to the proposed rule. The large database helps, but the relevant form of variation was built into the process before the later analyst opened it.

Now imagine a different file in which every visitor had been shown the same predicted favorite. The analyst might still learn much about that story and those visitors. What would be absent is the same coverage of choices under the randomized design. Replaying another rule would require further assumptions or further data. A more powerful computer could search the file faster without putting the missing article in front of a visitor who never saw it.

Yahoo recorded what visitors did after an article appeared, including nonclicks, as well as information about each visitor and article. A study of contribution has to use that evidence alongside the selection record. Counting only which story won the position would discard observations the experiment was designed to collect.

A replay based on yesterday's visits cannot, by itself, tell us how a new recommendation rule would change tomorrow's audience. Readers might return more often; publishers might offer different stories. Studying those changes requires a design or model that accounts for those responses, rather than assuming that the same encounters would keep arriving.

Imagine four groups of viewers, each always assigned its own particular image. Across an equally sized population, the images

would have equal shares of presentations. Yet within a group, the rule leaves no chance for another image. Aggregate diversity can hide a complete lack of alternatives within each group.

The converse is possible too. One image could dominate overall presentations while a carefully selected trial supplies useful evidence about a question involving another audience. The minority share alone would not tell us whether that evidence is weak or strong. We would need the selection rule, the groups involved, and the responses observed. Equal totals can conceal different processes; unequal totals can coexist with purposeful learning. Neither pattern in a chart substitutes for inspecting how each opportunity arose.

An allocation record can contain more than the winner. Task characteristics, eligibility terms, selection chances, and measured outcomes may supply additional information. Those observations should be included explicitly, even when they answer questions outside the simple allocation model.

A connection someone gets to make

A recommendation can also change a person's economic network. Karthik Rajkumar and colleagues studied randomized variations in LinkedIn's People You May Know recommendations. Different variants changed the kinds of connections suggested. The research, published in 2022, linked these experiments to subsequent job mobility. The effects of weak connections depended on how tie strength was measured; simply seeking ever weaker relationships was not a universal prescription.[42]

The connection makes the sequence particularly tangible. A recommended person can become a contact. That contact can help information about an opportunity reach a job seeker. A subsequent job change is then part of a history influenced by an earlier selection. The study's design allowed the researchers to investigate aspects of this process that an ordinary list of members and their later jobs would have left entangled.

Imagine studying a member's career after the connection has been made. The profile could show qualifications, previous employment, and a real new role. The question about the recommendation is whether changing who was suggested affected the path into that role. The person's abilities need not be fictitious for the route to matter. A useful match can involve both a qualified worker and information that reached that worker through a particular contact.

By varying connection recommendations and following later employment outcomes, LinkedIn's researchers could investigate how an introduction changed someone's opportunities. The useful evidence came from deliberately changing which connections people encountered.

What made a tie weak? The researchers used both the number of mutual connections and the intensity of bilateral messaging. These measures describe different relationships: knowing few of the same people and rarely exchanging messages need not coincide. Moderately weak ties did best by the mutual-connections measure; the weakest ties did best by interaction intensity. Even the practical meaning of trying a different connection depended on the measurement.[42]

The authors also examined the risk that treating one member could affect another member's outcomes. Connected people can affect one another even when their accounts appear separately in the data. Suppliers sharing a distribution channel can likewise share influences that a comparison has to account for.[42]

A network can improve access to opportunities while complicating attribution of their causes. References, connections, and observed performance need to be distinguished in an account of how the next allocation occurs.

Two long futures

Suppose the chance outside the favorite stays at one in a thousand at every allocation. Each opportunity adds a thousandth to the budget. A thousand opportunities add one; another thousand add another one. The process is heavily concentrated, but its accumulated outside chance keeps growing.

Concentration and a finite lifetime comparison budget are therefore different properties. A tiny chance that persists differs from a chance that vanishes quickly enough for its total to remain bounded. Nor does either calculation promise that awards arrive on schedule or reveal an informative input contrast.

Now consider a different, explicitly hypothetical allocation schedule. Start with a one-in-four chance outside the favorite. At every subsequent opportunity, halve that chance. The first few opportunities contribute most of what the entire future can add. Continuing forever adds successively smaller pieces, and all those pieces together amount to one half. There can be indefinitely many decisions while the accumulated outside chance has a finite ceiling.

The contrast concerns the whole sequence. A single photograph of either process could show a dominant favorite and a neglected rival. The later history depends on whether meaningful chances continue arriving or become scarce fast enough. To distinguish those possibilities empirically, we need evidence about the rules and their evolution. A concentration chart is a record of realized shares, not a measurement of that future sequence.

A third schedule makes the need for care clearer still. Let outside chances decline, but so slowly that the additions continue accumulating without limit. We do not need the arithmetic of that schedule to see the conceptual point. "Shrinking" is not a synonym for "finite in total." Before a claim about disappearing comparison does serious work, the rate of disappearance must be established in the setting being discussed.

The halving schedule is an accounting illustration, not a fitted model of a named company. An actual supplier's prospects could rise after a useful demonstration or fall after an incompatible product change. Its future chance could depend on who won earlier contracts. The budget follows the relevant conditional chances through those changes. Replacing that evolving process with a single fixed market-share number would remove precisely the history the object is meant to describe.

The schedules expose a distinction ordinary language can blur. A very small chance may feel equivalent to none when deciding whether to make one costly attempt. For what an indefinitely long record can reveal, persistence matters: a long sequence can accumulate what each opportunity adds only in a tiny amount.

A business, worker, or policymaker may need an answer long before an indefinitely accumulating process becomes helpful. An infinite lifetime budget could coexist with painfully little relevant opportunity over five years. A trial available after the purchasing decision cannot inform that purchase.

Nor can the opposite inference be made automatically. A finite amount of opportunity need not leave every practical question unresolved. A buyer might already have enough evidence to choose a service sensibly. The stronger issue is a promise of reliable, increasingly precise attribution throughout the admissible explanations of the model. That is a more demanding promise than choosing an option that works well with the information currently available.

Netflix might need to choose an image for a member; a public buyer might need to decide whether another supplier merits a pilot. Each can act with some uncertainty remaining. The useful comparison is one that could improve the decision enough to justify its cost.

How a lead can close a contest

There is a long mathematical history behind the difference between feedback that preserves comparison and feedback that eventually overwhelms it. One classic model begins with an urn holding colored balls. Draw a ball, return it, and add another of the same color. A color selected early becomes more likely to be selected again. Robin Pemantle's survey describes this family of reinforced processes and the many different behaviors produced by their rules.[43]

The urn isolates a feedback rule: an earlier draw changes the chances of later draws. Changing that rule can change how the sequence develops, even when every version can be described casually as success breeding success.

In the familiar urn, adding a ball of the drawn color can change long-run proportions while other colors continue to appear. Unequal histories can still keep producing comparisons. The rule governing each addition determines whether the outside chances have a finite lifetime total.

A stronger version gives accumulated success a much more powerful effect on the next selection. For a simple example, imagine weights that grow with the square of the accumulated score: doubling a score quadruples its weight. Research on appropriate strong-feedback models establishes a striking outcome. After some finite, random point, a single alternative receives every subsequent allocation. The result depends on the specified reinforcement conditions, not merely on a large lead.[44]

Roberto Oliveira studied the onset of such dominance, building on earlier monopoly results. Thomas Gottfried and Stefan Grosskinsky later distinguished several feedback regimes, including strong and weaker forms of monopoly. Their results explain how different reinforcement rules can produce different forms of dominance.[44]

What the comparison budget adds is an information question about the resulting record. In the strong-feedback model developed here, the remaining chances outside the eventual winner are

summable. The budget stays finite. Under the other conditions of the attribution result, indefinitely extending that single allocation record cannot supply a universally reliable recovery of the contribution effect. The prior dominance result and the information interpretation do different jobs.

There is a feature here worth holding onto. An alternative can retain a positive chance at every finite stage, yet receive no further awards along the realized history. A positive chance is not a promise of an eventual event. When those chances get small quickly enough, their continuing existence need not create an endless supply of informative deviations. Nominal eligibility and sustained practical contestability can therefore diverge within a precisely specified model.

Entry, new products, new tasks, or independent tests can reopen comparisons. Whether positive feedback closes them permanently depends on the allocation rule and which changes remain possible. Gottfried and Grosskinsky distinguish reinforcement processes with different long-run behavior; the rule determines which result applies.

The urn and Netflix's artwork system both use selection rules that change with earlier results. The urn makes the feedback easy to inspect. Netflix shows how a business can design those rules to keep learning.

The calls made to choose a policy

An experiment with rice farmers in India gives the selection question a different purpose. Maximilian Kasy and Anja Sautmann worked with Precision Agriculture for Development on ways to enroll farmers in a phone-based extension service. Six strategies combined morning or evening automated calls with different advance text-message arrangements. The recorded success was completion of five questions during the call. It was not a measure of a farmer's eventual harvest.[45]

The experiment proceeded in waves. Results from completed calls informed assignment to later strategies. The aim was to help select an approach for use after the experiment. That purpose matters: maximizing the number of completed calls during the trial is a

different objective from collecting the evidence needed to make a better choice for the much larger implementation that may follow it.[45]

The distinction is familiar once we put a calendar beside the experiment. This week's calls can deliver useful service. They can also teach the organization which method to use next month. Giving every remaining call to the current leader may improve the immediate recorded success rate while leaving a close challenger insufficiently tested. A reasonable allocation depends partly on whose outcomes, at which dates, the organization is trying to improve.

Kasy and Sautmann's exploration-sampling method retained substantial testing of plausible alternatives instead of simply concentrating on the strategy most likely to be best. Results from earlier calls helped determine which strategies received later ones.[45]

The case supplies an important counterweight to the idea that experimentation must mean equal shares forever. Once an approach looks clearly unpromising, continuing to assign it the same share may be a poor use of opportunities. A close challenger may deserve more investigation. Relevant comparison depends on which uncertainty remains, how much resolving it would matter, and the consequences of sending another call along each route.

Adaptive design ties opportunities to the uncertainty being investigated. A limited experiment on a relevant task can distinguish plausible explanations; a nominally balanced allocation on incomparable tasks may add little. The assignment process must be related to what the observer wants to learn.

The measure of success should remain visible as well. Completing five questions is something an organization can record promptly. It is useful for choosing among recruitment methods on that outcome. If the later decision concerns agricultural income or the quality of advice, additional evidence is needed. The outcome recorded for fast adaptation should not acquire authority over every consequence simply because it arrives in time to guide the next wave.

This is especially important when comparisons have different costs. An ineffective image may waste a small amount of a viewer's

attention. An unsuitable software deployment can demand weeks of staff time. A policy pilot may affect people whose welfare matters during the experiment as well as afterward. The agricultural study's stated objective helps identify the decision being optimized; other settings may require a different balance.

An institution can therefore preserve comparison while making increasingly selective choices. It need not suspend judgment until every candidate has been exhausted. The discipline is to state why the next allocation is useful, which possible responses would change the decision, and what costs are justified by that learning. Those questions turn a general preference for experimentation into something a buyer or program manager can actually implement.

What a large budget cannot buy

Equal chances at every allocation would make the budget grow, but still need not identify an input effect. If the measured input is identical across the alternatives, reallocating awards changes the recipient without creating the contrast needed to learn that input's effect.

For a less abstract picture, imagine testing two artwork files that display exactly the same image. More exposures can help estimate how that image performs with particular audiences. They cannot reveal the difference between the two visual treatments because the treatments do not differ. A system can create a great deal of nominal variation while leaving the intended comparison untouched. Counting chances is necessary for one bound; it is insufficient for a guarantee.

A change of name need not create an independent trial either. Products may share an underlying service, a bottleneck, or a faulty outcome measure. The budget does not certify independence or meaningful variation merely because a different alternative receives an award.

Yaneer Bar-Yam's distinction between large quantities of detailed data and information relevant to an intervention is useful here. His framework concerns how to identify what matters for a system-level

question, rather than treating detail as automatically sufficient. The comparison budget addresses a narrower allocation setting, but the shared discipline is to ask what the record can distinguish for the question under study.[46]

Yahoo's article evaluation depended on randomized logging and observed responses. The farmer calls depended on the measured completion outcome and adaptive assignment. In each case, the record's usefulness followed from how it was generated, rather than its size alone.

A buyer might obtain an independent performance test without awarding another full contract. If that test distinguishes effects the award history leaves uncertain, it adds information to the record. The allocation-only limit then describes only part of what the buyer can observe.

Likewise, known selection probabilities can be useful information, not merely quantities an analyst may assume into existence. A designed experiment may log them. An ordinary market may require estimates based on observed rules and behavior. If obtaining those estimates supplies additional information about contribution, that information has to be counted. An empirically defensible budget needs a defensible observation model alongside it.

Defining the object and measuring it are therefore separate achievements. The rice-farmer experiment could implement assignment probabilities deliberately; an analyst of an ordinary market cannot assume comparable access to its mechanism. Sometimes only a range of plausible budgets will be justified. That can still organize an investigation, provided the range and its consequences remain visible. Precision written into a definition should never be mistaken for precision supplied by observations.

The comparison budget cannot be read from a market-share table. A rival's sales tell us what happened, not its chance on the next job. Even a well-measured budget needs evidence about differences in what suppliers do and how well they perform. Frequent trials would

tell us little about engineering if they never compared meaningful differences in engineering.

A large company that keeps testing

Microsoft's Bing supplies a particularly direct challenge to the idea that the number of firms tells us how much learning remains. In a 2013 paper, Ron Kohavi and colleagues described more than two hundred experiments running concurrently inside the service. Some users were held out to assess the experimentation system's overall effect. The organization was deliberately maintaining differences in experience rather than relying solely on the record of the version everyone already used.[47]

One example involved adding links within advertisements so people could navigate directly to parts of an advertiser's site. The initial version increased revenue but worsened user metrics and page-loading time. Adjusting the presentation, including how many main advertisements appeared, changed that balance. The investigators were tracking more than a single increase in a commercial total. A result that looked attractive on one outcome still needed interpretation alongside others.[47]

Keeping only the revenue increase would lose part of what Bing's experiment revealed. Keeping only the company's size would lose the comparisons it arranged. The useful evidence combined an intervention, assignment, recorded responses, and a decision about which outcomes mattered.

The extra links offered a plausible convenience. Testing revealed consequences the designers had not settled in advance. Their skill helped create the proposal; comparison helped decide how to improve it.

A business can test many changes to its own product while leaving independent suppliers little opportunity to demonstrate theirs. Bing's internal trials can improve Bing without answering how another provider would perform with a route to users.

Sometimes the missing comparison calls for another supplier. Sometimes a company can learn more cheaply by changing its own trial or using records it already keeps. The question is which route will reveal something useful about the decision at hand, and what it will cost.

The price of keeping a door open

Keeping a comparison available has a price. Testing can require moving data, training staff, checking outcomes, and accepting disruption before its result is known. The hoped-for information must justify those costs. Some alternatives will sensibly remain unevaluated because testing them would cost more than the decision could gain.

Netflix's engineers described their artwork setting as favorable partly because exploration costs could be spread across a large membership. The cost still depends on what a person experiences during the test. A setting that makes outside scrutiny difficult can sometimes make internal experimentation economical.[40]

The practical question is which uncertainty could change a consequential decision. A small reliability difference may justify investigation when it affects years of expensive operation. A difference incapable of outweighing known switching costs may not. There is no economic instruction to maximize the comparison budget everywhere.

Existing records, a shared test environment, or a limited task can sometimes answer a question more cheaply than a full deployment. Whether they do so depends on their design and on the inference required. A lost route has no automatically adequate substitute.

The order matters. Start with the question, identify a comparison capable of answering it, and assess the cost of producing that comparison. Only then decide whether an access arrangement or purchasing rule helps. Beginning with a preferred number of suppliers and assuming that the resulting record will be informative skips the hardest part of the task. The directory can be long while the relevant test remains absent.

Learning can strengthen the case for an incumbent. An artwork experiment can confirm that the established image works better for the relevant audience. An informative test must allow a result to change our view, including by increasing justified confidence in an existing choice.

The comparison budget measures the accumulated allocation chance outside the current favorite. What those chances reveal depends on the inputs, observed outcomes, and response rules. Preserving more comparison is worthwhile when the resulting evidence improves decisions enough to justify its cost.

A different early award could produce another history under the same rules. Changing those rules could produce a different economy. Explaining a reward requires knowing which comparison is being made.

CHAPTER 3

Shadow futures

One economy, different histories

The same recording can have another history.

In MusicLab, Lockdown finished first in one experimental world and fortieth in another. The separate groups encountered the same recordings but different developing popularity records. Their charts supply observed alternatives to the idea that one realized ranking explains itself.[48]

A theoretical replay asks us to hold the underlying effects, starting conditions, and response rules fixed while allowing early allocations to differ. Later investment and behavior can then change because they respond to a different history. The economy follows the same rules without producing identical decisions.

Holding every later input fixed would describe another exercise. If success normally finances investment, suppressing that response changes the assumed economy. A replay preserves the rule connecting opportunity and behavior, rather than freezing behavior regardless of what happens.

A different early allocation need not permanently reverse the outcome. An initial lead can fade, be overcome, or compound. A model must allow the original rules and their uncertainties to determine which follows. Otherwise its preferred alternative has been written into the conclusion.

A shadow future is a feasible allocation history over a stated horizon under the same underlying input effects, starting conditions, and response rules. Different events can produce different subsequent behavior under unchanged rules. The object describes histories the modeled economy can generate, including those it never realizes.

MusicLab's groups contained different listeners rather than identical people sent down cloned lives. Its observable parallel markets therefore illustrate historical variation without constituting literal independent reruns of every feature of an economy.[48]

MusicLab recorded parallel markets. A theoretical replay describes histories implied by specified rules. The first supplies experimental observations; the second helps define what we seek to explain. Estimating the probability of an imagined history still requires evidence about the process producing it.

An ordinary archive contains the investments and outcomes that occurred. It lacks development that would have been financed by revenue never received. That omission becomes important when we ask the realized record to explain the effect of productive inputs on success.

What changes, and what stays

Changing an input before an allocation is an intervention. Changing the rule that allocates opportunities is another intervention. Both are legitimate economic questions, but neither is the same exercise as replaying the original process under unchanged conditions.

For the engineering question, we might specify an additional month of work on reliability, a particular starting date, and the contracts available over the next two years. We would then compare expected rewards with and without that extra work, allowing later investment and exposure to respond as the model says they would. The result is a contribution measure for that intervention and horizon. It is not simply the difference between whichever two histories happen to interest us.

Consider the customer-support study introduced in the prologue. Brynjolfsson, Li, and Raymond examined the introduction of an AI assistant and measured changes in agents' work, including issues resolved per hour. Their published study found average gains, with larger gains among less experienced and lower-skilled agents. The comparison concerned access to a particular assistance system in a

particular workplace. It did not measure each worker's share of the value created by the entire organization.[49]

A contribution question must specify its target. Does more engineering change reliability, whether customers observe reliability, or the chance of receiving revenue? An intervention on work and one on information about work can have different consequences.

Opportunity can finance investment, and investment can improve performance. Explaining the resulting reward requires tracing that chain. Recognizing the role of an early allocation does not make later productive work disappear.

MusicLab's separate markets help us see variation in success under a common experimental arrangement. They do not tell us the effect of rewriting Lockdown's chorus. To estimate that effect, we would need a design that compared the relevant recordings under suitable conditions. The song's different rankings and the effect of changing the song are related questions, but the first does not supply the answer to the second.

This is why contribution needs an explicit comparison. An additional engineer, a better recording, and a new distribution route are different changes. Their effects may unfold over different periods. A large reward could be sensitive to one and barely sensitive to another. Saying that success reflects contribution leaves these choices unresolved. Naming the change makes the claim something we can investigate.

This prevents explanations from becoming stories chosen after the outcome. Improving an alternative's initial quality, changing its starting position, and changing its early allocations are different operations. Each needs its own place in the analysis.

The fork in the laboratory

There is a striking predecessor involving an animal with no procurement department. Line Chloé Le Goff and Philippe Soulier studied reinforced choice using Argentine ants. In their laboratory setup, an ant entered a fork, chose a branch, and was removed. Another ant followed. The maze began without a pheromone trail. A trial ended after one hundred choices; the researchers recorded fifty trials.[50]

Earlier passages could influence later choices through deposited pheromone. The authors considered a model in which the response to the accumulated trail depended on unknown settings. They studied estimation from repeated trials and the limits of estimation from a single sequence. Their paper connected reinforcement with difficulties recovering parameters, even from an indefinitely extended path.[50]

The ants expose a statistical temptation common to reinforced processes: a longer record of the established route seems to promise an eventual explanation of why it prevailed. Yet continuing that route differs from observing the process begin again.

The fork was carved into white PVC and partly covered by Plexiglas. No food was present during the experiment. Only one ant occupied the apparatus at a time, and it could not return after choosing a branch. The researchers designed these restrictions to make their behavioral assumptions more credible: an arriving ant encountered the trail left by predecessors, rather than other ants at the junction. The apparatus helped define what a choice could tell them.[50]

The resulting paths did not all look settled. In fewer than ten of the fifty trials, the researchers described a branch as appearing to have been selected. In the others, selection was not obvious after one hundred passages. The same protocol had produced visibly different histories. Those finite records gave the researchers something that continuing traffic down one established branch could not: repeated observations of how a path developed from its beginning.[50]

Le Goff and Soulier had already shown that a long history of reinforced choices need not reveal the forces behind those choices. In a market, that raises a further question: can the opportunities left open to rivals tell us how much we can learn about why the winner won? The comparison budget gives those opportunities a precise meaning and connects them to the problem of measuring contribution.

The statistical question is how much opportunity the process retains to distinguish input effects. A shorter record can contain substantial relevant variation while a much longer one follows an established route. The design, assignment chances, and measured outcomes determine the difference.

A million entries are not a million beginnings

Imagine an ordinary coin with an unknown, fixed tendency to land heads. One explanation says heads occurs half the time. Another says it occurs three quarters of the time. Any particular finite sequence of heads and tails is possible under both explanations. Shared possibilities do not make the explanations equally persuasive.

A long sequence containing about three heads for every tail will increasingly favor the second explanation. Under the usual assumption that the tosses are independent with an unchanged bias, the proportion of heads supplies information that keeps accumulating. We do not need to watch every imaginable coin sequence. The one we observe can become enough to distinguish the two fixed biases reliably as the number of tosses grows.

A coin has unobserved possible histories too. One observed sequence can nevertheless distinguish its bias. A lasting attribution limit arises only when the actual process fails to separate the relevant explanations sufficiently as the record grows.

Yahoo's randomized article displays and recorded clicks were designed to support a particular evaluation. The logging mechanism mattered because the researchers did not observe how the same encounter would have responded to every other article. Their analysis

used the opportunities the design actually generated, with assumptions about those encounters and the evaluation procedure.[51]

The complete experiment must be specified. Awards alone supply one information set; measured reliability and independent tests can supply another. A laboratory result directly informative about the input effect cannot be excluded when evaluating what the record reveals.

In the model, the information limit concerns a specified family of candidate explanations for how productive inputs affect allocation. A candidate explanation is not an arbitrary story about the company. It supplies probabilities for the recorded outcomes at each stage. Two such explanations can disagree about engineering's importance while using the same purchasing procedure and the same rules for how suppliers respond to earlier contracts.

The comparison budget then asks how much conditional allocation chance accumulates outside the current favorite. The favorite can change as the purchasing history moves forward. As Chapter 2 showed, a small chance can persist long enough to accumulate without limit. The strongest result concerns the more demanding case in which the lifetime total is finite. It also requires restrictions on how different explanations separate as those chances diminish. Those restrictions specify how far the next-award probabilities can differ between candidate explanations.

To apply the result, an investigator needs to establish the feedback rules and the chances they leave for alternatives. Ant traffic, song charts, and market shares describe outcomes. Establishing a lasting learning limit also requires evidence about how the process generates them.

When waiting cannot finish the job

First, the candidate explanations must describe a common predictable experiment. Input and state rules are fixed functions of the information available before each award. If success finances investment, each explanation retains that response rule rather than changing behavior to fit the observed record.

Second, the candidate explanations retain the same possible next recipients after every history. If an award is possible under one explanation and impossible under another, that award could decisively distinguish them. The shared-possibility condition excludes this shortcut.

Think again of the coin. Both fixed biases permit heads and tails at every toss, yet the biases can be distinguished in the long run. Shared possible outcomes therefore do not suffice. Third, the model requires the separation between candidate next-award distributions to be controlled by the remaining chance outside the favorite, in each direction of the comparison. As opportunities diminish, the experiment's ability to separate those explanations diminishes with them.

For the bounded input differences in this allocation model, the required control can be established. A separate engineering measurement could nevertheless reveal an input's effect directly. Such evidence must be counted; the allocation record is only one potential source of information.

Fourth, under each candidate explanation the cumulative lifetime chance outside the favorite must remain finite, apart from events assigned probability zero. A dominant observed share or several years of declining rival chances cannot establish that property of the whole process.

The strong-reinforcement models discussed in Chapter 2 supply one sufficient setting under specified assumptions. Ordinary positive feedback does not. A persistent route to trials can preserve accumulating comparison; chances shrinking fast enough can have a

bounded lifetime total. Similar observed shares can arise from either arrangement.

Under all these conditions, the complete history cannot support a single contribution-estimation method that becomes reliably accurate under every candidate explanation. For a contribution target that differs across the explanations, no universal procedure eventually recovers the right value everywhere merely by continuing to watch this one history. This includes procedures more sophisticated than simply equating sales with merit. Better calculation cannot create the missing separation.[52]

The technical step concerns collections of histories. Under these conditions, a collection assigned probability zero by one explanation has probability zero under the other. Their probabilities need not be equal otherwise. A universally reliable method would have to separate histories enough to return different contribution values under different explanations. The model establishes that continuing this experiment cannot deliver that separation. A method could work under one candidate, but cannot promise accuracy under them all.

Under the theorem's conditions, continuing the specified history cannot deliver an attribution procedure whose error vanishes under all candidate explanations. Operational experience can remain useful for other purposes. The limitation concerns the information produced, rather than an analyst's patience.

The result also reaches an apparently modest alternative: avoid estimating a precise number and simply decide which of two different contribution values is correct. Under the same conditions, there is no sequence of such tests for which both kinds of mistake vanish. One cannot escape the attribution problem merely by replacing a detailed merit calculation with a binary verdict.

This is a conditional statement about a class of economic histories, supported by classical probability tools. It is not a finding that actual markets always exhaust their comparisons. The ambitious part is the connection between a defined allocation quantity and a persistent limit on learning input effects. Keeping the conditions visible makes

that connection stronger, because it shows what evidence would be needed to apply or challenge it.

What the record still reveals

The attribution limit leaves many facts intact. Recorded outages, completion times, and accepted work can establish performance. Contracts can establish payment obligations. The result concerns the recovery of a defined contribution effect, rather than a general absence of knowledge.

Even the allocation record can support one explanation more strongly than another. Awards that repeatedly coincide with an observed productive advantage may increase confidence in its effect. The theorem limits the ability to drive attribution error to zero across the specified alternatives; it does not require all explanations to receive equal support.

MusicLab showed substantial variation across worlds without finding that musical appeal was irrelevant. Social influence changed the relationship between appeal and success. Recognizing alternative histories does not require treating every song as interchangeable.[48]

Execution, predictability, and attribution deserve separate judgments. Demonstrating that work was completed, correctly predicting who receives the next reward, and estimating how much an input changes that reward are different achievements. Strong evidence for the first two can coexist with uncertainty about the third.

Prior information can narrow the range of plausible explanations. A separate engineering test may rule out others. A buyer may also have enough evidence to choose a supplier while remaining uncertain about why the business succeeded. The information needed for that decision can be less demanding than a complete account of contribution.

That uncertainty matters through the decisions it could change. An organization may have enough evidence to renew a reliable service despite unresolved attribution, while an investor assessing a different

technology needs the missing input effect. Complete explanation is not a prerequisite for every practical choice.

Two water supplies, one city

In London during the cholera epidemic of 1854, John Snow investigated a comparison created by two water suppliers. The Lambeth Company had moved its intake upstream in 1852, away from London's sewage. Southwark and Vauxhall drew from a more contaminated stretch. Their customers were intermingled: different houses along the same streets could receive water from different sources.[53]

Snow sought the water supply of houses where fatal attacks had occurred. This was harder than consulting a tidy customer list. Residents did not always know their supplier. He described using receipts and, when necessary, a chemical test that distinguished the waters. The eventual comparison supported his account of water's role in cholera. The practical work included finding out who had received what.

The inferential opportunity came from a change in supply and the intermingling of customers. Snow did not observe the same household through two literal epidemics, one under each water source. He investigated an existing contrast and argued that the groups' surrounding conditions made it informative. That argument about comparability carried the causal claim; the bare existence of two companies would not have done so.

Before concluding that a record cannot identify contribution, investigators should examine variation supplied by other customers, policy changes, and additional measurements. A result about a sparse observation scheme cannot justify ignoring a richer one that exists.

Snow's example gives verification a precise role. Establishing which water a household received defined the comparison; it did not eliminate the need to assess whether the groups were comparable. Measuring an input and identifying its effect similarly require distinct, complementary work.

An economy need not reproduce its entire past to learn a causal effect. It needs evidence that distinguishes the particular alternatives relevant to that effect. Sometimes naturally occurring variation provides it; sometimes a designed trial helps. The finite-budget argument identifies a setting where continuing the specified record cannot finish that job. It directs us toward better evidence, rather than away from causal investigation.

Agreeing about the next allocation

Two explanations can converge on the same prediction for the next allocation while assigning different roles to productive inputs and accumulated advantage. Agreement about what comes next need not settle why the process took its present form.

David Blackwell and Lester Dubins established a classical result in 1962 about conditional forecasts coming together as information accumulates. When the observers' probability laws satisfy their absolute-continuity condition, their conditional distributions for the future merge.[54]

In my finite-budget setting, the fixed candidate history laws have the relationship that Blackwell and Dubins require. Their classical result then applies: forecasts of the future allocation sequence merge, while the differing contribution values remain impossible to recover reliably under every candidate explanation.

Forecasting follows the process already in motion. Contribution asks what an intervention would change within a defensible account of that process. Each can be useful without answering the other's question.

Predictive agreement alone does not establish an attribution problem. Return to the coin with a fixed but unknown bias. Imagine two observers who begin with different reasonable probability distributions over possible biases. Watching the same independent tosses can bring their forecasts together while the observed frequency also reveals the bias. Their growing predictive agreement is perfectly compatible with successful parameter learning.

In the coin example, the fixed half-heads and three-quarters-heads explanations can be distinguished by a sufficiently long record. The observers' initial mixtures over possible biases and the laws generated by those two fixed biases are different objects.

Imagine a forecasting competition that rewards only predictions of the next supplier. It might identify who best anticipates the purchasing process. It would not automatically identify who best understands the effect of an extra engineering month. Those are separate tests. A forecast score becomes evidence about contribution only when the design establishes the relevant connection, rather than assuming that good prediction must explain every cause of the outcome.

A map with uncertain settings

The distinction between prediction and parameter recovery has another established home: spatial statistics. Hao Zhang's 2004 paper studied models in which observations become more densely packed within a fixed region. Under the conditions he analyzed, not all the relevant covariance parameters could be consistently recovered, even with increasing observation density. Nevertheless, certain predictive conclusions could converge and a particular combined quantity could be estimated.[55]

Imagine measuring soil across a field and using the readings to predict conditions between sampling points. Two models might increasingly agree about those predictions while leaving some of their settings uncertain. The forecasts could become useful before every detail of the underlying variation was known.

The sampling arrangement matters. Zhang distinguished filling gaps within one fixed region from collecting data over an expanding region. In reinforced allocation, continuing one established history likewise differs from observing additional informative beginnings. More data must be described by how they are generated.

Zhang showed how prediction could improve without identifying every underlying model setting. The allocation question has a related

form: whether forecasts of the next reward can become accurate while the effect of changing an input remains unresolved.

A prediction of tomorrow's workload can remain valuable while leaving the effects of more engineering, another supplier, or a different access rule unresolved. The conclusion is about the scope of an inference, rather than the usefulness of prediction itself.

Consider an AI buyer comparing forecasts of its current system's responses with estimates of the benefit of switching systems. Its logs may be excellent for the first problem. They may contain few cases in which another system was tried under comparable conditions. This is a hypothetical purchasing problem, not a conclusion about any named model. Its point is that the relevant comparison follows the decision being considered.

What a simulation can supply

Repeated runs of a simulation can produce different histories from specified starting resources, allocation rules, input effects, and responses to success. Their distribution reveals what those assumptions permit.

A simulation can expose contradictions, show that an imagined outcome is rare under the chosen rules, or compare the implications of persistent and diminishing exploration. Its conclusions remain conditional on the rules and effects entered into it.

The MusicLab researchers could compare actual recorded outcomes across their experimental worlds. A simulated procurement market does not acquire the same evidential status merely because it contains eight worlds, eight hundred, or eight million. Increasing the number of runs can reduce uncertainty about what the simulator implies. It does not independently validate the simulator's account of how engineering affects contract awards.

Suppose two candidate input effects remain compatible with the evidence. Run the simulation under one and it supplies one distribution of shadow futures. Run it under the other and it supplies

another. A precise average inside either simulation leaves the original uncertainty intact: which candidate governs the real procurement process?

Disagreement between defensible models can identify a useful experiment. Similar recommended actions may instead show that a decision is robust to unresolved attribution. Sharply different recommendations make that uncertainty material; displaying only a preferred simulation conceals it.

Yahoo's researchers specified a logging process and assumptions that allowed an existing record to evaluate another article-selection policy. Their replay method used logged encounters to estimate performance. It could answer that question without observing a complete alternative economic history.[51]

Simulation should distinguish historical variation, input interventions, and institutional changes. All three can be investigated, but their interpretations differ. Shadow futures first specify the economy being replayed.

A comparison with a purpose

An unobserved alternative could have been superior, ordinary, or unsuccessful. A missing comparison leaves uncertainty among those possibilities. It supplies no presumption that the unrealized outcome would have been better.

When eBay shut off brand-search advertisements, traffic largely shifted to unpaid search results in that setting. Paid clicks had overstated the number of additional customers the advertisements brought. Results for some other searches and users differed, giving the company reasons to change where it spent its advertising money.[56]

An informative experiment must be capable of disappointing the person who requested it. Results can strengthen an established arrangement or reveal costs that demonstrations concealed. A test

designed to vindicate a predetermined winner cannot perform that role.

Estimating the effect of an input also differs from deciding who deserves the resulting gain. Contracts, risk, incentives, and moral claims enter that further argument. Statistical attribution cannot settle them through the word contribution.

A valid contractual claim to payment can survive uncertainty about all the causes of commercial success. A promise and a causal explanation answer different questions.

For competition policy, the relevant question is whether preserving a comparison would improve a consequential decision enough to justify its costs. Users may bear disruption, and some experiments may be uninformative. A missing comparison creates a problem to investigate, rather than a reason to subsidize every alternative.

Beginning again

One way to learn more is to observe independent repetitions of an informative experiment from a common beginning. Under appropriate identification and regularity conditions, repeated finite experiments can support consistent estimation even when extending one reinforced path cannot.[57]

Separate organizations do not automatically supply independent repetitions. Shared staff, reputation, shocks, or development can connect their outcomes. Independence and a common informative beginning require a design, rather than a change in letterhead.

MusicLab separated its social markets so that popularity information developed within each world. The design created a way to observe differences a single market would not reveal. Applying an analogous design to suppliers would require identifying which influences actually cross between the proposed units. The useful lesson is about constructing a comparison, rather than giving every administrative subdivision the status of another world.

Repetition must still provide the relevant variation. If measured input profiles are identical across alternatives and no other observation separates their effects, beginning again cannot create the missing contrast. A large budget supplies opportunity; experimental design connects it to the target.

Nor does a repeated experiment have to answer every question. A trial could identify short-run reliability while leaving the consequences of years of accumulated experience unresolved. A buyer might deliberately choose that limited question because it is useful and affordable. The appropriate response is to state the horizon and scope, then use the evidence for what it supports. Modest, interpretable knowledge can improve a decision substantially.

Le Goff and Soulier compared what could be learned from one reinforced sequence with what could be learned from repeated trials. Changing the observation scheme can make a previously unresolved question answerable. In an economic setting, a new test or another measured outcome may supply evidence that ordinary transactions leave out.

The histories that get a chance

A productive economy can generate different histories, with real work and achievement along each. Shadow futures give those histories a precise place in an explanation of the one observed. Policy then poses the further question of which changes would improve the attainable outcome.

Existing evidence, additional measurements, new experiments, and redesigned allocation rules can support different decisions. The comparison budget describes a constraint within a specified process; shadow futures define histories left outside its realized record.

Bell's 1956 patent decree changed who could use existing inventions. The historical record lets us follow what happened after that opening, including which other barriers entrants still faced.[58]

The economic task is to identify which comparisons can teach us something valuable, how to preserve them, and when their benefits justify the cost. That task begins before the next winner is known.

CHAPTER 4

The histories that became infrastructure

How far can a judgment travel?

A market can grow while the number of independent places to try again shrinks.

Imagine a carpenter who disappoints a buyer in one town. Another town might offer a different buyer, unfamiliar with the dispute and willing to commission a small job. Now imagine that both towns purchase through one platform, which carries the carpenter's ranking from one to the other. The second town is still there. The second judgment may have disappeared. The carpenter's ability to start again depends on whether another buyer can make a fresh assessment.

Adam Smith began his account of the division of labor with a pin workshop. Specialization raised what its workers could produce together. Its reach still depended on finding buyers. In his chapter on the extent of the market, Smith contrasted the narrow opportunities of scattered settlements with the commerce made possible by water transport. A larger reachable market could support work that a small village could not.[59]

Distance therefore did two things. It limited the customers an artisan could serve, and it could separate the people deciding whether to give that artisan a chance. Removing distance could expand both income and competition. Whether it also joined previously separate judgments is another question. A new road connecting buyers is different from one authority deciding which sellers every connected buyer may see.

Guilds and settlement rules also obstructed movement in Smith's account. A newcomer could encounter a closed corporation in the

next town. Geographical separation offered a fresh assessment only where another buyer had the freedom to make one.[59]

Nor did formidable scale begin with software. By 1920, the Supreme Court's United States Steel opinion was recounting a huge industrial combination and disagreement over whether integration reflected the physical requirements of steel production or suppressed competition. Pig iron and finished steel were physical goods; that had not kept their producers small. Furnaces, transport, capital, and organization could already assemble power over a wide territory.[60]

The question is how far a common selection system can reach, and what remains outside it. One intermediary serving ten towns may offer customers more choice than ten isolated villages ever had. It may also carry one exclusion across all ten. Increased scale changes the institutional problem when the route to a fresh evaluation disappears.

In 1972, the physicist Philip Anderson called an essay “More Is Different.” Anderson worked at Bell Laboratories, and his subject was how new levels of physical scale and complexity could require new explanations. His title raises a useful economic question: when does an increase in reach change the kind of independence an economy can sustain?[61]

A bigger furnace changes the cost of producing steel. A railway changes where steel can be sold. A shared admissions rule changes which producer can reach those buyers at all. Digital distribution lets an evaluation travel with a seller and be reproduced across transactions at very little cost. The seller can move while the ranking follows. Power supplies and computing clusters remain physically constrained, but one judgment can govern opportunities throughout the network they serve.

Imagine a rival network that needs users before it becomes useful. Those users already communicate through the established network. If the two cannot connect, the entrant may have to assemble a very large customer base before it can offer its first customers a worthwhile service. A network that would work well once established need not

have a feasible route from a small beginning. Under those arrangements, a possible competitor can remain economically unreachable. Interconnection could change the arrangements and make a beginning possible.

Increasing returns can give early adoption a lasting influence over which technology develops, as W. Brian Arthur showed.[62] The question then extends beyond how much cheaper a large producer becomes. How long can its advantage keep compounding before an alternative receives a meaningful test? Where a network's reach also governs access, a successful position can last because the route to challenging it has become harder to build.

That changes what a continuing return tells us. An owner may keep improving the service and also benefit from the absence of a feasible starting point for rivals. No executive needs to cancel a competing project for that structural barrier to work. Capital earns another return, the return finances further expansion, and the alternatives that might have tested the arrangement never acquire a record. The history becomes richer on one side while remaining unwritten on the other.

The next buyer may have every reason to choose the established service. Testing a rival would cost time and might produce a worse result. Other buyers could benefit from what the trial revealed, without paying for it. When a common route closes, that potential learning can disappear across the whole network. The immediate transaction can remain attractive while the opportunity others needed is lost.

Bell's telephone network made permission tangible: a customer could buy an independent attachment, yet risk losing service by using it.

An invention leaves the laboratory

The transistor existed before the government opened Bell's patents.

In December 1947, John Bardeen and Walter Brattain made a semiconductor device amplify a signal at Bell Laboratories. Their point-contact transistor belonged to a research program drawing on physics, materials, instruments, and the work of several people. In his Nobel lecture, Bardeen emphasized the earlier theoretical work and improvements in materials on which the discovery depended. The invention was an achievement, and it had a history.[63]

On January 24, 1956, a different kind of event changed what others could do with Bell's inventions. An antitrust settlement required extensive licensing of Bell patents, with thousands of existing patents made available without royalties. The settlement also imposed obligations concerning technical information. It left the Bell System integrated. This was an opening of technology inside a continuing institutional structure, years before the later breakup.[64]

Between those events lie two questions that are easy to collapse. How did a remarkable invention come into being? What could happen when other people were able to build on it? Bell's accomplishment supplied the starting material for the second question. Recognizing the importance of that accomplishment does not tell us which arrangements would allow the material to produce the most useful subsequent work.

The semiconductor story makes that distinction tangible. A transistor on a laboratory bench is not yet a reliable supply of components, an affordable radio, or a workable design in someone else's product. Moving between those stages requires further decisions and further learning. Whoever gets to undertake that work can discover things the original inventors did not yet know. Whoever cannot undertake it leaves fewer results for later observers to examine.

Accumulated work leaves equipment, procedures, and skills that make some later projects easier. It can also leave permissions and restrictions that determine who may try them. Engineers using an established body of knowledge may work better because somebody has already paid to develop it.

Bell's records let us follow both routes. A technology could leave the laboratory through a license and a training program. A finished attachment could reach a shop yet remain barred from a customer's telephone. The invention, the right to build on it, and the right to use the result were separate economic opportunities.

Follow those opportunities rather than asking whether Bell was simply good or bad. One route led from a major laboratory to inventions others could use. Another led from an independent device to a customer whose telephone service could be threatened if the device was attached. The differences between those routes help explain why releasing knowledge and opening a market can have different consequences.

The price of a license

Bell had shared transistor technology before the 1956 decree. In April 1952, representatives of forty companies attended a nine-day technology symposium after paying a $25,000 patent-licensing fee. The program included a manufacturing visit. The Computer History Museum's account and collection identify this as part of a broader effort to spread semiconductor techniques. Access did not begin on the day of the settlement.[65]

That chronology prevents a convenient origin story in which the government released a wholly sealed invention and an industry immediately appeared. A licensing fee, a technical gathering, and compulsory licensing represent different arrangements for passing on knowledge. The relevant question is what each arrangement enabled, for whom, at what cost, and with what further restrictions. “Open” is too broad a description to answer all four.

For an engineer, access can require more than permission to practice a patented invention. There may be a difficult production process to understand, a material whose quality matters, or a component that performs differently outside a demonstration. A legal document can remove a restriction without supplying every skill needed to use the resulting freedom. The distinction is familiar to

anyone who has followed a recipe and discovered that instructions and competence are different possessions.

The symposium's combination of licensing and instruction is therefore revealing. Making an invention usable can involve transmitting practical knowledge alongside the rights. The economic object reaching another company is a bundle of opportunities and requirements. A cheaper license helps one part of the problem. It need not remove the other costs of turning an idea into something worth buying.

The government's contemporary account of the 1956 judgment recorded a requirement to furnish technical information at a reasonable charge. It also described annual patent lists and a mechanism for settling royalty disagreements. These were ways of making an access obligation operational. A general statement that firms should be allowed to innovate would have left the mechanics unresolved.[64]

The incumbent has a legitimate interest here too. Producing instructions, answering technical questions, and maintaining a stable interface require work. An access rule can shift those costs onto an organization that does not receive all the resulting benefits. Ignoring that work would make openness look cheaper than it is. Charging for every missing piece without scrutiny could make nominal access unusable. The design has to address both possibilities.

Bell's case thus begins with an economic sequence rather than a single unlocked door: permission, usable knowledge, practical implementation, and a route to a customer. Each stage can fail even when the previous one succeeds. The next case concerns a device that had already reached the stage of manufacture and sale. Its difficulty lay at the telephone itself.

A cup on a telephone

Hush-A-Phone was a cup-like attachment that fitted onto a telephone. It was intended to provide privacy for the speaker and reduce surrounding noise. The company had been selling it since 1921. By the time its dispute reached the federal appeals court in 1956, more than 125,000 had gone into use, according to the court's account.[66]

The telephone companies' tariffs prohibited attachments they had not supplied. They claimed the right to suspend or terminate service for violations. Vendors and users were told that Hush-A-Phone could not be used, and some distributors stopped selling it. The independent maker's ability to build a cup had not secured its customers' ability to use one.

The Federal Communications Commission had upheld the restriction. The appeals court examined the distinction between effects on the conversation and harm to the system. Using the cup for maximum privacy could reduce the clarity and volume heard by the other party. But the court saw no findings supporting the claimed injury to the telephone system generally. It set aside the Commission's order and sent the case back, leaving the antitrust questions open.[66]

This was a dispute about a mundane object, which is precisely why it is useful. We do not need to settle whether Hush-A-Phone was destined to transform communications. Its claim could be considered at the level of what the device did and whom it affected. A privacy aid might involve a trade-off that the two people in a conversation were capable of choosing for themselves.

A subscriber could cup a hand around the mouthpiece to seek privacy. The court asked why an attachment producing a similar effect should be treated differently. The cup also left the hand free to write. A restriction presented as protection of the telephone system reached down to an ordinary choice about how to hold a private conversation.[66]

A general network cannot function if every user is free to damage everyone else's service. That concern deserves a technical answer. The

question is whether a particular restriction follows from a demonstrated harm, a reasonable precaution, or a blanket preference for equipment supplied by the operator. Those possibilities lead to different rules. They should not be made indistinguishable by calling all of them protection of the system.

When a customer cannot use an independently supplied device, the market produces less experience of that use. Fewer people learn whether it is worth the inconvenience, whether a different design works better, or whether another application is possible. A prohibition can affect the record of performance as well as the maker's immediate sales.

The maker had already sold devices; this was no wholly untested invention. Its complaint went to the FCC in December 1948. Hearings followed in 1950, oral argument in 1951, and a final Commission decision only in December 1955. The appeals court noted the four-year delay after argument. Permission was a question about time as well as principle: an independent producer waited while the established system continued to operate. The court's record supports that chronology, not a calculation of the sales or discoveries the delay prevented.[66]

A footnote exposed the unequal starting positions. Telephone companies could offer equipment unless the Commission stopped them; an independent maker faced authorization before competing. The court called this unfair and said delay aggravated it, while leaving the legal question unresolved. A permission system can allocate time itself: one side continues accumulating customers while the other waits. That interpretation concerns a regulator's procedure as well as private control. It is distinct from the judgment on the cup.[66]

What happened after the decree

Decades later, Martin Watzinger, Thomas Fackler, Markus Nagler, and Monika Schnitzer studied invention after Bell's 1956 settlement. They compared follow-on patenting associated with Bell patents with carefully matched non-Bell patents. Their published finding was a lasting increase in innovation outside telecommunications, while the corresponding increase did not appear inside telecommunications, where customer access remained constrained.[67]

The design matters because a simple before-and-after count would be difficult to interpret. Semiconductor and other technologies were developing for many reasons. A comparison group provides a way to ask whether the patents affected by the licensing change developed differently from otherwise comparable patents. The credibility of that inference depends on the matching, the timing, and the absence of other differences that would account for the result.

Follow-on patents and their connections to earlier patents capture one dimension of invention. Products that went unpatented, their value to customers, and the achievements of companies that never developed remain outside that measure.

The contrast points to a practical question: can a business actually use the knowledge it is allowed to obtain? A patent license may be worth little if the route to customers remains closed. The Hush-A-Phone dispute makes the telephone network's rules about use tangible.

A buyer may have sound reasons to require reliability, security, and compatibility. A supplier may fail those requirements even with full access to the underlying knowledge. Testing whether an alternative meets a requirement produces evidence; barring the test leaves its performance unknown.

Bell's study measures follow-on patenting after a licensing intervention. To investigate the opportunities behind that result, we would also need to know which teams gained access, what work they attempted, and how their chances of reaching users changed.

The decree was an intervention in the economy's rules. It was not a replay of Bell's earlier history with a different lucky first award. That distinction helps explain what its evidence can do. It can inform our understanding of a change in access. It cannot show us every other route the same unchanged economy might have taken, or give each later inventor an exact share of credit for the resulting industry.

A list of patents is a record of inventions already made. A route to a customer can produce evidence about applications that have not yet been tried. Opening the first does not automatically open the second. This is why an institution can distribute a great deal of existing knowledge while leaving important opportunities for future comparison under the control of the same organization.

An advantage can become useful

One reason these choices matter is that success can improve the thing that succeeds. W. Brian Arthur's 1989 account of competing technologies made this mechanism explicit. Adoption can bring experience and improvement, attracting further adoption. Under the specified increasing-returns conditions, historical events can help select a durable technological outcome that need not have been the best available development path.[62]

Suppose, in a thought experiment, a widely used software system has trained operators, tested connectors, and a large collection of fixes. A rival's underlying design might be promising, yet the established system can be the practical choice today. Its accumulated advantages are services users actually value.

Could another history have developed a better set of capabilities? Changing a price at the end may not recreate the development that never happened along the way. A later subsidy can alter incentives; it does not automatically supply years of missing experience.

Bell's technology sharing makes the constructive side visible. Accumulated knowledge can become an input for other people's work. A strong incumbent need not keep every improvement inside its own boundary. An institution may preserve the benefits of a developed

technology while allowing others to learn from and extend it. Whether a particular rule achieves that depends on its practical terms.

Historical dependence, increasing returns, and technological dominance do not by themselves establish a finite lifetime comparison budget. To investigate that property, we need the allocation probabilities and an account of how productive inputs affect them.

The machine and the workforce

The keyboard offers a familiar version of the problem, and a warning about telling it too confidently. Paul David's 1985 paper used QWERTY to explain historical dependence through the interaction of machines, trained typists, and conversion costs. S. J. Liebowitz and Stephen Margolis subsequently disputed his empirical account, including the evidence offered for the advantages of switching. The keyboard is a contested case, not a settled demonstration of an inferior standard trapping the economy.[68]

The underlying coordination question remains useful. Imagine an employer choosing equipment that existing workers already know how to use. Choosing it can be sensible even if a different arrangement might have worked well had training developed differently. To establish that switching would improve matters now, we would need evidence about both the alternative's performance and the costs of getting there. Pointing to a different imaginable beginning would not settle the decision.

Transition costs change what a comparison measures. A technology can perform well in a demonstration while slowing work during training. An afternoon's performance and a year's performance may answer different questions. The evaluation horizon must follow the decision, rather than be selected afterward to favor a preferred technology.

Hush-A-Phone sharpens the contrast. Its case did not require proving that every subscriber should use the attachment. A subscriber's reasonable choice about privacy was narrower than a

claim that the whole network should be reorganized around the device. The scope of the requested opportunity mattered. Requiring proof of a total technological revolution before allowing a modest use would have set a peculiar standard for entry.

For an investigator of historical advantage, these examples impose a useful discipline. Identify the change being evaluated. Describe the cost of adapting to it. Look for evidence capable of distinguishing a promising alternative from an appealing story. That discipline protects the case for useful comparisons: its credibility should not depend on every famous account of lock-in turning out to be correct.

Finding out what nobody knew

Friedrich Hayek put discovery at the center of his account of competition. In a lecture delivered in 1968, published in the English translation cited here in 2002, he argued that competition reveals facts that would otherwise remain unknown or unused. Its point involves uncertainty about the particular results it will produce. If every relevant answer were already known, the discovery process would have little to discover.[69]

One outside learner was Tokyo Tsushin Kogyo, later renamed Sony. Its corporate history describes engineers studying a technical book and reports Kazuo Iwama sent from a 1954 visit to Western Electric. The team had to develop its own manufacturing equipment. Acquiring rights had not delivered a finished factory.[70]

Discovering a useful product and measuring each input's contribution to its maker's rewards are different tasks. A market can show that something works while leaving us uncertain about how its producer became the one able to offer it.

Licensing had not delivered a saleable radio. Sony's account describes a prototype, the TR-52, whose housing warped as the weather warmed in 1955. Plans to sell it were abandoned; the redesigned TR-55 reached the market later that year. A working transistor had left a practical problem unresolved.[70]

A licensed invention still needs manufacturing equipment, a casing that works, and a product someone can use. Development can uncover problems nobody priced into the original offer. Solving them takes further work.

At Hush-A-Phone's scale, a user could learn whether the privacy benefit justified the effect on a conversation. At a laboratory's scale, a company could learn how to turn a semiconductor technique into a product. These are different forms of discovery. Preserving the opportunity for them does not require a planner who already knows all the answers. It requires rules under which relevant answers can be produced and used.

There is no reason to expect that process to be costless or universally successful. A pilot can teach the buyer that a product is unsuitable. A development project can consume resources without producing an improvement. Competition's capacity to discover includes discovering disappointments. The economic justification for an opportunity concerns the value of what might be learned, assessed before knowing which result will arrive.

Who gets to accumulate a record

Merton described how scientific recognition brings resources and attention for later work. An early advantage can change what a researcher becomes able to do. The later achievement may be better because recognition helped make the improvement possible.[71]

Opportunity can produce the evidence used to allocate the next opportunity. Paid work finances development, exposes practical problems, and establishes a record. A weaker later application can therefore be both a consequence of earlier exclusion and a reason for another rejection.

The field experiments discussed in Chapter 1 isolated one part of this sequence. Van de Rijt and colleagues randomly supplied initial advantages and observed subsequent success in several settings. Their Kickstarter experiment found more later third-party support for projects given an initial donation. Larger initial advantages did not

produce proportionate increases without limit. The causal evidence concerned specified interventions, rather than a universal law of runaway success.[72]

Lacking a record differs from being unable to acquire a useful one. The first is an observation; the second concerns a development process. Treating them as interchangeable allows an existing allocation history to justify its own continuation.

Bell's licensing change can be understood as altering one input into such processes. A firm with access to a useful invention could attempt work previously made more difficult by licensing terms. The work would then generate its own results. This interpretation places the opportunity between existing knowledge and later achievement. It does not assume that removing one obstacle removes all the others.

Alchian's account of profit-based selection supplies another established perspective. A process can select surviving firms without requiring every firm to foresee or deliberately optimize the circumstances that favor it. In the Bell setting, the relevant environment includes the rules for using knowledge and reaching customers. Surviving within that environment and identifying the contribution of each input remain different analytical tasks.[73]

Selection and institutional analysis belong together. Success can reflect skillful adaptation to existing requirements. Whether those requirements were useful, whether alternatives could have developed under different arrangements, and what future changes would accomplish are additional questions.

Discovery inside the large organization

Bardeen described a field-effect proposal that initially failed. Investigating the failure directed research toward semiconductor surfaces; the point-contact transistor emerged from that program. Bell's laboratory could pursue an unpromising result until it opened another route.[63]

A patent list would not have performed that investigation. It required materials, instruments, theory, and people able to work together. Breaking those capacities apart might have made the experiment harder. Their value helps explain why opening a boundary and dismantling an organization are different proposals. The telephone attachment could be assessed without pretending the laboratory supplied nothing worth preserving.

The 1952 symposium adds a second dimension. Knowledge developed inside a large organization could also be transmitted beyond it. Internal research and outside experimentation can be complements. The question is how the arrangement connects them: what is shared, which uses are feasible, what costs are covered, and whether someone outside can produce a result that the larger organization would not have produced itself.

An internal experiment may reveal the input effect an outsider wants to understand. Its confidentiality can prevent that outsider from checking or using the result even though the evidence exists. Producing knowledge and making it accessible require different arrangements.

Likewise, an outside supplier is not automatically a fresh, independent history. It may use the same underlying components, draw on the same training, or depend on the same route to customers. These connections can be efficient. They also affect what a comparison can distinguish. Counting company names without examining the underlying experiment would give the institutional arrangement an informational meaning it has not yet earned.

Bell's record therefore supports a question more specific than whether large firms innovate. Which kinds of learning does a particular arrangement make easier? Which does it discourage or prevent? What could a feasible alternative preserve? Those questions recognize the laboratory's productive organization while leaving room to examine the restrictions around it.

The company's own case

AT&T's annual report for 1955 described an experiment in Americus, Georgia. Equipment using transistors carried five conversations over a single pair of rural telephone wires. The report also described experimental use of a Bell solar battery to provide part of the equipment's power. Here was discovery inside the system, directed at a practical service problem.[74]

The report made the company's argument for integration explicit: the people designing equipment, manufacturing and installing it, and operating the service belonged to a unified organization. In discussing the decree, it welcomed the preservation of those relationships. It predicted that major services would continue and that revenues and earnings would not be appreciably affected. Those were the company's contemporary claims and expectations, rather than a later independent evaluation.[74]

The Americus trials required a new component, its power supply, and an operating service to work together. Combining those kinds of expertise can make an integrated organization productive. An access rule has to preserve that capacity where it is valuable while allowing others to test work the organization would exclude.

At the same time, successful coordination inside an organization cannot establish that every outside restriction is necessary to achieve it. A test of rural equipment and a rule barring a telephone attachment concern different activities. Evidence that the first worked does not settle the second. Following the actual activity keeps a broad institutional defense from standing in for the narrower comparison a particular decision needs.

Integration and specialization create empirical tradeoffs. An interface may introduce failures an integrated service avoids, or permit a component to improve without replacing the whole system. The consequences depend on the technical boundary and must be tested.

Requiring proof that an entire replacement system works before permitting a component test can make that test prohibitively costly. Such a requirement needs a technical or economic justification, particularly when the component can be evaluated safely on its own.

A component's successful pilot can reveal little about the cost of replacing a whole system. Coordination that works in a limited test may become harder at a larger scale. Moving from one component to an entire service adds work the pilot never attempted.

Integrated production can make a useful component possible while restrictions prevent other components from reaching customers. Opening a patent, connecting an attachment, and breaking up a laboratory would change different parts of that arrangement. Their effects depend on which productive connections survive and which new work becomes feasible.

A bounded interface can preserve much of an integrated service while allowing a relevant component comparison. Reliability, coordination costs, and productive gains determine whether that arrangement is worthwhile.

A rule for connecting

In 1968, the FCC addressed another independently supplied device: the Carterfone, invented by Thomas Carter. It connected a private two-way radio system with the telephone network. The telephone handset rested in a cradle; the arrangement allowed a radio user and a telephone caller to speak through the base station. This was a practical connection between systems, rather than a new public telephone network.[75]

The telephone companies treated its use as prohibited under their tariff. The FCC found that the Carterfone met a need and did not adversely affect the telephone system. Its decision rejected a blanket prohibition covering harmless as well as harmful interconnecting devices. The Commission preserved the possibility of reasonable standards and restrictions on devices that actually caused harm.[75]

The cradle makes the economic issue visible. A person could possess the radio, the telephone, and a device capable of joining the conversation, yet face a rule forbidding that combination. The performance of the combination could not be inferred from a telephone's success on its own. A new use required an opportunity at the boundary between established systems.

The FCC also allowed the network to change. Telephone companies could improve their systems and revise reasonable standards. Independent manufacturers would have to comply; an owner might have to rebuild a device or stop using it. Permission to connect did not oblige the network to preserve every old attachment forever. It created a boundary with responsibilities on both sides.[75]

That rule changes the entrant's task. Instead of seeking permission simply because its equipment came from elsewhere, it must satisfy protective requirements. Those requirements can themselves be disputed and evaluated. The economic promise is a workable route to use, with a test of the claimed harm. Whether a particular implementation delivers that promise remains an empirical question.

A patent license gives an entrant permission to use an invention. Connection rules determine whether its device can reach the network. Technical standards determine whether it can operate there. Customers still have to want it. Clearing one stage leaves work to do at the others.

A right to connect becomes useful when someone can actually make the connection. The device must work with the network, and the terms must allow it to be used. Hush-A-Phone and Carterfone let us follow the gap between legal permission and practical opportunity.

Opening a route, paying for it

An experiment useful for future decisions can still be expensive for present users. Staff time, duplicate infrastructure, migration, and temporary reliability losses belong in the cost of preserving an opportunity.

There are costs on the innovation side as well. A policy that makes existing knowledge easier to use can affect the rewards expected from producing knowledge in the first place. That is a prospective incentive question. It requires attention to how research is financed and how the rule changes behavior. The usefulness of follow-on work does not remove it from the evaluation.

Bell's licensing intervention produced different innovation outcomes inside and outside telecommunications. A proposed policy in another industry needs to identify the knowledge at stake, the decisions it could improve, and the costs of preserving or replacing it.

The benefit is the improvement in decisions supported by additional evidence. An independent evaluation or better logging may answer the question more cheaply; an adequate result may already exist. Some uncertainty will be too costly to resolve.

Timing can alter those costs. Once infrastructure is organized around a single approach, an alternative can become expensive to test. Preserving a limited route earlier may retain an option difficult to recreate later.

A pilot can be badly chosen, poorly measured, or retained after it stops helping. Closing an unproductive program releases money and people for better uses. The reason to preserve an experiment lasts only as long as the prospective value of what it can teach.

The histories inside the system

Bell's transistor, its technology-sharing program, the patent decree, and the attachment disputes trace successive stages of economic possibility. Research produced something new. Licensing affected who could build on it. Rules about use affected who could reach an actual customer. At each stage, productive organization and institutional permission helped shape what happened next.

The patent study followed an actual change in access and the invention that came afterward. A shadow future asks instead what another history could have produced under unchanged rules.

Historical interventions supply evidence about changing an institution; the unrealized histories identify the alternatives missing from its ordinary record.

The cup and the cradle bring us back to the carpenter and the two towns. A boundary can let an independent invention reach an established network. It can also make the network owner the judge of what everyone connected to it may try. Digital markets extend that question: how many apparent destinations still lead to the same decision about who deserves a chance?

New technologies need feasible routes to development, delivery, and evaluation. Accomplishments deserve to be understood within the arrangements that produced them, including the rules governing who can attempt the next one. The histories that become infrastructure matter because later possibilities must pass through them.

Who gets to be tested

Before the first query

The choice can be made before you type.

Open a browser on a new device and enter a question into its address bar. Unless you have changed the settings, a search engine has already been selected to receive it. You can have a strong preference for that engine, no preference, or no clear idea how the selection happened. In each case, the question goes somewhere, a result comes back, and a little more experience accumulates around the service that received it.

In the Google search case, that apparently ordinary sequence became the subject of an extraordinary record. Browser developers, device manufacturers, and wireless carriers had agreements governing default search placement. Google's payments for distribution exceeded $26 billion in 2021, according to the district court's findings. Default access was valuable enough to command an enormous price even when a user paid nothing to enter a query.[76]

The court's August 2024 opinion credited Google's skilled engineers, continuing innovation, and high search quality. It also found that Google had unlawfully maintained monopolies in general search services and general search text advertising through exclusive distribution agreements.[76] Building a useful service and controlling a rival's route to customers are different activities. We can reward the first while restricting the second.

Distribution also shapes what a search engine gets to learn. A query is an opportunity to serve someone. It can also generate information about what people want and how well the service responds. A distribution arrangement can therefore influence both today's allocation of attention and the evidence available for tomorrow's

improvement. The advantage does not have to remain a matter of habit. It can help make the favored service better.

That possibility should make the question more interesting, not easier. If an engine learns through use, a rival may need use to develop. Yet a user can reasonably prefer the engine that has already developed. The established provider's present quality and the rival's difficulty acquiring experience can reinforce each other. We cannot resolve that sequence by announcing that quality matters, or that exposure matters, and refusing to consider the other.

The phone also brings us back to Bell. A route to the customer can govern which alternatives produce a record. For the telephone attachment, the route involved permission to use a device. For a search engine, it can involve the place where a question first arrives. The technologies and legal cases differ. The economic question travels: who gets a meaningful opportunity to be used, improved, and evaluated?

A query passes through several decisions. Which service receives it? Which service generates the results? Who observes the response? Who can use that response to improve? The answers can differ even when the screen offers the same number of brand names.

Firefox tries another default

The public record contains a useful change of route. From 2014 through 2017, Firefox's default search engine was Yahoo. Previously it had been Google. According to the court's account, Google's share of Firefox queries fell abruptly from roughly eighty to ninety percent to roughly sixty to seventy percent. Yahoo's share rose from around ten percent to around thirty. In 2017, when the default returned to Google, the query shares moved back.[77]

The users were not compelled to stay with either provider. Some evidently took a route different from the default. That did not make the default irrelevant. Its importance lay in how the arrangement changed the distribution of queries among people who retained the

ability to choose. A friction does not have to be an absolute prohibition to affect who receives an opportunity.

This was an observed change in an actual browser, rather than a clean randomized experiment. Search quality, advertising, and commercial terms also belonged to the history. The court described Yahoo increasing advertising to meet its payment commitment, worsening the experience and contributing to Mozilla's decision to change back. The switch cannot be turned into a timeless estimate of what every default does in every setting.[77]

What it does give us is a concrete contrast. A statement that Google received most Firefox queries has a different meaning once we know that changing the preselected engine changed the shares. The outcome reflects an arrangement as well as individual responses within it. A responsible explanation should investigate both.

Reaching users can expose a service's weaknesses. The additional use may reveal a problem, finance an improvement, or increase commercial pressure on the provider. Giving a rival a trial leaves room for it to disappoint as well as improve.

A successful experiment can strengthen the case for an incumbent. Conversely, a change of supplier can follow a poorly designed trial. The identity of the eventual winner cannot establish whether useful learning occurred. That requires examining the test, its outcomes, and the decision it informed.

Firefox's actual history also helps distinguish a changed rule from a replay. Changing the default changed the arrangement through which queries were distributed. It was an intervention. A shadow future under the old arrangement would hold the relevant rules steady and follow another feasible sequence. Both comparisons can help us think, but the evidence from a changed default answers a question about that change.

When a query teaches a search engine

The district court described several ways that user activity helped search. Queries reveal what people ask. Clicks, returns to the results page, and other interactions can help assess relevance. Query data can help decide what belongs in an index and which pages need to be revisited. More use can improve the work of finding and presenting information.[78]

An unusual query makes that advantage tangible. An engine cannot learn from a question it never receives. Nor does one encounter necessarily reveal which result was useful. A click, an immediate return, and a later reformulation can tell different stories. The court described rankings informed by accumulated interactions. Scale supplies more than a larger inbox: it can supply repeated occasions to learn which answers people find useful.[78]

More queries can improve the established engine's responses to the tasks it encounters. They leave a different service's development untested when that service receives no comparable opportunity to operate. The incumbent can learn more about its own performance while outsiders remain harder to evaluate.

A search result also reflects several decisions. The provider has chosen which pages to collect, how to rank them, and how to present the answer. A user's response concerns what was shown under those choices. It does not directly report the response to every result that might have been shown by another system. The record can be rich and still have an uneven relationship to the comparisons we want.

That unevenness gives the word “data” too much work to do. A catalogue of pages, a sequence of user interactions, and results produced by another company's engine are different resources. Each can help a competitor in a different way. Handing over one does not automatically hand over the capability to generate the others. The remedy proceedings eventually had to address these distinctions in practical detail.

Search share records where queries went. The comparison budget would accumulate their conditional chance of going outside the favorite, using a defined allocation unit and justified probabilities. Measuring contribution would also require a target and all informative observations. The court's search-share figures alone supply neither those probabilities nor their lifetime total.

The favorite here is the alternative most likely to receive the next allocation under the model. It can change with history. A service's share of past searches is an outcome, not that conditional probability; turning one into the other would require separately justified evidence and assumptions.

The historical mechanism and the model can nevertheless direct attention to a common practical problem. Opportunities to serve users affect what providers can learn and become. A challenger can be told to improve before receiving use, while useful improvement partly depends on receiving use. An institution concerned with competition has to examine how that loop works, alongside the benefits that existing scale supplies.

The comparison inside Google

The court record includes a particularly useful complication for a book about missing comparisons. Google evaluated its search quality against other engines using human-rated, side-by-side studies. It also compared its speed with Bing's. A 2017 comparison found a latency disadvantage and helped prompt Project Folly, an effort to reduce delays. The court described the project as successful.[79]

Inside the dominant organization, engineers could identify a weakness, examine a rival's performance, and improve. The same institution could produce useful comparisons internally while restricting competitors' access to queries.

A test of search speed can reveal how quickly a page loads. It leaves another provider's development under years of different distribution untested. Useful answers to the first question can coexist with missing evidence about the second.

For a user waiting for results, a faster response is a real improvement. For a potential entrant, a public demonstration of a faster prototype may still fall short of obtaining a route to users. Both perspectives matter. A system can support productive internal learning while making some kinds of external development difficult. That combination is more informative than a choice between declaring the incumbent all-knowing and declaring that it learns nothing.

Project Folly offers a more useful test than a count of experiments. Google's comparison identified a concrete disadvantage, and work followed to reduce it. An outsider appraising that evidence should ask whether latency is the uncertainty affecting the decision. If so, the test matters. If the question concerns another provider's development under different distribution, measuring response time cannot answer it alone.

A private organization can possess useful evidence that a buyer, researcher, or public authority cannot examine. The test exists, but others may be unable to verify the result or act on it. Opening the relevant evidence could resolve that dependence without requiring a new experiment.

A name for control over knowledge

Zahra Meghani called this kind of control an “epistemic monopoly.” Her 2021 paper examined regulatory agencies' dependence on product developers for the knowledge needed to evaluate technological products. She argued for access to credible sources beyond the sponsor and examined how broad patent rights could let a developer control conditions of independent research.[80]

For a particular question about contribution, epistemic monopoly means control over the opportunities needed to compare alternatives. An actor that controls who gets tested and who can inspect the results can help determine which explanations of the rewards will ever receive a hearing.

Distribution contracts can affect who receives queries. Data terms can affect who learns from an existing record. Confidentiality can affect who checks a test. These controls can overlap while obstructing different steps in an outsider's work.

Verifying invoices, hours, and completed work establishes inputs and execution. Identifying their effects requires suitable comparisons. Access to an existing informative experiment is a different problem from arranging an experiment that has never occurred. Those obstacles call for different responses.

This is why an information argument should begin with a question rather than a label. Which contribution are we trying to understand? Which observations could distinguish the relevant explanations? Who controls the route through which those observations would arise? The term becomes useful when it organizes that investigation. It becomes an accusation without content when it merely replaces “large technology company” with a more alarming phrase.

The product on the bench

Meghani's discussion has a physical counterpart to the search record: a product that outside scientists need to study. Her paper examined a dispute over the regulation of dicamba herbicides. The underlying court opinion, issued in June 2020, described Monsanto's field studies in support of an initial 2016 registration. The company had tested its reformulated herbicide in Georgia and Texas. At that time, it had not made the herbicide available for independent studies.[81]

There were results, then, and there were experiments. The problem was not an empty file. A developer had produced evidence about its own product, while another route to producing evidence was unavailable. That is different from saying the developer's measurements must have been false. It asks what could be learned by investigators able to choose and carry out their own tests.

The court later vacated the challenged 2018 registrations on grounds concerning the agency's treatment of risks. It did not base that decision simply on a rule that every product lacked satisfactory

data unless independently tested. The historical detail relevant here is the organization of inquiry before the earlier registration, not a general judgment about every herbicide or a statement of current regulatory status.[81]

A scientist with permission to read a company's white paper and a scientist with permission to test the product have different opportunities. The first can examine what the reported study says. The second can investigate another condition, reproduce a result, or find that the first study left a consequential question unanswered. The work may confirm the developer's conclusion. Independence matters because it permits a relevant inquiry, including one the developer did not select.

Without enough information about methods, an outside scientist may struggle to reproduce a study. Without the product, a detailed method may remain impossible to try. Disclosure and access have to meet the practical requirements of the investigation together.

Disclosure is useful when the existing record answers the question. Releasing another copy of a performance report cannot replace a failure condition that was never tested. The first task is to identify whether the missing element is access, interpretation, or another observation.

Expertise, instruments, and experience matter. An independent test can be incompetent, redundant, or badly designed. The institutional objective is to enable credible investigation of consequential questions and to assess its results, rather than to treat independence as a substitute for scientific quality.

More signs, one route

The historical concern is the reach of a judgment. A rejection by one independent buyer can leave other buyers free to form their own assessments. A shared ranking can transmit that rejection across a much larger market. Demand can expand geographically while the routes to independent evaluation contract.

Shared services can also reduce prices, improve reliability, and connect entrants to distant customers. The question is whether those gains preserve a practical route to another judgment. Multiple websites and separate owners do not establish that independence by themselves.

Search syndication gives this distinction a real institutional form. A service can obtain search results from another provider and present them through its own product. This can make entry feasible without duplicating an entire search infrastructure. It also means that several user-facing services need not represent several independently developed engines. Their relationship has to be described before their number is treated as evidence of independent discovery.[82]

The reverse possibility matters too. A single operator might maintain genuinely informative trials of different approaches. Project Folly is a reminder that a large provider can learn through comparison, although its particular tests do not answer every question. Counting organizations could miss that activity just as counting storefronts could exaggerate the independence of their underlying systems.

Separate firms can bring different incentives, technical judgments, and routes to users. Common ownership can make some questions less attractive to investigate. What a comparison reveals depends on those differences and on the tests the organizations actually conduct.

For a regulator or buyer, the useful map therefore follows functions. Who develops the service? Who distributes it? Who records outcomes? Who can change the experiment? Who can examine the result? The same company name can appear in several boxes. Different names can lead back to the same underlying process. The map tells us more than a count of badges on a screen.

Three different doors

The September 2025 remedies opinion and the December final judgment addressed several routes at once. The court did not require Google to sell Chrome. It restricted specified distribution conditions, required particular forms of data access, and provided for search syndication. These were distinct interventions with different objects. A headline announcing that Google must share would not tell a rival what it could actually obtain.[82]

One door concerned distribution. The judgment restricted tying the licensing of Google applications to the placement of specified Google services and restricted arrangements preventing partners from offering rivals. It also imposed one-year limits on certain agreements. These provisions concerned the route through which a user encounters a service. They did not themselves give the rival a finished search engine.

A second door concerned data. The judgment specified information from Google's web index and particular user-interaction datasets used in ranking models. An index is a structured record of what the service has found on the web. User interactions can inform relevance and training. Giving a rival access to these resources could help it develop a service, but the judgment did not simply hand over all Google's algorithms, ranking signals, or trained models.[82]

The web-index provisions named concrete fields: document identifiers, their corresponding web addresses, when an address was first seen and last crawled, a spam score, and the device type. They also included information identifying documents Google considered duplicates. These are useful ingredients for another provider's work. They are not an instruction to make the same ranking decision. Nor did the access order transfer the intellectual property rights of third parties whose material appeared in the index. An entrant could still need machinery for collecting pages, interpreting them, and delivering results. The list makes the distinction between knowledge access and a finished capability inspectable: we can ask which

specified resource removes which obstacle, rather than treat the whole search engine as a single indivisible secret.[82]

The third door concerned an operating service: syndication. A qualified competitor could obtain specified search results and features through an interface, subject to the judgment's terms. This allows a provider to use capabilities it has not reproduced for itself. It can lower the burden of serving users while the provider develops. It also raises a question about how the arrangement supports eventual independent capability.

The judgment limited reliance on that route. First-year Google syndication was capped at forty percent of a qualified competitor's annual US queries, with tapering over five years intended to encourage investment in independent search capabilities. The court left the pace and measurement of tapering for further determination. Access supplied a bridge; the terms sought to keep the bridge from becoming the competitor's permanent foundation. Whether that would work required evidence from implementation.[82]

Market access, access to knowledge, and permission to use a technical service change different opportunities. A measure that opens one route can leave the others closed. Evaluation needs to follow the resulting ability to develop and test alternatives.

This is where a comparison perspective becomes more demanding than a general endorsement of openness. Which door produces the evidence relevant to the decision? If the buyer wants to compare interfaces, access to the same underlying service might suffice. If it wants to compare independent methods of generating results, that arrangement may preserve the very dependence being investigated. A remedy must be connected to the question it is expected to help answer.

The court's findings of legal violations supplied the basis for intervention. Choosing a remedy raises a further practical question: will a rival gain a workable way to produce evidence that could change a buyer's choice?

The cost of opening them

Search payments also financed something outside search. The court's 2024 opinion reported that Google's payment to Mozilla exceeded $400 million in 2021 and represented about eighty percent of Mozilla's operating budget. The financial connection supporting Google's default position also helped support a separate browser. A change directed at one market could therefore affect the resources available in another.[83]

Mozilla's chief executive, Laura Chambers, welcomed the court's rejection of a proposed ban on search payments in September 2025. She argued that a ban would weaken independent browsers. She also welcomed greater flexibility to offer other search and AI services and called for careful implementation of data remedies.[84]

The position is economically intelligible. A browser is the software through which a person accesses the web; a search engine supplies answers to searches. More competition in one does not automatically finance competition in the other. A payment can help sustain a browser while also making a particular search relationship financially attractive. The arrangement has to be examined across both activities.

The court considered the risk that a broad payment ban would harm distributors or leave them using Google without receiving revenue. It declined to impose the requested blanket ban. That conclusion did not mean every distribution restriction was acceptable. It meant the remedy had to be assessed in the system where its effects would unfold.[85]

Removing a financial tie might open a search opportunity while weakening a browser through which alternatives reach users. Keeping the tie might preserve that browser while maintaining incentives unfavorable to rival search providers. One route into a market can depend on another, making the terms of access matter as much as their apparent openness.

A remedy can also remove useful experimentation financed by an incumbent's existing business. Any appraisal must count evidence lost

as well as gained, alongside changes in production, financing, and access.

This is not a demand that every alternative pay for itself immediately. A comparison can have benefits beyond its first transaction. It is a demand to identify those benefits and their costs. Hayek's discovery argument gives us a reason to value processes whose particular results are unknown in advance. It does not give us a reason to assume that every rearrangement of those processes increases the value of what gets discovered.[86]

Who can examine the test

Data-sharing disputes make the same problem concrete in a different way. A search record can contain sensitive information even when a name has been removed. In the remedies opinion, the court treated privacy as a valid concern and discussed techniques for reducing disclosure risks. The final judgment required safeguards, licensing terms, and information about the contents and preparation of the datasets.[87]

These requirements belong inside the informational analysis. Data made safer to share can become less useful for some purposes. A record whose preparation nobody understands can be difficult to interpret. A technically useful record might still be inappropriate for unrestricted circulation. The objective is a defensible opportunity to learn, with the people represented in the record included in the accounting.

The judgment's definition of a qualified competitor also involved more than asking for a copy. It included approved data-security standards, privacy and security audits, and a sufficient plan to invest and compete in or with the specified markets. Access was an administered relationship with obligations. It was not a public download button available to anyone who described themselves as a challenger.[82]

Conditions can make a remedy safer while raising the cost of using it. A small organization may struggle with requirements a large rival

can easily satisfy. The workable standard is one that protects the required security or quality without imposing avoidable barriers to a credible attempt.

Mozilla's response raised a related concern: data remedies should avoid simply transferring power from one major technology company to another. That was a stakeholder's warning about design. It deserves consideration alongside the potential benefit of making useful resources available. The name of the recipient cannot substitute for examining what the recipient will be able to test and how others will benefit.[84]

For a public evaluator, access to a test and access to its results are likewise separate. It may be possible to arrange an independent evaluation without releasing every underlying record publicly. It may be necessary to disclose methods so that a result can be interpreted. A confidentiality condition might protect legitimate information or prevent meaningful scrutiny. The right response follows the specific obstacle and purpose.

A published pass score may conceal performance on rare but costly failures. Detailed results could resolve that uncertainty without exposing every line of code. If the relevant condition was never tested, however, fuller disclosure cannot produce the missing result.

Control over what gets tried determines which evidence can be generated. Control over the resulting files determines who can inspect it. An institution can open its files while restricting new inquiry, or permit inquiry while keeping useful results difficult to check.

A screen that can be tested

Mozilla's 2023 study randomly varied browser choice screens during simulated device setup in Germany, Spain, and Poland. Information, list length, and timing were experimental choices. One comparison held the five-browser screen constant while changing when it appeared.[88]

When the screen followed a click on the preset browser's icon, roughly nineteen percent selected that browser, compared with eleven percent during setup. The preset was Samsung Internet or Edge. The preceding click was a proposed explanation, not a separately established mechanism.[88]

Chrome was an alternative here. Timing could favor a position occupied by another brand.

These simulated choices do not establish sustained browser use or independent search capability. A remedy's sequence can nevertheless affect its immediate outcome.

The choice-screen study changed a rule and observed responses. A shadow future would preserve the old rules while outcomes unfolded differently. The study supplies evidence about the intervention; a whole competitive history under other early events remains unobserved.

There is a constructive implication for the institution imposing a remedy. It can ask how to learn whether the remedy works. Which outcomes would show that a meaningful opportunity had opened? Which would indicate that users were merely clicking through another screen? What evidence would distinguish a temporary shift from a sustained improvement? Those questions should influence the record collected as the remedy operates.

They also create a route for revising an intervention that disappoints. A choice screen might need different information, a different timing, or a different role within a wider package. A trial can increase confidence in an incumbent or identify a problem with a challenger. An institution that values evidence should be able to learn both kinds of lesson without treating them as failures of principle.

The work after the judgment

The subsequent Google record brings implementation back into view. In their September 17, 2026 status report, the plaintiffs described the work of the technical committee, the development of processes, and efforts to implement the final judgment. The report discussed template licenses, requests for information, and exchanges over proposed terms. An obligation to provide access had become a sequence of organizational and technical tasks.[89]

This was the plaintiffs' account, rather than a court finding that every obligation had been fulfilled. It describes the practical route to access: who qualifies, which records can be used, how privacy is protected, what documentation is supplied, and which terms remain disputed. A rival's opportunity depends on how those details are settled.

A deadline in a status report is a plan, not a completed exchange. The proposed dates for templates and responses did not show that every later step had occurred or that a rival had received useful data. To judge the remedy, we would need to follow those steps through to their effects.

The legal dispute continued beyond the district court's decision. Google publicly announced an appeal in January 2026 and argued that the ruling discounted user preference, product quality, and competition. It also objected to the data and syndication requirements. In July, the plaintiffs' appellate brief defended relief and challenged the rejection of a payment ban. These were opposing parties' arguments, rather than an appellate resolution.[90]

The ordered remedy sets terms under which later evidence can emerge. As it operates, entrants' access, costs, and performance could support or undermine the expectations behind it. The parties' competing forecasts remain forecasts until then.

A formal promise of access becomes economically meaningful through specifications, a budget, security arrangements, usable

records, and a way to interpret outcomes. Those implementation requirements help determine whether comparison actually occurs.

An unused route could reflect a poor alternative, unnecessary conditions, or benefits that take time to develop. Those possibilities call for different responses. Evidence about the terms offered, attempts made, and work completed can distinguish them; the incumbent's continued success alone cannot.

Before a rival can grow

An alternative can disappear before it leaves enough evidence to judge. A database of completed transactions will contain no entry for a product whose development stopped before its first sale. Sharing the incumbent's records cannot recreate the engineers' unencountered problems, the customers' unmade requests, or the improvements that might have followed. Existing knowledge and opportunities to produce new knowledge require different remedies.

Matthew's Gospel tells a brutal story of preemption. Herod hears of a child described as a future king and seeks to eliminate the threat before it can grow. In Matthew's account, he orders the killing of Bethlehem's young boys, while the child he seeks has been taken to Egypt. The threatened successor is to be removed before he can become a rival. The economic question is what happens when an incumbent can close a challenger's route before there is a performance to compare.[91]

That question already has an economic literature. Colleen Cunningham, Florian Ederer, and Song Ma studied pharmaceutical acquisitions and subsequent project development. Acquired projects overlapping the buyer's existing drugs were less likely to continue development; the pattern was stronger where competitive pressure was weak. Their research investigates a mechanism through which buying a potential competitor can prevent competition from arriving. These are empirical findings with identification assumptions, not a randomized test of every acquisition.[92]

An acquisition can also finance development that would otherwise stop. Counting today's competing products can miss the consequences of either decision: continued funding may let a project mature, while closure may remove a chance to learn what the entrant could have built.

Peter Thiel makes a forceful case for monopoly in *Zero to One*, written with Blake Masters. In an essay adapted from the book, he compares competitive equilibrium to the "heat death of the universe." Durable monopoly profits, he argues, support ambitious research and long-term plans. His account anticipates new innovators replacing established monopolists.[93]

The difficulty begins with the route to becoming that replacement. A company can finance useful research while controlling the customers, interfaces, or tests a different research program needs. Its next invention will be visible. The rival's unrealized invention will not appear beside it in the accounts. How would we know whether protecting one laboratory accelerated discovery overall if the same protection prevented another laboratory from getting started?

Google's Project Folly makes the visible side concrete. A comparison exposed a speed disadvantage, and engineering work reduced it.[79] That improvement matters to users. It does not answer what a different supplier might have learned from a workable route to customers. The two developments could have occurred together, or access could have imposed costs large enough to change the research undertaken. Deciding between those arrangements requires a comparison that counts both development routes.

The mistake is to compare a funded innovator with a world of firms doing identical work on vanishing margins, then treat those as the available choices. An economy can preserve substantial rewards for an invention while requiring interconnection or bounded access that lets someone develop a different one. Commercial breathing room and authority over who may become a competitor are separable parts of the arrangement.

We can follow the dominant firm's research decades into the future and still leave a rival's possible development out of the calculation. The incumbent's acceleration then looks like the whole gain. A slower beginning for one large project might preserve several useful attempts; duplicated effort might also waste resources. The relevant comparison includes those costs and the independent learning they purchase.

A succession of innovators needs a route through which a successor can emerge. When the current winner controls that route, its freedom from competition can erode the process invoked to justify it. The decisive question is whether the next generation can acquire evidence strong enough to challenge the present one.

The pharmaceutical study concerns acquired drug projects; the Google judgment concerns search distribution agreements. Assessing either practice requires evidence about the development route affected, the benefits supplied, and the costs of a feasible alternative. A finding about one practice cannot establish another company's motives.

Requiring a mature product before granting the access needed to develop it protects the very condition preventing the rival from qualifying. Exclusive terms that close such an indispensable route should face a presumption against enforcement, subject to evidence that a narrower arrangement cannot protect legitimate interests. Existing antitrust already considers innovation and exclusion; an information-based remedy still needs an identified lost comparison, a useful decision it could inform, and benefits worth the remedy's costs. Legal liability depends on the applicable law and evidence.

Whose future is missing?

Return to the address bar. The question you type reaches an engine, and the engine returns a result. That useful act is real. So is the work that made it possible. The institutional question concerns the other routes by which useful acts might be attempted and evaluated, and the arrangements that govern who gets to learn from them.

For a buyer or public authority, the task is to make that question practical. Identify a decision worth improving. Find the comparison that could improve it. Ask who can produce and check the evidence, what doing so costs, and what existing knowledge can already resolve. Sometimes another route to users matters. Sometimes better access to a record or a well-designed internal trial supplies what is needed.

Rewards also carry personal consequences. A documented achievement can establish that someone made something valuable. How much they contributed, what they were promised, and what society has reason to reward remain separate questions.

CHAPTER 6

Who deserves the winnings

The song does not work harder in the world where it wins.

Lockdown, by 52Metro, is the same recording in the MusicLab market where it finishes first and the market where it finishes fortieth. The musicians have not composed a different chorus for the more successful world. Listeners arrive in different sequences and encounter different popularity records. The reward being distributed in this experiment is attention and downloads, rather than a documented royalty payment. But the experiment places an awkward question beside any argument that moves too quickly from success to deserving it.[94]

If we had seen only the winning market, what exactly would first place have entitled us to say?

That people downloaded the song, certainly. That it appealed to at least some listeners, also. That its makers deserved our admiration might be an entirely reasonable personal judgment. That the distance between first and fortieth measured an equally enormous distance in creative effort would require evidence the ranking does not supply.

This matters because we use the language of deserving to do several jobs. We use it to praise an achievement, enforce a promise, encourage a future undertaking, and justify the distribution of money. A sentence that sounds plausible in one of those jobs can become misleading when carried into another.

Consider a prize for building a machine. Its winner may deserve to be paid because the machine passed the announced test. A government might offer the prize because it wants more people to attempt the problem. An engineer might deserve particular scientific credit for the crucial design. Someone else may have supplied the financing that made construction possible. Those are four intelligible claims. They need not produce the same division of the proceeds.

The temptation is to find one number that settles all four. Profit is a convenient candidate. So is a salary, a patent award, or the size of a company. Each is already expressed in the language of measurement. Why not read the moral verdict directly from the account?

Because an account records what was paid under particular arrangements. Explaining what would have happened under different arrangements is a further task. Deciding which arrangements we ought to choose adds another. We can value the first record while asking the other questions.

The blue LED and the driverless vehicle were real achievements. Recognizing the ingenuity behind them still leaves a question about the reward. How much does making something valuable tell us about what its makers should receive?

The inventor and the bill

In his Nobel lecture, Shuji Nakamura describes buying a reactor to grow semiconductor material. The commercial machine had been designed for a different material from the gallium nitride he wanted. Some attempts produced no growth; others produced a black layer where the material should have been transparent. He spent a year and a half modifying the equipment. Buying the reactor was the beginning of the work.[95]

His two-flow design introduced another stream of gas that gently pushed the reacting gases toward the surface where the material grew. He reported uniform, high-quality growth over a two-inch surface. This was a physical change producing an observable result. The question of what the inventor deserved began with an achievement we can describe, rather than with the amount he was eventually paid.[95]

Nakamura's account also credits Isamu Akasaki and Hiroshi Amano, whose earlier advances helped establish gallium nitride as a workable route. His lecture explains how blue LEDs can produce white light by combining their output with a phosphor. The familiar bulb therefore leads backward into a history of materials, equipment,

and distinct technical advances. It does not lead to a single undifferentiated stroke of good fortune.

Someone who concluded from MusicLab that technical accomplishment is imaginary would have learned the wrong lesson. The reactor's performance supplies observations beyond a record of who received the next contract. Laboratory measurements, changed designs, and reproducible physical results can answer questions that a popularity ranking cannot. They belong in the evidence.

Now put a second document beside the lecture. In January 2005, Nichia announced a settlement with Nakamura, its former employee. The company's statement said it would pay approximately 609 million yen in remuneration and a further 235 million in interest damages. The agreement covered a broad collection of inventions, patent applications, and know-how from his employment, rather than just one blue-LED patent.[96]

The lecture and the settlement concern overlapping history, but they answer different questions. One explains technical developments through an inventor's account and cited research. The other records an agreement resolving an employment-invention dispute. Neither can simply be substituted for the other.

Interest makes the distinction particularly visible. Part of the payment concerned the passage of time. Adding interest changes what is owed without changing the semiconductor material, the reactor design, or the original experiment. It would be strange to conclude that the inventor's technical contribution grew by precisely the amount of interest accruing on a disputed payment.

The same point applies less visibly to the rest of an award. A legal settlement may reflect enforceable rights, litigation costs, compromise, and the scope of claims being released. Those considerations matter to the people involved. Settling their dispute does not require the payment to become a universal measure of the invention's social value.

Nichia's statement emphasized the risks carried by the employer and defended the settlement from its own perspective. That is an

interested party's account, not an independent allocation of all credit. Yet the underlying economic question deserves attention: what should a company receive for financing uncertain work? Praising an inventor does not supply an answer by itself.

Imagine, as a separate illustration, that a laboratory needs both a costly machine and someone capable of making it useful. Without the machine, the experiment cannot happen. Without the capable operator, it fails. Removing either may destroy the entire result. If we describe each indispensable input as causing all the output, we have not discovered a way to pay out the same proceeds twice. We have discovered that contribution needs a specified comparison.

Replace the operator with whom? Replace the machine at what cost? Allow how long for another laboratory to discover the method? Ask about the next batch of devices, the next decade of production, or the eventual availability of efficient lighting? These are different questions. An answer about one does not automatically answer the others.

For the hypothetical laboratory, the relevant replacement might be an available machine from another supplier. For a historical invention, it might be another research team pursuing a different route. The word contribution can sound exact while quietly moving between those alternatives. A responsible calculation must say what changes and what stays in place.

This is why shadow futures need discipline. Replaying history under the same underlying conditions and response rules lets us examine how outcomes vary. Removing an inventor, changing a financing arrangement, or replacing a technology is a separate intervention. We cannot establish the effect of removing a person merely by imagining a less fortunate version of that person's career.

Technical evidence can show that a reactor modification solved a production problem. Dividing the invention's full commercial and social returns among inventors, employers, suppliers and predecessors is a separate question.

Neither the original wage nor the later settlement can, by its size alone, measure an inventor's contribution. Letting actual payment establish deserved payment would allow every existing arrangement to certify itself.

Lockdown's two rankings expose a gap between a fixed recording and its reception. Measurements of a blue LED can settle important technical questions while leaving credit and reward unresolved. The uncertainty depends on the question: whether the device works, who helped make it, or what payment would be justified.

A prize waiting in the desert

On March 13, 2004, fifteen driverless vehicles set off on DARPA's 142-mile Grand Challenge. The best managed about seven and a half miles. Nobody collected the million-dollar prize. In October 2005, five vehicles finished a new, 132-mile course. Stanford's Stanley won the two-million-dollar prize in six hours and fifty-three minutes. The prize account changed from no payment to a large one; the history of work was longer than that entry.[97]

A prize waiting at the finish line has a useful property: its purpose can be understood before anyone reaches it. It invites people to attempt a problem. The reward can help justify expenses, attract collaborators, and concentrate effort on a testable objective. Those reasons do not depend on pretending that the eventual winner will be the sole author of autonomous driving.

Stanley's finishing time established something specific about performance on that course under those conditions. It did not measure the total contribution of every member of the team, every competing team, or every earlier researcher whose work entered the vehicles. A competition can deliver a clear winner while leaving that larger accounting unfinished.

There is no contradiction here. The prize organizer needs an administrable rule. The competitors need to know what qualifies. Paying the winner according to the announced rule can be justified by

the value of a credible promise. The rule need not claim to reproduce a complete moral history of the technology.

Nobody collected the first race's million dollars. The prize account recorded that no vehicle met the required threshold. What the teams learned along the way would have to be found in their tests and engineering records.

Equally, failure should not receive an automatic certificate of social usefulness. A vehicle that stops in the desert may reveal an important limitation or merely repeat a mistake that was already understood. Whether the attempt generated valuable knowledge depends on what it tested, what was recorded, and who could use the result. Failure is an outcome to investigate, not an achievement by definition.

DARPA announced the second challenge the very next day after the first ended. The decision preceded Stanley's victory by roughly eighteen months. It created another occasion on which teams could turn development into a public performance. We can inspect that institutional choice without assuming the organizer already knew which team would finish, or that every further contest would justify its costs.[97]

DARPA's documented sequence does not tell us what the optimal prize should have been. Doubling the headline amount between the two events was not a randomized test of prize size. The routes, technologies, participants, and time available also differed. The sequence demonstrates that an institution can announce a reward, observe failure, and arrange a further test without having solved every question of individual merit.

That is already a substantial lesson. Discussions of incentives often become arguments about whether a particular rich person is admirable. The more useful prospective question is what an offered reward induces people to attempt. The effect may differ across people, projects, and stages of development. Admiration is an unreliable substitute for investigating it.

Suppose a new contest promised a reward too small to cover even a credible attempt's costs. Increasing the prize might bring a valuable

team into the competition. Suppose instead that all capable teams were already participating and a larger prize changed only the amount transferred to the winner. The two increases would have different consequences. These are illustrative possibilities, not findings about DARPA's actual entrants.

In 2007, the task changed again. DARPA's Urban Challenge required vehicles to negotiate a staged city, deal with other moving traffic, and obey traffic rules. Six of eleven finalists completed it; Carnegie Mellon's Tartan Racing team placed first. A desert finish had not settled performance in traffic. Changing the test could expose a different set of capabilities and produce another winner. The institution kept asking questions beyond who had won its previous prize.[97]

The comparison budget adds the modeled chances of opportunities going outside the current favorite. A cash prize is not itself that budget. A million dollars can reward a contest with many informative attempts, or a demonstration whose outcome was effectively settled in advance.

The races left more than a winner's name. How far did a vehicle travel? Did it complete the course within the limit? Could it handle traffic under the new rules? Those observations distinguish failed attempts and successful ones more finely than the payment account does. Any application of the information argument must count that evidence. Discarding it would create an ignorance the actual test did not impose.

We can therefore defend a prize without treating its winner as the owner of all the knowledge it helped generate. We can credit unsuccessful teams for identifiable useful work without deciding that every attempt should earn the same payment. The practical task is to arrange rewards and tests so that worthwhile effort and usable learning can happen together.

The finish line resolves who receives a promised prize. It leaves us with the next design decision.

Paying for an attempt

William Kerr, Ramana Nanda, and Matthew Rhodes-Kropf describe entrepreneurship as experimentation. Their account begins before a startup has a market price that can supposedly reveal its worth. Financing an initial test can uncover whether a larger investment makes sense. The ability to stop after an unfavorable result can make an undertaking feasible when an immediate commitment to its entire cost would not be.[98]

That observation gives capital a role more interesting than waiting for luck. Someone must choose a test, pay for it, interpret the result, and decide whether to continue. A good arrangement can save resources by learning early that a promising idea will not work. The person who closes the project may have made a useful decision even though no triumphant company remains to display it.

Staged finance distributes the cost of discovery. Equity, customer-funded development, and grants can each support an initial test before a larger commitment becomes sensible. Their consequences depend on what is learned and how the parties share its costs and benefits.

Kerr and his coauthors also examined prediction within venture capital. In data from one successful firm, the initial partner scores did not clearly distinguish its later spectacular successes from weaker outcomes. All these ventures had already passed the firm's selection process. The comparison left the value of rejected ideas and the effects of later help from investors unresolved.

The successive races show why those abilities differ. A team unable to finish in 2004 might return with a better vehicle; a desert winner still faced a new task in a city. A backer could help make that development possible without knowing the final ranking. Requiring foreknowledge as a condition of financing would defeat much of the purpose of financing an experiment.

Expected returns can change which uncertain projects get funded. The effects depend on how people respond, which attempts disappear

under different terms, and whether alternative financing would support them.

Contractual entitlement has another basis. Investors and workers can act under agreed terms, complete the promised work, and acquire a claim to payment even when every causal contribution cannot be reconstructed. Reliable promises help people coordinate before outcomes are known.

That defense has limits of its own. A contract can settle what its signatories promised without settling every obligation to people outside it. If a hypothetical contract authorizes conduct that harms others, quoting the payment clause does not answer their complaint. Nor does a contract establish that its bargaining terms were the only possible terms under which the useful project could have proceeded.

The same financial share can have different economic explanations. Compensation for bearing risk differs from returns obtained through exclusive control of an indispensable route to customers. Investigating alternatives and behavior is necessary to distinguish them.

Useful tests may occur inside a large company, at a small entrant, or through a publicly supported project. Google's private experiments improved its service while restrictions on distribution affected what rivals could try.[99]

A defensible experiment specifies the uncertainty, the measured outcome, and the decision that follows. Hidden, uninterpretable, or irrelevant results are difficult to justify as public learning. A useful negative result can be valuable without producing a successful company.

This is a prospective defense of experimentation. It permits ambition and large rewards where they are justified by their effects. It also permits scrutiny of the arrangements surrounding them. We can ask for a future worth financing without first declaring that every past fortune was exactly proportional to its owner's contribution.

Different explanations for the same pay rise

A large pay rise does not identify its cause. It can accompany greater output, a different reporting arrangement, or a redistribution of an existing surplus. Those possibilities matter because changing incentives can affect them differently.

Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva make a related distinction in their analysis of top labor incomes. They separate responses to taxes into changes in productive activity, changes in tax avoidance, and changes in compensation bargaining. The same movement in reported earnings can carry different implications depending on which behavior produced it. Their tax analysis does not require an exact moral audit of every person's pay.[100]

A productive response changes useful output. An avoidance response can change the location or form of reported income without a corresponding change in production. Treating the same earnings movement as the same economic event would confuse those mechanisms.

Bargaining supplies another mechanism. In the authors' framework, bargaining can redistribute income without enlarging total output. An actual negotiation may also produce improvements, so applying that account requires evidence about the particular activity.

Historical position adds a separate attribution question. Productive work and an early advantage can operate together: opportunity supports development, and development improves later performance. Neither mechanism has to be fictitious for an exact division of the resulting income to remain uncertain.

Being helped by an early advantage does not establish misconduct. Historical position can matter alongside honest productive effort, without either accounting for the entire reward.

Likewise, lower earnings do not establish inferior effort or prove superior virtue. They are outcomes to explain through work,

opportunity, bargaining, and institutions before turning them into judgments about character.

Insurance against income loss, strong rewards for useful investment, and access to future evaluation pursue different purposes. Their effects and costs need evidence; none follows mathematically from the existence of alternative histories.

Casey Rothschild and Florian Scheuer provide another useful warning against an easy verdict. Their taxation model lets income combine productive work and rent-seeking that delivers private returns exceeding social returns. The authority need not observe each component separately. Changing taxes also changes the relative attraction of activities, so the effect depends on where the external costs fall and how people respond. A measure of rent-seeking alone does not determine the right corrective income tax.[101]

Resources devoted to useful engineering can respond differently to tax policy from resources spent pursuing privileges. A broad income tax can affect both activities, even when a narrower correction would distinguish them.

Other participants may change their behavior too. Altered returns can make productive work more attractive, or encourage new attempts to capture privileges. A description of waste alone cannot determine which response dominates.

Chapter 3 explains when an expanding allocation record can remain insufficient for universally reliable contribution recovery. A revenue chart cannot establish those conditions. Public finance can still analyze policy without settling every part of an income's history.

The practical implication is to state what information a proposal actually needs. An income-insurance rule might need earnings and eligibility records. A policy encouraging experiments might need evidence about costs and responses. An exact retrospective division of a fortune into contribution and position asks for something more demanding. These projects should not be treated as though they must all pass the same impossible examination.

The risk that gets counted

Risk falls on more than investors. Researchers commit time they cannot recover; employees depend on continued work; competing projects may lose support. Counting only capital exposure can settle a moral argument by excluding other consequences before examining them.

An investor can lose capital; a worker can lose a livelihood. Their exposure differs, as do the promises made to them. A judgment about rewards has to consider whose consequences are being counted, including the people whose risks appear nowhere in the winner's investment account.

The DARPA contest supplies a clean separation. A prize offers an upside for reaching a defined objective. It need not fully insure every entrant against failure. If we were designing another contest, we might consider grants for qualified attempts, shared testing facilities, or different prizes at intermediate milestones. Those are proposals to compare, with costs and possible distortions. They are not conclusions established by Stanley's victory.

An initial grant could let a capable but poorly financed team enter. It could also support a weak team whose attempt teaches little. A large final prize could encourage demanding engineering. It could also concentrate effort on the scored task while neglecting an unmeasured concern. Choosing the arrangement requires an account of the behavior we want and the evidence we can obtain.

The Kickstarter experiment from Chapter 1 provides a useful check on our intuition. Van de Rijt and his colleagues gave an initial donation to a randomly selected half of two hundred new, unfunded projects. Treated projects became more likely to receive subsequent support from other donors. That is evidence about the effect of an initial advantage on later support, rather than evidence that recipients had suddenly become more deserving.[102]

Now imagine using such a result to design a program supporting new projects. We would want to know which projects the program

brings forward, what they produce, and whether the additional donations finance useful work. A finding about subsequent support does not answer all those questions. It gives us a reason to investigate a mechanism that a simple comparison of successful and unsuccessful fundraisers could miss.

One defense of a first payment is that the recipient has already earned it through superior work. Another is that the payment could make a worthwhile undertaking feasible. The second can be persuasive even when the first is uncertain. A public subsidy would still need a worthwhile project and an advantage over other uses of the money.

The value of a proposed experiment is prospective. It should be judged using the information reasonably available when it was funded, including what failure could teach. Later commercial success alone cannot answer whether the original test was worth attempting.

Security can affect participation in experimentation. People unable to meet ordinary expenses after failure may decline a promising attempt. Whether a particular insurance arrangement makes useful experimentation more feasible depends on its financing and behavioral effects.

Claims about risk-bearing deserve specific evidence: what was financed, which losses were possible, which alternatives existed, and what behavior the terms induced. An investment can be genuinely risky at the beginning and later benefit from an arrangement that makes challengers hard to develop. The initial exposure does not establish how competitive every subsequent return remains.

A risk taken once cannot, by itself, justify permanent control over the opportunities future competitors need. A promise of payment can be kept without giving its beneficiary authority to close indispensable routes to independent evaluation. An incentive to attempt the first experiment should leave room for someone else to attempt the next.

Nakamura's account separates the purchases from the learning. The reactor existed before it could grow the material he needed. Its owner financed an uncertain undertaking; modifying it required work

that the original purchase did not supply. Nichia's statement emphasizes employer risk, while the lecture describes technical changes. Putting the two documents together makes each claim more concrete without making their eventual settlement a scientific measurement.

There is also a difference between bearing uncertainty and creating it for someone else. In an illustrative procurement system, a supplier might invest in reliability and accept the chance that buyers will prefer a rival. Or it might arrange to make rivals' performance impossible for buyers to assess. Both activities can involve expense. Expense by itself does not establish equal social usefulness.

A charge for destroying valuable comparison would address conduct and its external costs. Its justification would depend on the loss caused and the instrument's effects, rather than the successful supplier's wealth or a rival's complaint alone.

Decent treatment and an entitlement to a contract remain different matters. An opportunity can be humane to preserve yet too unsafe, uninformative, or costly to purchase. Respect for workers includes taking the needs of those relying on the service seriously.

The risk argument becomes more constructive once it stops serving as a badge attached only to winners. We can investigate arrangements that enable useful attempts, allocate losses tolerably, and preserve reasons to work well. We can abandon arrangements that perform badly. The choice is between particular institutions with particular consequences, rather than between reverence for a fortune and indifference to enterprise.

What a reward should do

What should a reward accomplish? It can encourage useful work, compensate investment, and make a valuable experiment possible. Security against ruin can help someone attempt that work in the first place. A system can pursue those aims while keeping practical routes open for the next capable alternative.

Lockdown can be admired without treating its position in one market as a measure of its makers' worth. A payment promised under an agreement can be respected without insisting that the agreement measured every contribution perfectly. A system that encourages more music need not produce an exact ranking of effort among its winners.

Income support can protect households without generating evidence about their work. An experiment can generate evidence without protecting their incomes. Identifying the intended purpose prevents one benefit from being treated as if it automatically supplies the other.

A popular musician or creator can earn a great deal without controlling anyone else's access to an audience. A platform that decides which creators receive meaningful exposure occupies a different position. The extreme tail of an income distribution does not identify that difference. The relevant question is who can prevent another person's work from receiving a credible hearing.

Changing promises after people have relied on them can damage future cooperation. Prospective rules let investors and workers know which obligations attach to the returns they seek. They also allow the costs and benefits of a change to be assessed before people commit their resources.

Credit needs similar care. Robert Merton's account of the Matthew effect showed how recognition can accumulate around already eminent scientists. Recognizing that mechanism does not require denying the accomplishments of eminent people. It invites us to examine how acknowledgment is assigned and how accumulated standing affects access to further work. His analysis is a predecessor to these questions, not a discovery made here.[103]

In the LED story, accurate acknowledgment can name particular advances and their authors. It need not wait until economists settle the division of every dollar earned from lighting. When more detailed evidence distinguishes contributions, use it. When it cannot sustain

an exact division, praise can remain specific rather than expanding into a claim that all rewards have been morally calibrated.

A similar restraint helps with criticism. A disappointing reward can be evidence of bad rules, weak bargaining power, an unfortunate sequence, poor performance, or several of these together. The possibility of a shadow future does not tell us which explanation fits a particular person. It asks us to investigate before translating low earnings into a verdict on character.

Would loosening the connection between income and merit weaken the motivation to do difficult things? The answer depends on how people respond to a particular change in rewards. A payment can be worth promising because it encourages useful work, even when the eventual income cannot measure exactly how much its recipient contributed.

The driverless-car prize promised payment for completing a demanding course. It did not require a belief that the winner authored the entire technology. The question was whether the incentive was worth its costs. A reward that also grants control over future testing adds another cost: the learning that control can prevent.

Another objection points toward equality. If the rankings could have differed, why not give everyone the same reward? Because the premise does not establish equal contributions, equal costs, equal needs, or identical responses to incentives. Lockdown's divergent rankings unsettle an easy inference from outcome to worth. They do not hand us a complete theory of distributive justice.

Substantial incentives, reliable contracts, and redistribution that gives people security and access to productive assets can be judged by what they make possible. The relevant questions are whether useful work continues, whether people can survive a failed attempt, and whether a capable outsider can acquire the means to begin. None requires an exact moral calibration of the existing income distribution.

Agreement about each person's exact contribution may remain beyond our evidence. Agreement about the next opportunity is still

possible. We can choose rules under which accumulated rewards help finance further attempts, while limiting their owners' ability to decide who is allowed to challenge them.

From the prize to the power cable

An AI supplier can do valuable engineering and commit substantial capital while its eventual rewards also depend on which systems get deployed first, which users encounter them, and which alternatives can afford a credible trial.

Before a computing facility is built, investors need an expectation of returns, workers need compensation, and customers need a service that performs. Society may benefit from capabilities no single payment fully captures. Those needs give reasons to finance construction without making the eventual fortune a complete measure of what caused it.

Keep the light bulb in view. Whatever we decide about the division of credit, semiconductor material still has to be produced and a useful device still has to work. Keep Stanley in view as well. Whatever we decide about the prize, a vehicle still has to complete the course. An economics attentive to possible histories must remain attentive to physical accomplishment.

The economics of an AI fortune begins among chips, cooling systems, and power cables. Long before anyone debates the winnings, someone must build a machine that works.

The intelligence factory

In the International Energy Agency's 2026 account of artificial intelligence, one of the consequential technologies is a connection to the electricity grid.

The report describes delays that have little to do with a chatbot's ability to write a sentence. Transformers can take years to arrive. In many jurisdictions, a grid connection can involve a wait of five to ten years. A company may have money, a plan, and customers eager to try its service while still lacking a route by which enough electricity can reach the machines.[104]

This is an unusual setting for a discussion of something described as intelligence. We expect intelligence to concern ideas. Here it also concerns a delivery schedule for electrical equipment. A better algorithm cannot, by itself, carry power down a transmission line that has not been built.

Nor can the phrase data center tell us what a facility will accomplish. A building can house many kinds of computing. The useful economic questions concern the equipment inside it, the work assigned to that equipment, and the arrangements through which its output reaches people. The same physical capacity could support different services and different opportunities to test them.

In its April report, the IEA estimated that all data centers together consumed about 485 terawatt-hours of electricity in 2025. A terawatt-hour is a billion kilowatt-hours, the unit familiar from electricity bills. Its central projection put consumption at roughly 950 terawatt-hours in 2030. That is an outlook, not an observation from the future, and both totals cover more than AI alone. The seemingly weightless answer occupies a material economy.

Someone must obtain the chips, install them, keep them working, and finance the interval before a useful AI product earns anything.

There are strong reasons for some of that work to occur in large organizations. There are also reasons to ask whether the resulting arrangements let other approaches demonstrate what they can do. Understanding both requires following the power cable into the building.

The machine needs a building

At Oak Ridge National Laboratory, preparations for the Frontier supercomputer involved a cooling plant, elaborate piping, and foundations capable of carrying cabinets weighing four tons apiece. The laboratory's published construction account records a failed joint in a large pipe during a pressure test. Water poured out; the team identified the failed weld and repaired it. Before the computer could do its work, the building had to survive the plumbing.[105]

Buying processors had not completed the job. The supporting systems had to pass their own tests. A joint in a pressurized pipe had to hold before the planned operation could proceed. Its usefulness was clear long before anyone could divide the value of the discoveries the machine might support.

Frontier is a scientific supercomputer with that proper name. A frontier AI model is a system near the leading edge of relevant capabilities at a particular time. Such a model is produced and operated on physical systems; it is not the building itself.

Begin with the chips. An accelerator is a processor designed to carry out suitable computations efficiently, often by performing many operations in parallel. Graphics processing units, or GPUs, became central tools for this work. A collection of them does not automatically behave like one enormous, dependable computer. They need data, memory, connections, and software that assigns and coordinates the work.

Imagine a team whose members must repeatedly exchange intermediate results. Adding people may help, but only if they can communicate quickly enough to keep the work moving. A training cluster presents an engineering version of that problem. A useful

processor waiting for data or another processor is an expensive object accomplishing less than its specifications suggest.

Cooling creates another coordination problem. The electricity used in computation becomes heat that must leave the equipment. At Oak Ridge, an account of the cooling system describes circulating water warm enough to avoid the refrigeration required by older arrangements. The design uses the fact that water can carry heat away even when it would feel warm to us.[106]

That small reversal of intuition matters economically. Cooling is a design problem with alternatives, rather than a fixed surcharge attached to every computation. Improvements can change how much useful work a facility gets from a given expenditure. They can require knowledge that is embodied in installation, controls, and operating practice as well as in the chip.

At LUMI, the European supercomputer in Kajaani, Finland, the operator describes another destination for the heat: the city's district-heating network. Its account says excess heat warms hundreds of households annually. The physical by-product of computing becomes an input into another service.[107]

The important feature is the connection between activities. A facility in one place may be near a usable heat network; another may have no comparable outlet. A household's demand for heating can vary with the season while the computer's workload follows a different schedule. Evaluating the arrangement requires following those actual uses and alternatives.

LUMI's location offers a practical route for moving waste heat to people who need it. Its wider economic benefit depends on the heating it replaces and the equipment and operating costs of the transfer.

That is relevant to a comparison of AI factories even when they offer similar services. Their private accounts can omit different benefits and costs falling outside their walls. A meter reading alone does not identify those effects. Neither does a claim that all

computation is virtual. The heat has to go somewhere, and in Kajaani some of it goes into homes.

Counting processors can miss those differences. Counting electricity can miss what the equipment accomplishes. Revenue records what a company receives. Each measure answers a question; none alone explains the whole productive system.

The Oak Ridge construction record makes the missing work visible. Someone had to discover the defect during testing and correct it. That is a contribution we can describe concretely. We do not need an exact division of the supercomputer's lifetime scientific value to acknowledge that preventing a failed pipe from interrupting operation is useful.

Engineering competence can determine whether an AI facility works. Its share in the winning company's eventual fortune is harder to recover. The operating machine demonstrates an accomplishment; explaining the income requires further evidence.

Keeping the chips working

Meta's technical report on the Llama 3 family provides a view inside large-scale training. For its largest language model, the researchers describe a fifty-four-day period containing 419 unexpected interruptions, in addition to planned ones. Their account also reports more than 90 percent effective training time. A long list of things went wrong, yet the operation continued doing useful work for most of the elapsed time.[108]

The two facts belong together. Reporting the interruptions alone would make the exercise sound like a collapse. Reporting the useful training time alone would hide the organizational and technical work needed to achieve it. Automation, diagnosis, and recovery turned a collection of fallible components into a functioning undertaking.

A single failed GPU could require the entire training job to restart. A processor that still worked but ran slowly could hold back thousands of others. Meta's report describes tools for finding those

stragglers. The coordination problem was physical: many expensive processors depended on the pace and reliability of the components around them. Their number alone did not establish the useful work they could perform.[108]

The Llama team reported significant manual intervention was needed only three times during that period; automation handled the other issues. An entrant renting well-run computing might obtain the benefit of those procedures without developing them afresh. The service would still need to permit the experiments its product requires.[108]

Scale can help in several different ways. The work of developing a useful system can be spread across many uses, provided the resulting service remains useful to those users. An operating team can support more than one project. A costly improvement in diagnosis may pay off repeatedly, rather than needing to be rediscovered for every customer.

A second possibility concerns idle capacity. Imagine a shared facility serving customers whose compatible workloads arrive at different times. Combining their demand could keep equipment occupied more consistently than separate facilities would. This is a conditional illustration of an advantage from pooling, not evidence that every large operator uses its machines efficiently or that every workload can be scheduled interchangeably.

Centralizing production need not make downstream services identical. Shared infrastructure can support different research methods and applications. The comparison question follows what can vary above that common layer.

Renting access to well-run computing can reduce the need to duplicate physical infrastructure before attempting a product. The provider's expertise can make more projects feasible. Access terms determine which of those projects can actually proceed.

Requiring each supplier to reproduce the entire cooling plant, network, maintenance system, and staff would buy organizational independence at a considerable resource cost. The extra comparison would have to improve decisions enough to justify that duplication.

An exclusive capacity arrangement can restrict downstream experiments while preserving useful operating expertise. The productive value of a facility and the informational consequences of its allocation rules therefore require separate assessment.

We would want to examine the reason for the reservation. A long commitment might finance construction that would otherwise be impossible. A priority rule might protect work that cannot easily tolerate interruption. Or a restriction might exclude an alternative despite feasible capacity being available. These possibilities require evidence about the particular arrangement; the existence of a large cluster does not choose among them.

Large shared computing is more than a hypothetical possibility. For 2026, the Department of Energy's INCITE program allocated access to seventy-five computational science projects at Argonne and Oak Ridge. Its published account describes scientific peer review alongside assessments of whether proposed work is technically ready for the machines. The organization operating the facility and the teams using it need not be identical.[109]

The allocations proposed distinct uses of the shared machines, including William Collins's large ensembles of weather extremes using machine learning and research on foundation models for materials. Different teams could apply for resources to pursue different questions; completed results would come later.

INCITE's policy document distinguishes technical readiness from scientific impact. Readiness alone is insufficient for an award. A proposal needing preparation can be considered for preliminary access through another program. A team can need experience with the resource before it can demonstrate readiness for a large allocation.

Shared access replaces one expensive hurdle with another, smaller one: a team can apply for time instead of financing an entire facility. Applications still take work, reviewers can make mistakes, and capacity is scarce. The practical question is which promising projects can now be attempted.

The Llama team learned about its own operation. A failing component, a useful recovery method, or a bottleneck in communication could generate actionable evidence inside the company. Those internal tests improved the operation while leaving other questions about competing products unresolved.

The more demanding question is which question the learning answers. An improved recovery procedure can establish how to keep this cluster operating. It does not, without further work, establish what a differently financed rival would have built or which product a buyer would prefer after a fair trial. A factory can become better understood while its place in the wider economy remains harder to explain.

Three different jobs

The equipment performs different jobs at different stages. In pre-training, a language model's numerical settings are adjusted using large collections of examples. One common task is predicting text from preceding text. The point is to acquire patterns that support later performance, rather than to write a useful answer for a particular customer at that moment.

Post-training then changes the system further. It can use demonstrations and preferences to improve instruction following, usefulness, or particular capabilities. Meta's Llama report distinguishes these stages explicitly. A system that has learned patterns in text is not automatically the assistant its developer wants to offer.[110]

Inference is the use of a trained model to produce outputs. A person asks a question; computation produces a response. The work may include additional steps, such as retrieving information or using tools. That use has a cost each time it occurs. Training a model does not eliminate the equipment, electricity, and operations required to serve it.

For an ordinary analogy, designing a machine differs from operating it to make the next item. The analogy has limits because a

trained model is a collection of numerical settings that can be copied. But it helps prevent a common accounting confusion: the expense of developing a capability and the expense of using it are different parts of the undertaking.

DeepSeek's report on its V3 model separates pre-training, extending the context the model can handle, and post-training in its cost table. That division tells us that even a reported training figure can combine several activities. It does not represent a timeless price for producing intelligence in every setting.[111]

Entry can mean training a model, adapting one, or building a service using an existing model. These undertakings require different resources and permissions. A claim about the cost of one cannot stand for the cost of all three.

A supplier that cannot finance a large general training run could still develop a valuable application. A supplier with a powerful model could still build a poor application. The difference might involve reliability, the customer's information, the surrounding software, or the way a worker uses the output. Counting large training runs does not count every opportunity for useful innovation.

Access to inference does not grant control of model development. A product requiring changes to the underlying model may need other permissions, technical capabilities, or a different design. The ability to submit a prompt does not supply every input needed to compete.

The word frontier also needs a task attached to it. A system can lead on one evaluation and be unsuitable for another use. Its position can change as methods and competitors improve. Treating frontier as a permanent title bestowed on a few companies obscures the very discovery process we want to understand.

In its 2026 report, Stanford's AI Index attributes more than 90 percent of the notable models in its 2025 dataset to industry. The dataset is curated for notability; it is not a census of every model or application. It gives evidence of industry's prominent role in this part of development, while leaving open the reasons for that pattern and the consequences of particular access arrangements.[112]

An assessment of entry must identify the layer being entered and the experiments possible there. A training strategy can be blocked while another useful application remains feasible, or application-level access can coexist with dependence on a single development frontier.

Two budgets

In 2022, Jordan Hoffmann and colleagues at DeepMind investigated how to spend a fixed amount of training computation. Their Chinchilla model used fewer parameters and more training data than the larger Gopher model, with the same total training compute. The researchers reported stronger performance across a range of evaluations. The study challenged the idea that making the model larger was the best use of the available resource.[113]

The concrete choice makes the word budget useful. A research team has access to computing for a period. It can spend that resource on different combinations of model size, training duration, and data. Experimenting with those choices can reveal a better recipe. The factory supplies a constraint; it does not dictate a single design.

The Chinchilla work also involved multiple smaller training experiments. This is a case of comparison produced inside a large organization, directed at a particular technical problem. It is evidence against an argument that scale must always suppress experimentation. Resources can make demanding comparisons possible.

Now distinguish the computing budget from the comparison budget. The first concerns resources used to perform computation. The second, in my allocation model, adds the chances that successive opportunities go to someone outside the current favorite. They are different objects. A very large computing budget could support useful rival tests, or mainly serve a product already selected for deployment.

Spending on competing training recipes produces different evidence from spending on repeated deployment of a fixed recipe. Both can be productive. Only observations relevant to a defined

comparison belong in an account of what the spending establishes about that comparison.

Neither description can be inferred from the electricity meter. The meter tells us about energy use. To understand the evidence, we need the designs, measurements, and decisions attached to it. A low electricity bill can accompany an informative test; an enormous bill can accompany repetition. The economic value of either depends on what is being accomplished.

Chinchilla also changed how its settings were adjusted, how text was divided into units, and the proportions drawn from different parts of the training dataset. The paper records those differences. Its result therefore compares two training packages; it does not isolate model size as the sole cause. The productive lesson is the researchers' ability to investigate a prevailing recipe. Future data, hardware, and uses could make another recipe preferable.[113]

Tests conducted before an award can reveal differences the award history alone leaves unresolved. Their measurements add to the evidence available about competing services.

Where deployment mainly reproduces the record of an established service, the questions concern how alternatives obtain use, which outcomes are checked, and whether tasks are comparable. A market share cannot answer those questions.

A million generated answers can reveal performance on encountered tasks without supplying the missing comparison. Applying Chapter 3's result requires specifying the observation process and counting every additional informative outcome and independent test.

Putting the two budgets beside one another helps us ask a constructive question. Can we use some of the computing resource to produce a more informative comparison at tolerable cost? Chinchilla shows the importance of technical experiments. A buyer's pilot might address a different uncertainty about practical performance. Neither requires treating useful scale as a mistake.

The five-million-dollar number

DeepSeek's V3 report contains a number likely to attract anyone interested in the economics of the intelligence factory: approximately 5.6 million dollars. It is the report's calculated cost for specified training work, using an assumed rental price of two dollars per H800 GPU hour. The authors explicitly exclude earlier research and experiments on architectures, algorithms, and data from that total.[111]

The qualification is part of the number's meaning. It does not turn an achievement into a failure. Producing an effective training run with fewer resources can be important. But a reader asking what it costs to create a capable AI organization needs a different account from a reader asking about the computing attributed to one completed training pipeline.

The distinction resembles the difference between paying for a successful reactor run and financing the attempts needed to develop the reactor. The earlier LED chapter showed how repeated modifications could make later production possible. An efficient final operation can rest on costly discovery. Measuring the operation accurately does not make those earlier costs vanish.

Published methods can reduce the cost of reproducing a successful approach. That is one benefit of knowledge transmission, and one reason reproduction cost differs from discovery cost.

Incomplete methods, unavailable data, and different operating conditions can still require fresh experiments. An expenditure figure in a research report does not, by itself, establish the cost at which another organization can reproduce the capability.

A company may spend heavily on research that never improves its final product. Later commercial success cannot show which of those expenses were necessary. Comparing development routes can help distinguish indispensable work from costly attempts that added little.

Kerr, Nanda, and Rhodes-Kropf's account of entrepreneurship supplies the established economic starting point. Early tests can inform later commitments, and the option to stop can matter.

Bringing that account into AI asks us to examine the sequence of experiments and financing decisions. It does not give the final winner exclusive ownership of the concept of experimentation.[114]

DeepSeek's table covers a specified part of the work. Comparing its figure with another budget requires matching the activities included and the way computing time is valued. The cost of one completed training pipeline and the cost of developing the organization that produced it answer different questions.

What can be copied

There is an asymmetry inside the factory. A trained model's numerical settings can be distributed without building an additional power station for each copy. Running the copied model still requires suitable equipment. Reproducing all the research that led to it is a further undertaking. Sharing the artifact, sharing access to computing, and sharing the means of discovery are different actions.

Kenneth Arrow's 1962 analysis of invention explains an older version of this difficulty. Knowledge can be costly and uncertain to produce while being relatively cheap to use or transmit once available. The social benefits of using it can exceed the rewards the producer can capture. Arrangements that create incentives to invent can also restrict its subsequent use.[115]

This is an established economic problem, not a new property discovered when AI appeared. What changes are the particular artifacts, costs, and control points. A model's settings might be available while its training data are not. A technical paper might describe a method while the equipment needed for a competitive implementation remains expensive. Different forms of access answer different constraints.

A common released model can support different services, customer problems, and error-reduction methods where its terms permit. Shared inputs can lower the cost of useful experiments without making their results redundant.

Suppose instead that our question concerns the training process itself: would another dataset or architecture have produced a better model? Two applications using unchanged copies of one model do not perform that comparison merely by having different company names. They may answer important application questions while leaving the training question unresolved.

That is why independence needs an object. Independence of ownership, computing provider, model ancestry, task assignment, or evaluation can each matter differently. Demanding independence in every respect could be prohibitively expensive. Assuming it exists because a storefront has a new name could be equally misleading.

Arrow's appropriation problem applies here. If immediate unrestricted access prevents recovery of the costs of valuable research, some research may never be undertaken. A route for later experiments is valuable only if the knowledge and equipment those experiments need also get produced.

The incentive argument still requires a comparison of feasible arrangements. Terms permitting returns and downstream experiments can differ from terms closing a customer route without enabling additional research. Both can be described as protecting investment while having different consequences.

Public support, contractual access, released artifacts, and competing private facilities all require resources. Each can make some experiments feasible and others expensive. The useful choice depends on the research or service being developed, including the cost of creating its inputs.

The terms attached to a model, a dataset, or a computing service help determine which later experiments are feasible. A release can make a new application possible while leaving another training strategy beyond reach. The factory creates possibilities. Its institutional arrangements determine who can act on them.

From computing access to a useful comparison

Suppose a team has a different model design and enough money for a useful small trial. It still needs suitable computing, permission to connect its system to the intended service, and access to tasks whose results a customer will accept. If the only provider of an indispensable route refuses that use, the team can fail before its model produces a result worth comparing. The missing company is then absent from the market-share table as well as the performance table.

That absence can be structural. The resources needed to get started may depend on customers who will arrive only after the entrant reaches a scale it cannot finance. A new entrant could be viable once established while lacking a feasible path to establishment under the current rules. Interoperability, shared infrastructure, or different financing can change that path. The question is which change enables a credible attempt at a worthwhile cost.

AI scaling laws and this institutional problem concern different relationships. Evidence that a larger training run improves a measured capability does not establish that independent entry is impossible. The stronger concern arises when scale advantages combine with control over the resources, distribution, and evaluation needed to challenge them. An incumbent can then reinvest returns while rivals cannot reach the stage at which their work would generate useful evidence.

The duration of that advantage matters. A return protected by a development bottleneck can continue after an initial research investment has paid off. Its persistence may reflect continued invention, control of the starting conditions for competitors, or both. To understand the income, follow what the next investment enables and whose experiment the access rules prevent.

Computing access is one stage in producing evidence. A system must also reach relevant tasks, return interpretable outcomes, and permit those outcomes to affect a decision. These requirements connect infrastructure to discovery.

Task selection, comparable difficulty, valid outcome measures, privacy, and operational risk determine whether a service comparison is useful. The evaluation design has to justify its cost.

Distribution can close the route even when computing is available. A default that receives nearly every request can accumulate a record while alternatives obtain little ordinary-use evidence. Availability of an input does not guarantee access to an informative use.

Bell's patents and search distribution expose the same gap. A firm can possess a useful input while lacking a practical route to a customer. Computing for sale does little for an entrant barred from the tasks through which its service could be evaluated.

Performance can nevertheless justify a strong default. An investigation must include measured quality, actual switching, available comparisons, and the costs of alternatives. Access-dependent evidence is something to interpret carefully, rather than to dismiss.

Limited, well-measured comparisons can sometimes retain the convenience of a default while supplying useful evidence. Their value depends on the uncertainty resolved, implementation costs, and the decisions they can change.

The comparison budget prescribes no universal spending ratio. Small persistent chances can accumulate indefinitely, and large nominal opportunities can fail to identify a relevant effect. Evaluation concerns the process and its information.

Discovery also needs a route to action. A demonstration has less practical value if its result cannot influence subsequent adoption. Switching terms and interoperability help determine whether learning can change an economic arrangement.

An alternative training strategy needs capacity and permission to use it. An alternative service needs tasks and credible outcome checks. A buyer needs evidence relevant to the purchase. Several firms can depend on the same provider at one of those stages while remaining independent at another.

Nor does having many firms guarantee that all three are open. Several services could depend on the same model and the same route to users. They could still innovate meaningfully in their own domains. The useful question is what uncertainty their differences can resolve, and what important uncertainty remains outside their experiments.

Taking counterfactual economies seriously requires specifying feasible alternatives, what their tests could establish, and the full cost of producing that evidence. The resulting comparison can justify preserving a route without treating every proposed competitor as entitled to infrastructure.

Who pays for the connection

The grid now returns to the story. On October 14, 2026, the IEA published another report on integrating data centers into electricity systems. It described connection queues containing duplicate or speculative proposals. A large queue can therefore present planners with both an urgent problem and an uncertain measure of how much demand will materialize.[116]

Imagine, solely as an illustration, one developer exploring sites on several grids while intending to build only one facility. Adding every request as if it represented a separate, certain future load would exaggerate the developer's eventual use. Ignoring the requests entirely could leave the winning location unprepared. The planner needs evidence about commitments and timing, rather than faith in a single headline total.

Here is another economic record that does not explain itself. A proposed project, financed construction, installed computing equipment, and sustained customer demand are different stages. Moving from one to the next can change the risks facing everyone involved. Treating a proposal as completed capacity confuses a possible future with an observed one.

The IEA's October report also describes the importance of allocating connection costs and risks. If new infrastructure is built for a large customer that consumes less than expected or leaves early,

somebody must absorb the cost. Contractual commitments and tariffs can determine how much falls on other users of the electricity system. These are institutional choices alongside the engineering.

The IEA's comparison gives AEP Ohio's large-data-center arrangements a concrete form: a ramp-up period, minimum capacity payments, a long commitment, and exit fees. The customer reserving infrastructure cannot assume its obligation ends whenever its appetite for computing falls. That shifts some risk of unused capacity toward the customer requesting it. The terms are a way to allocate risk, not proof that the resulting investment is optimal or that households bear no remaining costs.[116]

Operating commitments matter as well as construction commitments. In another illustrative arrangement, a customer could accept occasional interruptions in exchange for earlier or cheaper access. A computing job able to pause safely might fit that bargain; a service promising immediate responses might find it costly. The relevant unit is the particular workload and obligation, rather than an undifferentiated category called AI demand.

That choice connects the contract back to the engineers. Recovery procedures, stored progress, and scheduling can change which interruptions are tolerable. A physical improvement might make a more flexible arrangement feasible; an inflexible promise to customers might rule it out. The institution and the machine have to be understood together.

We need evidence to distinguish those cases. A large electricity demand is not inherently a social loss. Nor does calling the demand innovative establish that every associated cost ought to be paid by others. The public question concerns useful output, displaced uses, environmental effects, and the allocation of risk as well as the private business plan.

A pilot using scarce capacity competes with other uses of the machines. What it might teach has to be worth the electricity, equipment, labor, and service forgone. A smaller test or a suitable existing record may answer the question at lower cost.

Long commitments can finance investment while restricting access to scarce capacity. A different arrangement could retain the financing benefit and preserve useful independent experiments. Whether it improves on the commitment depends on the experiments' value and the costs of making room for them.

This is the tension that a price or concentration statistic alone cannot resolve. A low price today can coexist with a poor route to testing tomorrow's alternative. A large producer can supply the very capacity that makes new experiments affordable. The economic task is to trace the mechanisms rather than assign one moral meaning to size.

Leaving the factory

A working facility has overcome physical obstacles: pipes that could fail, interrupted training runs, equipment shortages, and difficult design choices. Those achievements make the next request possible.

But the completed training run is a beginning for the economic story. A model can be served repeatedly, incorporated into different products, evaluated on changing tasks, and routed toward some users rather than others. Those arrangements affect which observations accumulate and which alternatives develop a usable record.

Follow the next request. Someone chooses a system, the machines produce an answer, and a person or another program uses it. Who checks the answer? Can the buyer try another system on a comparable task? Which failures remain invisible if every request follows the same route?

One answer becomes a million, then a billion. Each can be useful. How many of them tell us whether a different system would work better?

A billion answers

Inference turns a trained model into a stream of outputs. Each interaction can also create evidence, but what it reveals depends on which requests arrive, how the model is selected, and whether the answer is checked.

The records can include the request, selected model, answer, time taken, and a user's acceptance. A check against an independently known outcome adds another observation. These records answer different questions because they arise from different measurement choices.

The trained model need not change its numerical settings during this exchange. A later development process might use collected examples to update it. The service could also retrieve a newer document or apply a different instruction without retraining the underlying model. Improvement can occur in the model, the surrounding software, or the people evaluating the result.

Now repeat the exchange a billion times across many workplaces and tasks. The records can become immensely useful. They can expose recurring mistakes, reveal delays, and help developers improve a service. But the number of completed requests does not, by itself, tell us which questions the resulting evidence can answer.

One question concerns this particular order: did the service find the correct deadline? Another concerns the alternative model: would it have done better on comparable work? A third concerns the development process: which investment produced the capability? A fourth concerns an economic arrangement: what would have happened if another supplier had received the early opportunities and resources to develop?

We can learn a great deal about the first question without settling all the others. We can also design the service to collect evidence about

more than one. The choice matters before the billionth answer arrives. It matters when we decide what happens to the first request.

Two answers to one question

Chatbot Arena offers a concrete departure from an ordinary exchange. In the platform described by Wei-Lin Chiang and colleagues in their 2024 paper, a user enters a prompt and receives responses from two anonymously selected models. The user compares them and votes; the identities are revealed afterward. The interaction creates a paired preference observation, rather than simply another conversation with a named favorite.[117]

The two columns on the screen do useful work. They bring alternatives into the same encounter, expose them to the same prompt, and delay the information that a famous brand might provide. We can argue about the design while recognizing what it adds. A record of which service people already use supplies a different kind of evidence from a vote after they have seen two responses.

Preference and correctness remain different. A polished wrong answer can be more convincing than a correct but awkwardly expressed one. Assessing accuracy may require evidence beyond the preference judgment.

Arena's paper itself distinguishes benchmarks based on human preference from those based on a known correct answer. That distinction helps us resist treating one ranking as a certificate for every use. A buyer interested in pleasant conversation and a buyer extracting deadlines may need different measurements. Neither has to dismiss the other's question to insist on measuring its own.

Prompts and users arriving at a public platform can differ from those in a specialized application. Long documents, damaged scans, or technical vocabulary can change performance. A general benchmark is evidence to assess for transportability, rather than an entire decision basis.

Unequal task assignments complicate performance comparisons. A service handling clear documents can have a higher acceptance rate than one handling damaged scans even if its underlying capability is no better. Evaluation must examine the assignment process as well as the score.

A paired test can help with that particular problem. We could send suitable examples from each category to both services and assess their outputs using a consistent procedure. The cost includes additional computation and evaluation. The benefit is evidence directed toward a decision that a single unbalanced stream would have made harder.

The Arena design also reminds us that comparison has to be produced. Someone builds the interface, selects model pairs, collects responses, and maintains the analysis. Those activities are part of the institution. The platform's methodological choices affect what a vote means, which differences it can reveal, and how confidently a ranking can be interpreted.

Suppose a purchasing department then uses a ranking to select its default supplier. The ranking can influence subsequent exposure. That possibility does not invalidate the original tests. It creates a new question about how the tests and the resulting deployment interact. A good measured result can become an input into a process that later generates a different kind of evidence.

Service and evaluation can be designed together. Delivering an accurate answer at an acceptable cost remains the productive aim; a suitable comparison can reveal when another system better serves it.

When the judge likes the packaging

Yann Dubois and colleagues tested how an automatic judge responds to presentation in their length-controlled AlpacaEval study. They prompted models to give concise or verbose answers. For one GPT-4 version, the unadjusted win rate ranged from about twenty-three to sixty-four percent as the instruction changed. The underlying model had not received a new training run; the prompt and resulting answers differed. Their statistical length adjustment reduced the swing. The experiment exposes sensitivity to presentation, while leaving us to investigate which changes also improve the answer.[118]

Each response faced a reference answer to the same instruction. The judge's preference supplied the score. It was a measure of which answer the evaluator favored, rather than an independent check against a separately established correct solution.[118]

Arena records users' comparisons of two answers. AlpacaEval uses the judgment of a particular automatic evaluator. An adjustment to that evaluator's scores tells us about its measurements; human voting needs evidence of its own.

Think of the deadline in the purchase order. One service returns the date and identifies the relevant passage. Another returns the same date inside a long explanation of the document. A buyer might prefer either presentation for a legitimate reason. An evaluation intended to measure the extraction task should nevertheless understand how much of its ranking depends on the surrounding prose.

Now imagine that the longer response contains an error elsewhere. The extra detail has changed more than the length of the answer; it has created another opportunity to mislead the worker. Alternatively, the longer explanation might expose an ambiguity that the short answer concealed. Length by itself cannot tell us which happened. The measurement must remain connected to the purpose.

The length-controlled study also makes a counterfactual question explicit: how would the estimated preference compare if the responses had the same length? Its statistical adjustment supplies an

estimate under a model. The authors are not claiming to have observed every possible rewritten answer or eliminated every source of evaluation bias. A defined counterfactual can be useful without becoming an omniscient reconstruction.

For a developer, the practical stakes concern which improvements an evaluation rewards. If a score rises after a formatting change, the developer should investigate whether users actually complete their work better. If better reasoning raises the score, that is worth distinguishing too. The same final ranking can emerge through changes with different consequences for the customer.

Who defines the test helps determine who receives the next opportunity. An accessible benchmark can support discovery, but its convenience is insufficient: its design must expose differences relevant to the intended use.

Choosing before paying

The next design decision concerns which models to call. Sending every request to the most expensive available system may waste resources on tasks a cheaper system handles adequately. Calling every model for every request can produce comparisons, but also an unnecessary bill. The problem invites a selector: software that routes the request according to the work it appears to require.

Isaac Ong and colleagues' RouteLLM research investigates such a selector. Their framework learns from preference data and routes between a stronger, more expensive model and a cheaper alternative. They evaluate the resulting tradeoffs on specified benchmarks and compare their methods with random routing. Their reported improvements concern those models, data, and evaluations, rather than a guarantee for every future workplace.[119]

The purchase order makes the motivation easy to understand. A short document with a clearly labeled date may require less capability than an ambiguous revision spread across several pages. A well-designed router could reserve the more costly service for the

cases where it adds enough value. That is a productive improvement even if the user never sees the selection happening.

A router needs evidence about which model performs well on each kind of request. The strongest overall model may offer little advantage on a simple task. Paying for it on every request can waste resources a task-specific choice would save.

The RouteLLM paper also reports limits to transferring a router across tasks. Data resembling the eventual evaluation can improve performance. Correspondence among training examples, evaluation cases, and actual use therefore matters to the interpretation of a score.

Lingjiao Chen, Matei Zaharia, and James Zou explore a related route in FrugalGPT. Their system can call a model, assess the response, and decide whether to stop or call another. The answer already obtained becomes an input into the next decision. The study compares cost and performance on defined tasks, including classification and reading comprehension.[120]

In the workplace illustration, the system might first ask a cheaper model to extract the deadline. If its assessment of that response is unfavorable, it could ask another model. This differs from selecting once before seeing an answer. It also differs from obtaining an independent human check of correctness. The assessment mechanism itself needs evaluation.

FrugalGPT's paper includes cases where additional calls are made even though the models' answers agree. Its discussion identifies the selector's uncertainty as a source of avoidable expense. Delegating a decision to a model leaves another judgment to evaluate: when does that model decide to stop?

An economical cascade might lower the cost of trying alternative services. If the interface reports only one answer and one brand, the buyer may be unable to inspect those comparisons. Several logos can conceal dependence on one underlying service; a simple interface can conceal useful tests of several.

Routing and cascades can make it cheaper to compare services as well as to use them. A buyer who can afford only one expensive

answer may be able to check several cheaper ones. Whether that happens depends on what the selection system tests and records.

One distinction is essential before we try to measure a comparison budget. A router's estimate that one model will produce the preferred answer differs from the chance that the request will actually be assigned to that model. RouteLLM uses a predicted preference and a threshold to make its selection. The prediction and the assignment rule do different jobs.

A predicted seventy percent chance of producing the preferred answer is not a seventy percent allocation probability. A selector could send every request to the predicted winner. Adding the remaining thirty percent figures would then mismeasure the process that actually assigns work.

A different rule could deliberately randomize some assignments. Its allocation probabilities would describe that design, while its predicted performance scores would describe the selector's expectations. Both can be recorded. Their relationship matters to the analysis, but they are not interchangeable numbers simply because both can be expressed as percentages.

Checking answers reveals performance on the tasks a system encounters; independent evaluation can supply information about alternatives. Those measurements belong alongside the assignment record when we assess what can be learned.

The work of checking also has a production process. Someone may need to open an amendment, resolve an ambiguous field, or determine which document version governs the task. AI could assist with that work, but an automated score remains something to validate against the relevant standard. A service can lower the cost of producing an answer while leaving costly uncertainty about whether the answer is useful.

The relevant chain runs from the observations training a selector through assignment rules, checked responses, and decisions using those checks. Following it distinguishes efficient production from useful comparison and unsupported precision.

The objectives still need to be named. A router optimized for the best current cost/performance tradeoff need not be optimized to discover a new supplier's potential. It may already have enough evidence for its operational decision, or it may need occasional new observations. We cannot determine which merely by noting that the routing is automated.

A record worth reusing

Long before today's language models, a news page posed a related problem. The Yahoo Today Module could highlight an article for a visitor and observe whether the visitor clicked. It could not observe that same visit's response to every article left unshown. Lihong Li, Wei Chu, John Langford, and Xuanhui Wang investigated how previously collected data could help evaluate another recommendation rule.[121]

Their work used records gathered through randomized article selection. Under their stated conditions, an evaluator can retain suitable events where a proposed rule would have selected the article that was actually shown. The retained record then includes a real response to that choice. The researchers compared offline evaluations with online results in the Yahoo application.

A historical record can contain deliberately created variation. It can then support questions that simply repeating the current favorite would leave unanswered. Collecting the right record expands what later analysis can discover.

Randomized request logs with checked outcomes can support evaluation of another selection rule under the relevant coverage and identification assumptions. A literal parallel economy is not required for every counterfactual question.

That last qualification matters. If a service was never available for a particular kind of order, the record may contain no relevant responses from it. Knowing that the probability was zero does not supply the missing performance. An estimator needs suitable observations or

additional defensible assumptions, rather than an increasingly elaborate way of describing their absence.

The Yahoo authors call their method replay. Their use of the word concerns evaluating a recommendation policy with suitable logged events. My shadow futures concern alternative histories generated under specified economic conditions and rules. These are distinct uses. A procedure that reuses article records is not a time machine for recreating the entire industry with a different first winner.

Evaluation of current systems, variation across alternative development histories, and the effects of changing a development policy require different comparisons. Randomized request logs can help with the first without establishing how capabilities would have evolved after different early opportunities.

A record can justify its collection by helping a buyer choose a better service. That decision can be worthwhile even when the evidence leaves each supplier's past contribution unresolved.

This gives us a practical reason to preserve details that a success count omits. Which alternatives were available? Why was this one selected? What was measured afterward? A usable assignment record and meaningful outcome can make a small pilot worth more, for a specific decision, than a much larger archive lacking those features.

The pilot also has costs. Someone receives the less suitable answer when the comparison goes badly. Someone checks the outcome. Some data may be inappropriate to retain or share. Useful evaluation requires an observation design that fits the task and evidence worth those costs.

The same ancestor

Our hypothetical buyer opens three services and receives three answers to the order question. Three storefronts are visible. What has been compared?

Perhaps each service has independently trained a different system. Perhaps each has adapted a shared model. Perhaps all three pass the

request to the same supplier, with different surrounding instructions. The answers may still differ. The number of interfaces does not identify which part of the production process varied.

Meta's Llama 3 report supplies a concrete example of a relationship between models. It describes using its flagship model to improve smaller models during post-training. Different members of a model family can therefore be connected through the development process. Treating their outputs as evidence from wholly unrelated research histories would miss that connection.[122]

A smaller model can behave differently, require fewer resources, and fit a particular application better. Testing it against its larger relative can guide the purchase. Both can still inherit a training assumption the comparison never varies.

Imagine that our three services extract the same wrong date because the underlying document representation loses the same footnote. Asking all three may produce reassuring agreement while leaving that error untouched. Now imagine that one service uses a better document parser and finds the footnote. A common language model has not prevented a meaningful improvement elsewhere in the service.

These are illustrative cases, but they identify an empirical task. Follow the chain far enough to understand which components are common, which vary, and where the outcome is measured. Independence is valuable relative to a question. It should not be a ceremonial word attached to a company name.

A thousand carefully checked orders can expose failure patterns that one order could not reveal. New requests and new checks can add information about a system even when the same model handles them all.

But generating a thousand responses without a new check does not create a thousand independent confirmations of the underlying claim. If the uncertainty concerns the actual delivery date, the order and its amendments remain relevant evidence. A longer conversation among

programs might help interpret them. It cannot make an unread amendment irrelevant by agreeing without it.

Yaneer Bar-Yam's "From Big Data to Important Information" distinguishes detailed records from information useful for understanding an intervention.[123] For the delivery order, the decisive detail may be an amendment that thousands of generated answers have never checked.

The workplace can act on that warning. It can measure whether the extracted date is correct, whether the answer helped a worker complete the task, and whether a different service changes the result. Each measurement has a purpose. We gain clarity by naming those purposes before admiring the size of the resulting database.

Predicting the next choice

Forecasts of the next allocation can converge while explanations of productive contribution remain different. Chapter 3 states the conditions under which that distinction persists.

David Blackwell and Lester Dubins showed in 1962 how conditional forecasts can merge under an appropriate relationship between probability laws. Hao Zhang later established an example in spatial statistics where some underlying parameters cannot be consistently recovered even though important prediction can remain effective.[124][125]

Forecasting which supplier wins an allocation is a different task from checking whether an AI answer is true. The answer needs evidence of its own.

Acceptance, correctness, and causal contribution are separate targets. Predicting that a user accepts an answer does not establish that it is accurate, or identify which investment caused the capability. A common interface does not make the underlying inference identical.

For the buyer, the next action could be straightforward. If errors matter, check a suitable sample against the original orders. If supplier choice matters, preserve a meaningful comparison. If a historical

attribution claim matters, identify the intervention, horizon, and evidence that could support it. The inquiry becomes more useful as it becomes more specific.

Sixteen agents and one compiler

A swarm sounds like a crowd. On a computer, it can mean several copies of a model pursuing assigned work, using tools, keeping records, and returning results. The economic question begins with what those activities accomplish. Sixteen running programs can perform sixteen useful investigations, or spend the afternoon rediscovering one mistake.

In February 2026, Nicholas Carlini at Anthropic reported setting sixteen Claude instances to build a C compiler, which translates code into instructions a computer can run. Compiling the Linux operating system, they stalled on the same bug and interfered with one another's fixes. He redesigned the tests: GCC, an established compiler, handled most files while the new compiler handled varying subsets. Failures became easier to isolate. The reported achievement still had limits, including inefficient generated code and dependence on other tools. This was an attributed capability experiment, not a measured replacement for a human workforce.[126]

The revised tests helped separate failures. The checking arrangement belonged to the productive machinery as surely as the model did. A candidate program needed reliable tests that could expose its mistakes.

There are different ways to organize that machinery. Anthropic's June 2025 research system used a lead agent to distribute searches among others. Carlini's compiler account explicitly described a setup without a lead agent assigning tasks. In either arrangement, ask who directs the work and supplies the feedback.[126]

Agent systems can organize candidate generation, execution, and checking, as the compiler account demonstrates. Their productive value depends on the full arrangement: human-defined objectives,

tools, tests, and judgments about errors, as well as the cost of model calls.

Retained code, test cases, and usable project records can support later comparison when a model changes. The rights to those assets therefore affect the cost of developing and evaluating another approach.

Failed attempts can matter as much as final outputs. Preserving their records may prevent repeated mistakes, while requiring storage, documentation, and protection of confidential material.

Control of tools and records can make switching expensive even when software generation becomes cheap. This is an institutional question about portability and ownership, distinct from the capability demonstrated by an agent experiment.

Carlini's account describes checks within one firm using related models. A legally independent project could still use a defective test. Validity depends on feedback that can expose an error, wherever the test is organized.

A thousand agents can reproduce one mistaken interpretation; a single check against a relevant document can reveal it. Abundant software labor can multiply attempts faster than it produces trustworthy judgments of their results.

A town throws a party

In a 2023 study, Joon Sung Park and colleagues populated Smallville with twenty-five software agents. One received the intention to organize a Valentine's Day party. The authors report invitations spreading and five invited agents arriving at the cafe. Another resident mistakenly identified her neighbor, Adam Smith, as the author of The Wealth of Nations. Knowledge inherited from a language model had intruded into the neighborhood.[127]

These characters supply a tempting contrast with a compiler. A simulated town can be rerun, with rules changed and histories recorded. But what would establish that its response to a tax change

predicts the response of people? Believable conversation and a program's successful execution do not supply that validation. The study investigated believability and agent architecture, not the accuracy of economic policy forecasts.

Simulated demand can reveal implications of assumptions and help design an empirical test. Its preferences may also reflect the instructions and training inherited from a common model. Multiplying simulated participants reduces some sampling noise without correcting a mistaken demand mechanism.

A shadow replay keeps underlying parameters, starting conditions, and response rules fixed while the random history changes. Altering procurement or training policy is a separate intervention. Either simulation needs evidence connecting its rules to the actual economy. A generated conversation can reveal how programmed agents interact. An executed program with independent checks can also expose a proposed design's errors.

When a generated answer can be tested

There is a more encouraging route from generation to evidence when a proposed answer can be checked directly. Google DeepMind's 2025 AlphaEvolve report describes a system that proposes computer programs, runs evaluations, and uses the results to guide further proposals. The language model helps generate candidates; executable tests supply feedback about what those candidates do.[128]

One application concerned scheduling jobs in Google's data centers. A machine can have spare computing capacity but be unable to take another job because another resource, such as memory, is already exhausted. The report describes searching for a better rule for assigning work to available machines. This is the factory's coordination problem turned into an object of automated investigation.

The researchers used a simulator based on historical workloads, tested on unseen recent data, and then measured operation after deployment. They reported recovering an average of 0.7 percent of

fleet-wide computing resources that would otherwise have been stranded. That measures available computing capacity, rather than an equivalent reduction in electricity use.

The sequence matters as much as the headline. A proposal entered a test with a defined outcome. A promising simulated result faced another check. Deployment supplied further observations. The program was not considered better simply because another language model produced a fluent explanation of why it ought to work.

Smallville's party unfolds under programmed rules. A scheduling simulation can also misrepresent a real system. In the AlphaEvolve account, measurements from subsequent operation provided a way to check the simulated gains against actual performance.

For software generation, useful evidence comes from executed programs and checked outcomes. Candidate code can help organize a search, but its correctness needs testing beyond the model's ability to produce it.

The test set still needs attention. A program might exploit a quirk of the examples, perform well on clear documents, and fail on revisions. A good development process would investigate such limits. Automating the proposals makes the quality of evaluation more consequential because many candidates can now be optimized against whatever test we provide.

This possibility strengthens the constructive part of the comparison argument. An AI system can help us propose and assess alternatives at a cost that changes which experiments are worth attempting. It can do so inside a large firm or in a smaller undertaking. The organizational label does not tell us whether the tests are informative.

Evidence that a scheduling rule uses machines better can justify adopting that rule. The experiment compares ways to schedule the available machines. It leaves the contributions of earlier investors and the effects of alternative ownership arrangements to other evidence.

Generated proposals can help us test ambitious ideas when we can define an outcome that establishes improvement. The question is how

to connect each proposal to evidence that can distinguish success from a plausible story. That connection lets us use AI's capabilities while examining whether the resulting system accomplishes its purpose.

What the billion requests leave behind

Record design determines what inference can follow. A final answer and acceptance signal supply different evidence from a record also containing task characteristics, assignment rules, checked outcomes, and appropriate alternative-system tests. Equal request volumes need not produce equally informative records.

Acceptance can reflect verification, difficulty checking, or time pressure. Comparing outcomes with independently checked evidence helps distinguish those explanations and identify specific failures. The feedback signal must be interpreted before it is treated as correctness.

Production can be concentrated while a limited experiment preserves a relevant comparison. Assignments justified as learning must expose informative differences and be capable of affecting a decision, rather than merely adding a second name to a record.

The favorite can vary by task. Aggregate request totals can conceal a lack of within-task comparison, while a dominant service can host useful internal experiments. Logged assignment rules and checked outcomes matter together.

Adding agents can multiply work without giving another supplier a chance at a task. The comparison budget accumulates the conditional chances of allocation outside the current favorite, given the history so far. Checks on the team's completed work supply further evidence about its performance. Both the assignments and those checked outcomes affect what can be learned.

Stanford's 2026 AI Index distinguishes training from inference in discussing energy use. Each extra attempt still requires computing after the model has been developed. More attempts are worthwhile when what they produce or reveal is worth those resources.[129]

AI can lower the cost of generating and testing alternatives. It can also operate within institutions that concentrate control of records, distribution, and evaluation. Its useful output and its effect on independent discovery both belong in the economic assessment.

The constructive choice is to identify a valuable uncertainty and the observation that could reduce it. A paired vote, checked document, randomized assignment, reusable log, or independent deployment can help under suitable conditions. Testing today's extractor also differs from testing whether its use teaches workers to recognize tomorrow's difficult cases. The next question follows the person who has to do that work.

The person at the other end

A date in a report depends on more than the model that supplied it. The computing, service selection, evaluation, and person checking the answer all help determine whether the date can be trusted and used.

Someone supplied examples used in development. Someone expressed a preference in an evaluation, prepared a document, checked an answer, or corrected an error. Arena's paired votes make one such contribution visible. In a workplace, the corresponding judgments can be less conspicuous, folded into the ordinary activity of getting a task done.

If an AI service improves the worker's performance, that is a real possibility worth measuring. If it changes which tasks the worker gets to attempt, how expertise develops, or what an employer can observe about ability, those are further questions. An answer that helps finish today's task can also change the conditions under which tomorrow's worker learns.

Behind the answers are people doing work, people whose work is changing, and beginners still waiting for a first assignment. The system that handles today's task also helps determine who gets to learn tomorrow's job.

CHAPTER 9

The first rung

On September 10, 2020, an engineer ran a load test and an AI assistant stopped making suggestions. The interruption appears in an unusual place: an economics paper about customer support. Workers continued answering customers while the software that had been helping them was unavailable. A breakdown had created an opportunity to ask what remained after assistance disappeared.[130]

Usually, a company wants to know whether its new tool works. Does it resolve more complaints? Does it shorten the queue? Does the customer leave satisfied? An outage raises another question. Has the worker acquired something that can survive the disappearance of the machine?

A worker and a computer can become a more productive team without the worker becoming more capable alone. They can also become a more productive team because the worker learns from the computer. The difference is easy to miss while both are working.

The customer-support study supplies a hopeful case. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond examined the introduction of a generative assistant among 5,172 agents. In the published study, access raised issues resolved per hour by an average of 15 percent. The gains were concentrated among workers with less experience and lower initial performance. Experienced, highly skilled workers gained much less; some quality measures deteriorated slightly for the strongest workers.[131]

These were live text conversations about business software. An agent had to understand the customer's problem, find an appropriate response, and communicate it. The assistant suggested language and information. The human could use, change, or ignore its suggestions. The measured improvement belonged to that arrangement: a person handling customers with help available.

But the outage evidence adds something. Previously assisted workers retained improvements in handling speed when suggestions became unavailable. The pattern was consistent with learning, and it became stronger with more prior exposure. It was imperfect evidence: outages were not a clean randomized training experiment, and the conversations handled during them could differ. Still, the machine's temporary absence helped the researchers investigate a benefit that its continuous presence might conceal.[132]

The first rung of a career need not be protected from AI. In this workplace, AI appeared capable of helping people climb it.

That possibility should make us more demanding about what we measure. If the company had recorded only assisted output, it would have missed part of the question. If an observer had looked only at the reduced gap between beginners and experts, the observer might have concluded that expertise was becoming unimportant. Yet an assistant can transmit useful practices precisely because someone developed those practices in the first place.

Consider the customer's next unusual problem. A quick answer is valuable if it is right. Someone must recognize when a familiar response does not fit, when the customer has supplied a misleading clue, or when an apparent solution creates another problem. A system that helps more people develop that judgment has a different economic future from one that merely makes their ordinary replies sound equally accomplished.

The study cannot settle that future. It did not observe compensation, and it did not establish the long-run response of total employment. Better service might reduce the labor needed for a fixed volume of contacts. It might also make support useful enough that customers seek more of it. Employers could change targets, hours, staffing, or pay. A productivity estimate does not select one of these responses for us.[133]

The missing wage record matters for a second reason. Suppose a beginner's improved reply draws on patterns learned from excellent workers. The resulting output does not arrive with a receipt dividing

its value among the beginner, those earlier workers, the model developer, and the employer that organized the service. The customer has received help. The division of credit remains a further question.

A narrowing performance gap can mean useful knowledge travels farther. Experts might still be crucial for diagnosing unusual failures, developing practices, or improving the system that distributes the old ones. The strongest support workers' smaller gains concern that deployment. They do not establish that a business can dispense with the expertise on which future assistance may depend. A firm needs to examine that continuing work before turning the average gain into a staffing plan.

The machine can create real value and improve human learning while the workplace's reward history leaves the contribution of each input unresolved.

The answer and the ability

A mathematics classroom offers a sharper separation between successful assistance and acquired skill. Hamsa Bastani and her collaborators studied nearly a thousand high-school students in Turkey using two versions of a GPT-based tutor. One resembled a standard chat assistant. The other had instructions designed to support learning, including guidance to offer hints instead of supplying the complete solution.[134]

The students practiced, then took an assessment without the tool. During practice, both assisted groups performed better than the group using ordinary resources. Once assistance was removed, the students assigned the standard assistant performed worse than the control group. The tutor designed around learning largely avoided that harm. Its striking practice gains did not translate into a clearly established advantage on the subsequent unassisted assessment.

Getting through a problem with help and solving a related problem afterward are different achievements. A completed worksheet can look like evidence of either. Testing without the assistant made the students' independent performance visible.

The experiment measured short-run learning with particular tools and tasks. How assistance was organized changed what students could do when it was withdrawn. Following their later working lives would require a different study.

The tutor's design required human work. Teachers supplied problem-specific information, correct solutions, and guidance about common mistakes. The assistant was instructed to ask for the student's attempt and respond to the difficulty. Behind the apparently effortless exchange stood a choice about what the student should still have to do.

The experiment preserved an occasion to find out. Students encountered a teacher's review, practiced under assigned arrangements, then completed a closed-book, closed-laptop assessment graded with a common rubric. Observing later performance prevented the study from ending when an interface first made the task easier. Workplace assessments face the same question: when should the next observation occur?

Buying an answer and supporting learning are different uses of assistance. A tool can complete a task or help a user develop the capacity to handle later tasks. Immediate deliverables can look similar while the consequences for later ability differ.

Neither approach is always preferable. A customer facing an urgent deadline may reasonably want the repair. An employee already familiar with the problem may gain nothing from another lesson. Instruction takes time, and a tool that insists on teaching every experienced user could become a nuisance. The relevant choice concerns the purpose of this task for this person at this stage of development.

The classroom researchers also examined students' messages. With the standard assistant, students often simply asked for the answer. With the learning-oriented tutor, they exchanged more messages and increasingly asked for help or attempted solutions. The authors saw this contrast as consistent with students using the standard assistant to obtain solutions instead of working through problems. It helps us

see a possible mechanism. The message patterns do not separately identify the contribution of each safeguard, since the tutor changed several things together.[134]

Deliverable quality, transfer to an unfamiliar task, and the ability to detect an incorrect recommendation require different assessments. Passing one does not establish the others.

There is no reason to make every assessment an exercise in working without computers. The relevant future skill may be choosing, checking, and combining powerful tools. An accountant need not abandon a spreadsheet to demonstrate professional judgment. A programmer may reasonably use an assistant in the evaluation of programming skill. The test should preserve the ability the employer actually needs to know about.

Assessment should follow the future ability that matters, including recovery when a familiar assistant fails, checking sources, and reviewing output from another system. Preserving a test that answers no useful question can become an expensive ritual.

The first rung is therefore more than a collection of simple tasks. It is an arrangement that lets a beginner attempt something, receive feedback, improve, and demonstrate the improvement. AI can change every part of that arrangement. It can make practice cheaper, feedback quicker, and difficult work accessible earlier. It can also complete the activity from which the beginner was supposed to learn.

The disappearing control group

The experienced programmers in METR's early-2025 experiment present another surprise. Sixteen developers worked on 246 tasks in open-source projects they knew well. Tasks were specified before assignment to conditions allowing or disallowing AI. With the tools available in that study, allowing AI made completion take 19 percent longer.[135]

That result had a precise setting. These were experienced contributors in mature projects, facing real tasks and established

standards. The tools were those used during the early-2025 study, primarily a coding environment with then-current models. Nineteen percent longer describes task completion time under the experiment's measurement. It does not mean that every programmer became 19 percent less productive, or that later tools must have the same effect.

The developers' expectations pointed in the other direction. They had expected assistance to save time and afterward still believed it had done so. The discrepancy makes a practical point. A tool can feel helpful during individual moments while adding time across the complete job. Waiting, supplying context, examining suggestions, and correcting mistakes all belong in the cost of producing an acceptable result.

METR collected screen recordings and examined a subset in detail. The records made activities such as prompting, waiting for generation, reviewing output, and writing code separately visible. That is a useful antidote to judging a coding assistant by how rapidly text appears. The desired product is an acceptable change in a working project. A suggestion that arrives quickly can still create work before it belongs there. The recordings do not establish a single cause of every slowdown, and the researchers considered multiple possible explanations. They do show how a better measurement can move the discussion from an impression about assistance to the actual sequence of actions required to finish a task.

Set this study beside the customer-support deployment. One finds substantial average gains, especially for novices. The other finds a slowdown among experienced developers in familiar repositories. The interesting question becomes which workers, tasks, tools, and working arrangements produce which result. A single number for the effect of AI would conceal the decisions an employer actually has to make.

In its February 24, 2026 account of a follow-up effort, METR reported difficulty obtaining a trustworthy comparison. Some developers disliked the prospect of being assigned to work without AI.

Some tasks were kept out of the experiment. The available observations were changing along with adoption.[136]

One participant's published comment included the simple explanation, “I really like using AI!” That is an ordinary preference. It is also a research problem. If a developer avoids entering the experiment because a particularly suitable task might land in the no-AI condition, the measured comparison no longer represents the full set of tasks the researcher hoped to study.

Random assignment can compare tool use among eligible, submitted tasks. Tasks kept out of the experiment remain absent. If developers submit only work they are willing to do without AI, the results describe that selected work.

Study pay also fell from $150 to $50 an hour, potentially altering participation. Developers could run agents concurrently, complicating the meaning of time spent on a task. Unfinished work and differences in task selection created further difficulties. These were reasons to reconsider the design, rather than declare that the original slowdown had cleanly reversed into a precisely measured speedup.

The attraction of the tool could be evidence that it had become more useful. It could also reflect convenience, enjoyment, expectations, or the kinds of work people now chose to attempt. METR believed the newer tools likely helped more, while emphasizing how weakly the follow-up identified the magnitude. The preference to use a product and a credible estimate of its productivity effect answer different questions.

Quality, completion time, and workers' experience should be measured separately. A slower method can be preferred for other reasons. Calling every benefit productivity obscures the choice, while disregarding preference discards something workers value.

This problem is familiar outside software. A worker who has reorganized a day around a tool may reasonably resist surrendering it for research. An employer may dislike paying for deliberately slower work. A researcher cannot assume that preserving a clean comparison costs nothing merely because the comparison would be informative.

Adoption can change what a later comparison means. Users develop habits, forget old procedures, and change which tasks they attempt. Returning to an earlier tool then changes more than a suggestion's availability. An evaluation must specify which stage of that adjustment it measures.

Staged introduction, comparison of assisted training methods, and assessment on new tasks can still supply evidence. Each answers a different question. More observations of the adopted workflow cannot automatically recover every earlier comparison.

A company might accumulate millions of AI-assisted edits while losing the practical ability to recruit a comparable group for a particular no-AI trial. The edits could reveal errors, quality, and day-to-day performance while leaving the alternative working arrangement untested.

Adoption can make a tool more useful while making a particular experiment harder to run. METR had to redesign its study because the way developers worked had changed.

From tasks to jobs

Bank tellers offer a historical warning against equating an automated task with an eliminated job. James Bessen reported that the number needed in an average urban branch fell from twenty to thirteen between 1988 and 2004, while urban branch numbers grew by 43 percent. David Autor drew on this account to explain how lower operating costs and expanding demand can change employment. The remaining teller work also shifted toward customer relationships and other services.[137]

Autor also identified deregulation as part of the branch expansion. The figures do not isolate the cash machine's causal effect. They show why counting automated withdrawals cannot by itself tell us how many people a bank will employ. The business can change the number of outlets, the services it offers, and the work it assigns to people.

Autor's argument concerns relationships among tasks. Automation can increase the value of complementary work, but the employment response depends on demand for that work. Saving time can expand production, improve quality, or reduce labor hours. Identifying an automated step is the start of an employment analysis.

There is harder evidence on the other side. Daron Acemoglu and Pascual Restrepo studied exposure to industrial robots across American labor markets over 1990–2007. They found adverse effects on employment and wages in more exposed places. A machine's productivity benefits did not ensure that the affected local workers would gain.[138]

Acemoglu and Restrepo studied industrial robots performing work such as assembly, welding, and painting. Their analysis used differences in local exposure based on industry composition and robot adoption. Its causal interpretation depends on accounting for other forces that could affect employment in those places.

Places entered the period with different industrial structures, making their exposure to robot adoption different. The researchers used that variation and investigated competing explanations. Comparisons across labor markets supplied evidence within one American history, under the assumptions of their design.

The teller account and the robot study should make us impatient with easy reassurance as well as easy catastrophe. Automation can reduce the labor needed for particular work. Lower costs and new activities can create demand elsewhere. The balance, location, timing, and distribution of those responses matter. An aggregate gain can coexist with a damaging local loss.

The first rung adds a further adjustment. Firms need workers today, but they may also need people capable of doing more demanding work later. If the tasks that once introduced beginners to a profession become cheaper to automate, the employer can redesign training, recruit experienced staff, hire fewer entrants, or develop another route into expertise. Those choices affect the supply of skilled labor that future firms will encounter.

Preserving obsolete manual work solely because it once trained beginners can be costly. The constructive question is whether an alternative route develops the needed expertise, and who has an incentive to finance it.

A simulation might provide repeated practice without risking a customer's records. A supervised assignment could expose a beginner to a more demanding diagnosis sooner. The classroom tutor offers a real example of deliberately changing assistance around learning. The customer-support deployment offers evidence consistent with skills spreading through assistance. These possibilities deserve trials alongside the fear that the entry route will disappear.

It is possible, however, for an employer to discover that it can buy experienced judgment elsewhere more cheaply than it can cultivate it. If many employers make that choice, the next generation's opportunities may change. That is a hypothesis about incentives and collective adjustment. It requires evidence about hiring, training, mobility, and demand. It cannot be established by pointing to a machine that writes a good report.

Who gets the difficult assignment?

Responsibility itself can reinforce opportunity. A successful assignment produces a record that supports the next selection. Employers can rationally rely on that record while earlier allocation decisions partly determine whose ability becomes observable.

Absence of a record differs from a record of failure. Neither guarantees future success, but the evidence available in the two cases is different. Evaluating skill requires distinguishing a poor result from a missing occasion to demonstrate it.

AI could widen access to supervised practice by reducing the cost of checking routine steps. It could also concentrate scarce responsibility among fewer already-tested workers. Which effect occurs depends on hiring, training, assignment, and demand.

Platform work provides documented evidence of selection through a record. The International Labour Organization's 2021 report describes ratings affecting task allocation and access to work, with negative ratings contributing to deactivation.[139]

Geographic mobility offers less independence when the same ranking system governs work across locations. If practical alternatives are also unavailable or reuse the same record, moving does not create a new evaluator. This is a hypothesis about dependence, rather than a description of every platform worker.

The ILO report also records surveyed workers using another platform after permanent deactivation from one. Platforms can open distant demand as well as close accounts. A worker's ability to recover depends on whether another platform offers usable work and how much it relies on the same record.[139]

Europe's 2024 Platform Work Directive addresses one part of the problem. It provides for portability of personal data generated through platform work in automated monitoring and decision systems, including ratings and reviews. Its national transposition deadline is December 2, 2026. At the time of writing, the deadline remains ahead; the existence of the directive is not evidence of uniform implementation or successful remedies.[140]

A portable reputation can help a good worker leave without starting from nothing. It can also make an unfair rating harder to escape. Carry the same flawed score everywhere and portability spreads the original judgment. My proposed safeguard therefore pairs voluntary export with correction, an explanation of context, appeal, and opportunities for independent assessment. A fresh evaluation is different from a right to conceal fraud.

Independent assessment, correction of disputed ratings, and context-sensitive records can make reputation contestable. Each costs something, and an existing score may often be useful. The relevant freedom is a workable way to challenge the evidence and obtain another meaningful observation.

A supervisor's assessment, a practice task, or a second employer's trial can supply evidence missing from the original rating. A reputation system should leave room for those observations to change its verdict.

The employer still needs deadlines met and people paid. It need not solve perfect merit before organizing work. The first rung can disappear because a task was automated. It can also disappear because one institution's early verdict travels farther than the worker's practical ability to find another judge.

Buying a future workforce

Employers can automate routine work, redesign entry routes around assistance, or investigate the consequences before changing staffing widely. These choices trade immediate output against the cost and future value of developing judgment.

Automation may be economical where routine work is reliable, demand for human review is limited, and experienced judgment remains available. A thinner entry route does not make the production saving imaginary.

Assisted training may instead support growing demand and develop future reviewers. The customer-support findings make that possibility worth investigating; they do not establish a universal response across occupations.

Evaluation itself consumes supervision, time, and operational resources. Protecting customers and giving participants a fair basis for taking part belong in its cost. A relevant comparison must justify that expense.

The horizon must match the staffing question. A week of faster output cannot establish whether reducing supervised entry positions creates a shortage of reviewers two years later. Skill development and progression require longer observation.

The measurements should fit that decision. Immediate output and error rates matter. So do the ability to handle unfamiliar cases, the

supervisor time required, the quality of later reviews, and whether participants progress into roles the business needs. Retention and the movement of trained workers to other employers may affect the private return to training. Measuring everything indiscriminately is less useful than observing what distinguishes the feasible choices.

Intermediate measures can help explain that development: unfamiliar problems handled, errors caught before delivery, and repeated supervisor corrections. They can reveal an impressive early performance that fails to transfer, or a costly beginning that develops valuable judgment.

A useful study defines what evidence would support expansion, revision, or stopping. The classroom experiment demonstrates why polished output and subsequent capability should remain separate outcomes.

METR's experience also warns that selection can change once users develop strong preferences about tools. A credible comparison may require compensation for extra time, different randomization, or a narrower question. A control-group label alone establishes no comparability.

Employees should understand what the test is meant to establish. A trial intended to compare training methods can become distorted if participants reasonably believe that one condition marks them as candidates for dismissal. A firm has to account for the behavior its evaluation creates. That is part of making the evidence useful, as well as part of treating people fairly.

Training can benefit later employers as workers move. That creates a possible gap between private and wider returns, which requires evidence about mobility, costs, and skill formation before it can justify a subsidy.

The same reasoning cuts against preserving a program simply because it trains people. Training can teach obsolete practices, select poorly, or consume supervision needed for more valuable work. An alternative employer, professional course, or well-designed AI tutor might supply a better route. The comparison has to include plausible

replacements, rather than treat the existing arrangement as the only imaginable source of skill.

There are limits to what one firm can settle. It cannot observe the entire labor market's response to simultaneous adoption. It may not know how another employer's staffing choices affect the future pool of reviewers. This is a reason for research across firms, careful labor-market evidence, and institutions that can evaluate shared problems. It is not a reason to substitute a confident story for missing observations.

An employer choosing the fastest tool today may be entirely sensible. A workforce, an industry, or a government may have a longer horizon and different interests. Making those interests explicit is more productive than asking whether AI is simply good or bad for labor. The important choices concern who gains, who bears the cost, what can still be learned, and which arrangements remain feasible.

Who owns the workforce?

The rise of agent work also changes which assets are accumulated. Model calls and checking are operating expenses. Equipment, retained code, reliable tests, and durable productive capabilities may support future production. A large stream of generated work need not leave its purchaser with independently usable assets.

Workers can benefit through wages, ownership claims, and cheaper goods or services. Those are different channels, and their distribution cannot be inferred from a count of agent sessions or completed tasks.

Investment can expand the capacity to attempt new projects or strengthen exclusive control of routes through which others must be tested. Both may involve productive expenditure. Their consequences for future comparison depend on what the resulting assets permit.

Automated production and reinvestment could help entrants develop or let well-financed incumbents advance faster. Expansion

still needs customers, finance, scarce resources, and reliable checks. A successful agent experiment supplies one part of that process.

Accumulation also faces depreciation, imitation, failures, and rivals. Productive integration can improve services while making independence harder. The relevant assessment includes both those benefits and the feasibility of another development route.

Anton Korinek and Donghyun Suh examine automation and capital accumulation together in their March 2024 scenario model. Investment can complement work that remains with people. Its pace relative to automation and the assumed range of human tasks can produce very different wage paths. Their model also separates reproducible capital, including compute, from fixed scarce factors. It gives no unconditional forecast that wages or monopoly profits must move in one direction.[141]

Software cannot instantly create an electricity connection or manufacture servers. A shortage can delay every supplier; control over access can prevent particular rivals from developing. Increasing capacity addresses the first problem. The second also depends on who can obtain and use it.

If machines perform more of the work, access to the assets behind that work becomes a more consequential route to income. Employee ownership, jointly financed equipment, and broadly held investment funds can distribute a claim on future output. Public computing access and independent testing facilities can give people resources with which to develop something different. Broader ownership and access should be assessed by who can actually use them and what they enable.

A dividend from an incumbent and the means to challenge it accomplish different things. A worker might own shares in an AI supplier while lacking the money, computing, or customer access to start another business. An ownership policy should therefore be accompanied by workable entry routes, voluntary portability, and opportunities for independent assessment.

The same requirement applies to public provision. A computing facility with one official evaluator can become another gate. Published eligibility rules, more than one credible route to assessment, and appeal against exclusion make access answerable to a challenge. Financing an alternative should buy an opportunity to demonstrate useful work; continued support should respond to what that work shows.

The future of work includes the future of assets built through work: who can retain them, redirect them, and use them to attempt something new. Affordable tools and usable routes to demonstrate judgment can change together or move in opposite directions.

Investment can coexist with closed access, while publicly financed access can become another gate. The terms governing a computing allocation, a work record, or an interface determine who can turn a promising design into a credible competitor.

Competition for future evidence

In January 2023, Britain's Competition and Markets Authority announced that six major banking providers had completed the remaining items in its Open Banking roadmap. The common standards still had to remain usable, with continuing monitoring of performance and availability. Completing implementation did not complete the obligation.[142]

Open Banking let customers share account information securely with authorized third parties and initiate payments through new services. Banks had to support common interfaces through which other businesses could actually use permitted information. An administrative right had to become a working connection.

Receiving your records in a form another service can interpret is more useful than an unexplained file. Being able to use that service reliably is another achievement again. A box marked data access can conceal a working route to an alternative or an interface that fails when needed.

The banks had been required to create and fund the organization maintaining the standards. Those duties continued after implementation was declared complete. Access depended on an institution someone remained responsible for maintaining.[142]

Open Banking illustrates a broader question for competition policy: can a feasible access arrangement improve what participants are able to discover, while preserving productive benefits and protecting sensitive information?

Established systems can be valuable and expensive to replace. The economic case for a remedy depends on the information it supplies,

the decisions that information can change, and the costs of making the route usable.

The comparison budget identifies a feature of an allocation process. Assessing a remedy requires a further account of the value of its evidence and its consequences for production.

What would the evidence change?

Evidence has economic value through the choices it improves. A precise ranking of alternatives can leave the best available action unchanged; a modest observation about a costly failure can matter greatly. Decision relevance precedes a demand for more data.

David Blackwell supplied a foundational way to compare experiments. Under the classical conditions, if one experiment's observations can be produced from another's by adding a suitable kind of noise, the richer experiment can do at least as well across decision problems. Someone with the richer evidence can imitate the poorer procedure.[143]

Imagine a documented record of every eligible trial request: suggestion, checked outcome, and time required. A second report contains only a summary computed from those records. Someone with the interpretable full record can produce the summary; someone with the summary cannot generally recover discarded detail.

Collection costs, privacy, and implementation enter after the gross comparison of experiments. Different experiments can also answer different questions, such as operating accuracy and integration costs, without fitting one universal ranking.

Evaluation should specify feasible actions, the period under study, relevant outcomes, and how errors and costs enter the decision. A measure that rewards one dimension while ignoring another can encourage behavior that improves the score without improving the service.

Average outcomes can conceal different burdens across users. The appraisal must make consequential differences visible and compare actions an institution can actually take.

The baseline must also be feasible. Removing safeguards from the assumed status quo exaggerates a remedy's value; comparison with a perfectly informed decision maker sets an unattainable benchmark.

A buyer can learn enough to choose a contract while the exact contributions of programmers, models, training sources, and earlier customers remain uncertain. The trial earns its keep through a better decision. Further testing is worthwhile only if the remaining uncertainty could change a consequential choice.

Opening the existing record

An audit of existing evidence can precede another experiment. Private tests, independent evaluations, and operational records may already resolve the question. All informative measurements belong in the analysis; disregarding internal experimentation would manufacture a deficit.

Protocols matter alongside results: eligibility, task selection, withheld cases, version changes, and whether evaluators knew whose output they were checking. Detailed records can still answer the wrong question.

Google's December 2025 search judgment makes this practical concern visible. Its specified data-sharing provisions required enough information for qualified competitors to understand the datasets, including descriptions of contents, sampling, and privacy-related transformations. It did not simply order delivery of an unexplained pile of records.[144]

A million query records and an explanation of how they were selected are different assets. An entrant may need both to interpret the data it receives. The order specifies that access; its usefulness depends on what recipients can do with it in practice.

A record limited to accepted outputs describes that selection, rather than performance on rejected outputs or tasks never encountered. An audit can identify the gap or establish that existing evidence is adequate. The analysis must be capable of supporting no further intervention.

Zahra Meghani's work on epistemic monopoly examines dependence on developers for evaluation evidence and control of research conditions.[145] Accurate reporting can coexist with unanswered questions when the producer chooses which investigations occur.

Scrutiny requires expertise, secure handling, and time. An authorized evaluator can inspect protected records and report methods, uncertainty, and conclusions. Accountability concerns what can be checked and challenged without necessarily disclosing personal data.

Disclosure can also change competition. The 2023 Merger Guidelines discuss how easier observation of rivals' terms can increase susceptibility to coordination. Giving every competitor every business record need not improve the strategic environment. An independent evaluator and a rival sales department have different uses for confidential information.

Open Banking's completion decision distinguished service-quality indicators and underlying data from current-account transaction records supporting authorized access and payments. It specified services and continuing obligations rather than publication of every internal file.[146] That specificity matters to remedies intended to create usable comparison.

A file that can travel

Portability can lower switching costs when useful records would otherwise have to be rebuilt. Open Banking's standards, conformance checks, and performance obligations provide a documented reference: compatibility was treated as a service requiring maintenance.[146]

Ipsos's February 2026 publication supplies another kind of banking evidence. Its independent service-quality survey, conducted during 2025 under a regulatory requirement, asked customers about recommending providers and particular services, including online banking and branches.[147] These experiences matter, but customers were not randomly assigned to banks. Their needs can differ.

Customer experience can inform a decision without isolating the causal effect of each input into that experience. Survey evidence and randomized assignment answer different questions; neither should acquire the other's interpretation.

Copying a record to more organizations does not create independent histories of the process that produced it. Operational gains from portability and new evidence from experimentation should be assessed separately.

Interpretation matters too. Imagine a field marked positive means successfully resolved in one record version, but problem requiring action in another. Exported files omit the version. A reader could reverse the results while processing every row correctly. Under one interpretation, following the signal helps; under the other, it hurts.

Knowing that someone with the correct interpretation could use the records does not supply the actual buyer with a usable procedure. If the records cannot distinguish the meanings, more equally ambiguous rows may not help. A verified definition can let the buyer use one procedure that reads the version and interprets the result.

Definitions alone reveal no outcomes. Outcomes with unknown meanings may be unusable. Together they can support a decision; their joint value need not equal the sum of their separate values. Freely available, understood evidence need not harm a buyer who can ignore it. Cost and lost interpretation create the problem. Retaining the trusted record preserves the earlier procedure.

What survives migration should therefore be specified: category definitions, corrections, and the link between a result and its service version. Counting exported fields is an inadequate test of usefulness.

Interoperability concerns whether services can connect to the systems and users needed for a relevant test. A bounded interface may suffice. Security, maintenance, capacity, and changes to integration costs determine what scope is feasible and who should bear the burden.

Compatibility can require security work, additional capacity, and continuing maintenance. It can also alter incentives to invest in an integrated service. A workable access duty has to assign those costs and preserve the productive benefits the connection relies on.

Making the missing comparison

An audit cannot recover an outcome never observed. If that outcome matters sufficiently, a designed experiment is needed, with comparable tasks, documented assignments, checked results, and records of missing observations.

Assignments can interact through shared resources and participants' responses. Giving one supplier a task may change another supplier's capacity or behavior. Comparing periods or groups, or modeling those interactions, can make the outcomes interpretable where balanced assignment alone cannot.

On June 3, 2019, a field experiment began calling rice farmers in India. Precision Agriculture for Development wanted to choose how to recruit farmers to its telephone service. Maximilian Kasy and Anja Sautmann helped compare six combinations: morning or evening automated calls, with no text alert, an alert an hour ahead, or one a day ahead. From a government partner's list, the service set aside ten thousand valid numbers outside the do-not-disturb register. Seventeen waves of calls followed, with assignment changing as results arrived.[148]

The morning call preceded by an hour's warning received nearly four thousand assignments. The evening call without a warning received fewer than four hundred. Every option was tried, but the assignments were unequal. At the end, the authors put the probability

that the morning call with that warning was best at just over three quarters, under their statistical model. Uncertainty remained.[148]

Success meant answering five recruitment questions, rather than improved harvests or income. The evidence could guide the next recruitment campaign without establishing the agricultural service's total value. The outcome must fit the decision.

It could adapt assignments as results arrive, giving a promising service more requests while preserving informative alternatives. Serving today's requests and learning what to buy next year need not recommend identical allocations.

The comparison budget prescribes neither equal shares nor a universal quota. A small persistent chance can accumulate indefinitely; large nominal opportunities can be wasted on tasks that fail to expose the relevant difference.

A declared decision rule helps prevent changing the criterion after results arrive. The test must be capable of changing an action on grounds specified before its outcome is known.

Transportability also matters. A result obtained under unusually clear tasks, expert staff, or an obsolete version may not travel to ordinary operation. Broader testing, calibration, a narrower application, or rejection of the proposed change can each be appropriate.

The assurance concerns expected performance across the retained accounts. Some residents or contracts can still fare worse. A sound decision uses the available evidence and justified assumptions before the outcome is known. A fortunate result cannot repair a method that ignored relevant costs.

A feasible policy must use a procedure available with the information actually observed. Imagining a decision maker who is told which explanation is true does not supply such a procedure.

When every road leads to the same judge

Multiple purchasing organizations can inherit one ranking or one defective test. Separate encounters then reproduce a common judgment. Independence matters when it changes the relevant selection or measurement process.

This is the historical possibility running from Smith's towns through Bell's network to platform work. Wider markets can support more specialization, better service, and more customers. They can also extend one institution's reach over the decisions determining who may develop. A market can grow while the number of independent places to try again shrinks. That is a proposition about institutions to investigate, not a measured law of history.

Consider an explicit hypothetical. Suppose OpenAI became the only foundational AI supplier. Add a further condition: it also controlled indispensable routes to customers, the evaluations deciding admission, and access needed to develop competing systems. These are assumptions about a possible arrangement, not a description of OpenAI today or a forecast. The question concerns the structure, whichever company might occupy it.

Even a sole model supplier would not automatically possess total epistemic monopoly. Independent laboratories could retain ground truth, buyers could test alternatives, and other technologies could answer the question at issue. The stronger scenario requires control over the production and checking of relevant comparisons. Its permission becomes necessary to test the alternatives to its authority, with no workable independent route remaining.

The service might be excellent and cheap. Its owners might have invested wisely and lawfully. Those facts belong in the appraisal. But a satisfied present customer cannot report the performance of a rival that never received a feasible development route. The concern reaches beyond a price increase: which alternative capabilities will be permitted to become observable?

Saying that markets clear does not answer this. Clearing concerns offers that can be made under existing conditions. It does not guarantee that an undeveloped supplier acquires the inputs, customers, or evidence needed to make a competitive offer. Nor do a million agents necessarily provide a million independent judgments if access to all of them passes through the same gate.

The urgent question is who controls the chance to become a competitor. A merger, a default contract, or an infrastructure license can help determine which alternatives are ever developed enough to be compared. When the incumbent controls an indispensable route to that development, a low price and a satisfied customer leave a further question unanswered: what could another supplier have become? Competition policy should protect workable routes to answering it, including routes a challenger cannot create entirely on its own.

A concentrated future should be evaluated through identifiable mechanisms, reversibility, and feasible safeguards. If independent evaluation could become prohibitively costly, waiting until lost skills and institutions must be rebuilt can itself be consequential. That possibility requires evidence rather than dismissal or assumption.

Hayek treated competition as a way to discover facts that could not be known in advance.[149] The question now is how a rival obtains the computing, customers, and tests through which something unexpected can be discovered.

Bell's experience shows why the route has stages. Opening patents supported follow-on innovation outside telecommunications, while the study found a different result within the sector. Permission to use knowledge was not the entire route to a customer.[150] Publishing an incumbent's records and allowing an entrant to generate new records can therefore accomplish different things. A thousand copies of one history remain copies of one history.

Scale also supplies research, reliability, and lower costs. One large supplier can conduct informative internal tests; several firms can repeat a shared error. Private testing counts. A public trial can displace it. The aim is to preserve consequential independent learning

where its benefits justify its full costs, rather than keep every proposed competitor alive.

Precaution without closing discovery

Competition policy already reaches beyond prices. The United States' 2023 Merger Guidelines discuss products, features, innovation, wages, benefits, and research competition.[151] Control over future evidence gives those concerns another practical dimension: an entrant can lose the chance to demonstrate an improvement before customers can judge it. Any legal remedy still has to meet the applicable statutory requirements.

Who should bear the burden when closing a route could destroy the evidence needed to judge that closure? Requiring the excluded challenger to supply the missing demonstration makes the obstruction its own defense. When a gatekeeper controls an indispensable route to useful independent evaluation, and credible evidence shows that closure would be hard to reverse, access should therefore be protected by an enforceable, rebuttable duty. The party seeking the restriction should have to demonstrate why a narrower arrangement cannot protect legitimate interests. A challenge to the gatekeeper's own judgment must be assessable by someone other than the gatekeeper.

The timing matters. Requiring a challenger to arrive with a mature product can protect the very restriction preventing it from developing one. The evidence needed at this stage concerns a credible development route, the obstruction to it, and the value of keeping it available. Waiting for the last effective challenger to disappear can destroy the evidence needed to appraise the decision.

Keeping an option to learn has a cost, as does losing it. Arrow and Anthony Fisher examined how waiting for more information can affect the choice to make an irreversible investment. Their analysis treats that future information as part of a cost-benefit appraisal.[152]

The rule could require documented interfaces, workable interoperability, voluntary data portability, or access to specified

knowledge needed for a credible test. Requirements should follow the obstruction. Search data access cannot supply a missing electricity connection; model access cannot repair a dishonest outcome measure. A regulator should identify what the entrant could actually do under the proposed terms.

The EU's Digital Markets Act gives this boundary a concrete form. For designated messaging services, gatekeepers must connect specified functions they already offer their own users: individual messages and attachments first, group messaging within two years of designation, voice and video calls within four. Access is upon request and free of charge. The law also requires preservation of the security provided to the service's own users, including end-to-end encryption where applicable.[153]

That is an obligation to build and maintain connections between privately owned services. A separate provision concerns access to specified search data on fair, reasonable and nondiscriminatory terms. The two duties have different designs. They demonstrate that ownership can coexist with participation rights, without establishing that every implementation succeeds or that an AI company automatically owes the same duty.[153]

For AI, an obligation might concern an interface or a secure evaluation route rather than publication of every model weight or training record. Fees may be justified; privacy, security, research incentives, and genuine technical constraints matter. Claims of unavoidable harm should receive independent scrutiny. Letting the gatekeeper alone decide whether a competing test is safe would make the safeguard depend on the permission it is intended to examine.

The public authority also needs limits. It should specify the affected decision, plausible lost comparison, legal power, expected benefit, costs, and less burdensome substitutes. A secure audit may suffice where existing evidence answers the question. Prospective trials matter when no file can contain the missing contrast. A time-limited evaluation power would require legislation wherever existing

authority is insufficient, with a defined scope, an evidentiary threshold, and rights of challenge and review.

Property rights should reward invention while leaving competitors a workable route to test and develop their own ideas. Where those routes depend on essential infrastructure, the owner should face a duty to provide appropriately bounded access on workable terms. An incumbent's private tests can satisfy a particular information need when their evidence is usable by the affected decision makers. They do not settle whether an excluded entrant could develop a capability those tests never allow it to attempt.

Government can also become the single judge. A mandatory benchmark, favored consultant, or official platform could synchronize mistakes across an economy. Independent appeal, alternative methods, and revision must constrain the evaluator as well as the evaluated. Precaution should keep discovery possible on both sides of the public-private boundary.

Who can keep the next entrant possible?

The rise of agent production makes access more consequential. Cheaper software work can still depend on computing, usable records, distribution, and credible evaluation. Ownership of generated output does not automatically provide those complements.

EuroHPC offers computing allocations, NAIRR organizes access to research resources, and China's AI Plus policy calls for coordinated computing and investment. For an applicant, the differences appear in practical decisions: who can qualify, what work is permitted, and what can be done with the results.

EuroHPC's Playground call, opened in April 2025, offers an entry route for European startups and smaller businesses with limited computing needs. Applications arrive in order, with eligibility and technical checks rather than a competitive ranking. Allocations last one, two, or three months on a single system. The published call also warns that demand can leave particular capacity unavailable. The door has a procedure and a queue.[154]

Its Industrial Innovation modes offer eligible AI small businesses free computing for innovation. Crucially, the March 2026 terms say generated data and models remain the user's property and can be commercially exploited. They also require publication of results outside confidential portions and a final report. Public provision and private accumulation coexist in these stated terms.[154]

For a team building a business, a finished allocation is still some distance from a finished product. Appropriate data, continued testing, and access to customers determine whether its retained model can become part of a viable service.

An access program's value also depends on whether work can continue after an allocation ends. Reproducible tests, transferable records, waiting time, support costs, and displaced private purchases help determine whether it enables additional useful experimentation.

Property protection leaves choices about how resources may be used. Article 17 of the EU Charter protects possessions and intellectual property while allowing regulation of use under specified conditions.[155] The practical choices concern who can use a resource, on what terms, and how investment and entry can continue. Common institutions and member states must still make the different parts of an arrangement work.

Support for private enterprise leaves room for public provision of research infrastructure. The July 2025 AI Action Plan advocates deregulation, but also encourages open models and wider access to computing. It identifies long commitments to large cloud suppliers as an obstacle for many startups and academics, and recommends developing computing markets and the National AI Research Resource, or NAIRR.[156]

In September 2026, the National Science Foundation announced an operations center for NAIRR, the public-private research initiative. The center would coordinate providers, integrate computing and other resources, operate a portal, and support users. A researcher could reach resources through an organized public route rather than negotiate every component alone.[157]

Research access and commercial entry remain different routes. Eligibility, permitted use, and portability of results determine how far a research initiative can support later independent development. A portal can simplify access without removing dependence on its participating providers.

American authorities have also examined the private arrangements. In January 2025, FTC staff reported on partnerships between major cloud suppliers and AI developers. The report identified potential concerns involving input access, contractual and technical switching costs, and privileged information, without making a determination of illegality.[158]

Such partnerships can finance valuable capabilities. A developer may obtain the computing it needs while accepting terms that make a later move expensive. An access initiative has to leave a workable route beyond the initial allocation if it is to support an independent business.

Deregulation and access protection can affect different constraints. Removing an unnecessary permit can support entry, while leaving an indispensable interface at one provider's discretion can preserve a bottleneck. A count of regulations removed does not measure the ability to conduct an independent test.

China's August 2025 AI Plus policy starts from a more explicit program of coordination. It calls for shared and interconnected computing, agent services, open-source development, and patient capital. It addresses new forms of work and employment risks as well as industrial adoption.[159]

Patient capital and shared computing could support redesign before commercialization. Their usefulness still depends on admission rules and the ability to try unfamiliar approaches. Coordinated resources can coexist with a common gate that rejects departures from an established method.

China's 2023 interim generative-AI measures encourage innovation and shared infrastructure, while imposing political content requirements on covered services offered to the mainland

public. Their scope expressly excludes development and applications that do not provide such public services.[160]

An economy could make it easier to try competing industrial designs while restricting which politically unwelcome conclusions a public service may generate. A test earns little independence if its commercial or public sponsor can suppress an inconvenient result. Researchers need to be able to report that the favored system failed.

An excluded applicant's rights and the ability to report failure therefore belong in the comparison alongside computing and finance. Europe and China cannot be grouped into one alternative to America simply because both may intervene. Nor does a larger public budget settle the question. Follow the attempted product, its tests, and the grounds on which somebody can refuse the next step.

Limiting exclusion at a bottleneck can sometimes enlarge the practical scope for private enterprise. A bounded, secure interface can leave firms privately owned while changing the terms on which they compete. Whether it does so is a question about the particular design and its costs.

Public intervention can create a gate of its own. Certification costs may exclude smaller firms, or an evaluator may suppress inconvenient failures. Contestability requires limits on official control as well as private control, and reliable expectations about the assets successful experiments create.

To choose between arrangements, return to the same sequence. Can a qualified outsider obtain resources? Can it reach customers and receive informative work? Can someone check the outcomes independently? Can useful records travel? Can an exclusion or a failed official test be challenged? The answers must survive ordinary use. A published entitlement that nobody can exercise is the institutional equivalent of the exported file nobody can read.

The society with the first leading model need not be the society best able to recover from a mistake in it. When the established system fails, a rival needs a route to computing, customers, and independent testing. Public resources can keep that route available, or absorb

scarce engineers and money without producing usable alternatives. Recovery depends on what another team can actually investigate, demonstrate, and put into service.

The price of finding out

Audits, portable records, interfaces, experiments, and independent evaluation address different gaps. There is no permanent ranking among them. The appropriate remedy depends on what evidence already exists, what is missing, and which feasible route can supply it.

Appraisal connects the possible mistake prevented to its probability and consequences over a stated period, then counts testing, migration, supervision, privacy protection, and lost production. A cheaper adequate substitute belongs in the comparison.[161]

Preventing a rare serious error can justify costs spread across many routine transactions. The decision depends on whose burdens and future benefits count. A payment transfers money between participants; an opportunity cost concerns what scarce capacity could have produced elsewhere. Counting forgone production once, and including consequences outside the purchasing department, makes the tradeoff clearer.

Testing can improve current service and future evidence at once. Its benefits should be counted without duplication, and its continuation should depend on whether remaining uncertainty can still change a consequential choice at an acceptable cost.

Adequate choice among feasible actions can coexist with a limit on universally reliable contribution recovery. Institutions do not need an exact division of all economic merit before acting.

Arrow's analysis of information use and appropriation supplies a further question: do those deciding whether to investigate capture its wider benefits?[162] Evidence benefiting other participants can create a gap between private and social incentives. Concentration alone cannot establish that gap.

Both incumbents and entrants have interests in the design of a remedy. A trial can subsidize sales, and an audit can be too narrow to challenge an established position. The appraisal must explain the improvement sought, who benefits, and who bears the cost.

Open Banking's continuing obligations provide the concluding reference: a route maintained after the formal opening. Competition policy concerned with discovery must assess whether usable records and workable access allow meaningful comparisons over time.

Closing a useful route to comparison can leave costs with the people who needed the evidence. Taxation could make the actor bear some of that loss. Such a charge needs a causal link between conduct and harm, an affected decision worth improving, and a practicable way to measure and address the cost.

Taxing lost comparison

Sweden introduced its nitrogen-oxide charge in 1992. Covered energy-production facilities paid for emissions, while receipts were returned in relation to useful energy output. The design separated a harmful by-product from the productive activity that generated it.[163]

A business decision can improve current production while closing a route to discovering alternatives. If that closure prevents useful learning that others cannot replace, its private return omits a cost borne elsewhere. Making the decision maker face that cost changes the incentive to close the route. A corrective charge is one instrument. An access obligation may preserve the comparison more directly.

There are two tax questions here. One asks how confidently income can be divided between productive contribution and historical position. The other asks whether particular conduct imposes an external cost by preventing valuable comparison. Explaining a fortune and changing future incentives are different tasks.

Income and its explanation

Payroll records, completed work, and measured output establish meaningful facts. They do not necessarily identify how much an input changed the probability of receiving a reward. A causal question about revenue also differs from one about a product's benefit to users. Each requires its own target, observations, and comparison.

Consider the fixed-input version of the allocation model. An admissible adjustment to hidden position can offset a change in the productive-input effect, leaving every observable allocation probability unchanged. Two candidate explanations can then assign the same law to all recorded histories while giving different answers about what changing the input would do. A rule based solely on that

record has no observation on which to distinguish the exact contribution divisions.

This is an observational-equivalence argument. The underlying effects differ between candidate explanations; a shadow replay instead preserves those effects and varies the realized history. It is also separate from the finite-lifetime-comparison result. Input variation that the hidden position cannot absorb, or additional informative outcomes, may break the equivalence. The limitation concerns the specified experiment, rather than every possible form of economic evidence.

Taxation can proceed without an exact retrospective causal audit of every reward. Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva distinguish responses involving productive activity, avoidance, and compensation bargaining. The different behaviors have different implications for tax policy.[164]

Casey Rothschild and Florian Scheuer analyze taxation when income combines productive and rent-seeking activities that the authority cannot separately observe for each person. Changing a tax can alter relative returns across activities, strengthening or weakening a direct correction. Averaging the hidden components does not automatically produce the right rate.[165]

These approaches assess consequences using assumptions about behavior and welfare. They leave room for observable bases and plausible ranges without requiring exact knowledge of individual merit. The comparison-budget argument challenges a stronger confidence in what a reward record must eventually reveal; it does not make public finance impossible.

Moral desert and contractual entitlement require further judgments. A valid promise of payment can matter while its recipient's complete causal contribution remains uncertain. Security against income loss, rewards for valuable investment, and practical routes into new work pursue different aims. Evidence about alternative histories does not turn them into one policy or dictate equal incomes.

What would a charge measure?

The Swedish charge depends on a verification chain. The environmental agency describes instruments, declarations, and audits. Facility equipment must be compared at least annually with independent measurements from accredited laboratories, and the agency coordinates checks of the laboratories themselves.[163]

Lost comparison has no corresponding physical meter. A prevented assignment could have revealed a serious failure, repeated something already known, or produced an unreliable measurement. Its value depends on the decision it could change and the evidence available elsewhere. Counting missing assignments cannot establish those differences.

The comparison budget accumulates modeled allocation chance outside the current favorite. To move from that statistical object to money requires an account of how conduct changes opportunities, what observations those opportunities produce, and what their absence costs. A finite budget supplies no monetary damage estimate by itself.

Removing a restriction may leave an experiment infeasible if participants still lack resources, technical access, or a valid outcome measure. Another source may already supply the evidence. The loss caused by the restriction depends on what a feasible alternative would have made possible.

Scale can make replacement harder. Losing one local route may leave independent institutions elsewhere. Losing access through a gate shared across a large network can remove many practical routes together. That is a hypothesis about institutional dependence to investigate, alongside the productive gains the shared system supplies.

Beneficiaries matter too. A firm already has an incentive to buy information whose benefits it captures. Other users can benefit from an investigation without paying its producer. Kenneth Arrow examined the difficulty of appropriating the returns to information,

whether generated deliberately or as a by-product of ordinary activity.[166] This established information problem supplies a foundation for asking what useful evidence an exclusionary action prevents.

The proposed correction must count both sides. Productive integration can reduce operating costs, while secure evaluation may preserve a useful test without unrestricted disclosure. Privacy, security, disruption, and the cost of maintaining access belong in the comparison. The economic objective is a better attainable arrangement, with all its resource requirements included.

Production and private testing

A firm's response can occur on more than one margin. It can change a restrictive activity and change how much evaluation it purchases. Private testing is real evidence. If it supplies the same usable information to the same decision makers in time, treating the original loss as wholly unreplaced overstates that loss.

Equivalence needs to be demonstrated. An internal AI benchmark can use synthetic tasks while a field study observes real users; their errors may overlap, and their results may reach different people. Internal findings unavailable to an affected outside decision maker cannot automatically replace that person's missing evidence.

In their model of information acquisition, Luca Colombo, Gianluca Femminis, and Alessandro Pavan show that privately chosen information can differ from the socially efficient amount. Depending on the setting, the discrepancy can run in either direction.[167] Underinvestment therefore needs a specific account of benefits that private incentives insufficiently protect.

Different instruments affect those incentives differently. A charge on a restrictive term changes the return to imposing it. Payment for verified evaluation changes the return to producing evidence. Opening access may reduce the need for purchased tests; subsidizing tests may finance work that would have occurred anyway. Those interactions determine what a correction achieves.

Convenient tax bases can combine unlike activities. Computing produces ordinary outputs and also supports experiments investigating failures. A charge on all computing reaches both. Electricity, emissions, revenue, and distribution can justify separate tax questions, but an information-based correction needs evidence connecting its chosen base to the relevant loss.

Profit is no more direct a meter. Useful engineering, scarcity, historical advantage, and restrictive terms can all contribute to earnings. A corrective proposal must identify conduct and consequences, rather than use the size of a return as a substitute for them.

When public information displaces private effort

Colombo, Femminis, and Pavan also establish that greater public information precision reduces privately acquired precision in their specified model.[168] The result concerns an endogenous response, rather than a universal one-for-one cancellation of tests. Evaluating public provision requires counting what private participants change afterward.

A public result equivalent to one a firm planned to purchase can alter who pays without adding evidence. It may still save resources if public provision is cheaper. Different public measurements can instead expose failures private testing missed, reach other recipients, or supply work beyond private capacity. Announced test counts cannot distinguish those outcomes.

California's Electric Program Investment Charge provides a documented way to finance shared investigation. Utility customers fund research administered by the Energy Commission and participating utilities, supporting electricity-system research, development, and demonstration.[169]

During the December 20, 2022 Ferndale earthquake, the Redwood Coast Airport Microgrid supplied nearly fifteen hours of backup

during a regional outage lasting nearly sixteen hours, according to the project team's January 2026 report. Customers included an airport and a Coast Guard air station.[170]

The installation also had a persistent fault. Converters connecting the large solar array to the batteries repeatedly triggered protective trips. The report identified an unwanted current path to ground, particularly in humid conditions. Replacement equipment was eventually installed and tested. Another part of the system allowed backup operation despite the solar-converter problem.[170]

A demonstration can therefore provide service, expose a costly failure, and potentially inform later projects. These benefits have different recipients. The team found that readily monetizable benefits did not cover assessed costs, while the value of resilience depended on the community and services involved.[170] Establishing a worthwhile subsidy requires assessing wider benefits against the full costs, including what private activity it replaces.

Knowledge transmission is another step. A published diagnosis can help others only if it reaches them in a usable form and applies sufficiently to their problem. AI evaluation faces the same requirement: findings must be interpretable, relevant to subsequent tasks, and capable of changing a consequential decision. A large collection of scores can be less useful than a small test that resolves an important uncertainty.

William Kerr, Ramana Nanda, and Matthew Rhodes-Kropf describe entrepreneurship as experimentation, with staged commitments allowing learning before full investment.[171] A negative result can save resources by supporting abandonment. Its value comes from what was credibly learned, rather than failure itself.

Replacement and the feasible alternative

David Blackwell compared experiments through the decisions their information permits. If one can reproduce another's observations through a known conversion rule independent of the hidden state, a decision maker can imitate the second experiment's choices.[172] This supplies a discipline for evaluating whether evidence is an adequate substitute; acquisition and implementation costs require an additional calculation.

A lost field comparison may be replaceable by a cheaper independent evaluation. That route can reduce the sacrifice while preserving useful information. It must reach the relevant recipients in time, fit the decision, and account for capacity and delivery costs. A laboratory measurement that omits the development and learning enabled by real participation may not supply the same opportunity.

Replacement also changes the institution. A shadow replay preserves starting conditions and response rules; a new laboratory or access program changes them. The evidence such an intervention produces can be valuable without being equivalent to the lost history.

Secure infrastructure, qualified staff, independent checking, and documentation create fixed costs before the first result appears. Existing evaluators or shared facilities can reduce duplication, while common errors and centralized control can create new dependencies. A limited commission may establish feasibility before permanent provision is justified.

These considerations favor comparing whole arrangements: preserving access, buying private evaluations, supplying shared tests, or combining narrower measures. Useful institutional invention can retain the benefits of scale while reopening a consequential comparison. Preserving every old channel indefinitely is no substitute for that appraisal.

Financing and incentives

Britain's 2017 retail-banking order required providers to support implementation arrangements and fund independent service-quality surveys, with contributions linked to active accounts in the relevant survey market.[173] The account rule allocates a bill for specified duties. It does not measure each provider's responsibility for lost evidence.

Financing useful evaluation and correcting harmful conduct are different achievements. General revenue, supplier contributions, and charges tied to a demonstrated restrictive action produce different incentives. Raising enough money for a laboratory does not establish that the funding instrument targets the cause of the information loss.

The payment is also different from the resources consumed. A transfer can finance staff time, computing, and supervision; counting both the transfer and those inputs as destroyed output would double-count resource cost. Who pays still matters for distribution and behavior, including changes in prices, wages, investment, and entry.

Public provision has an opportunity cost as well. Money and expertise used for evaluation may displace another useful activity. Even a good laboratory can be financed through a charge that discourages valuable production or testing. Its funding and corrective effects require separate examination.

Verification is part of that expense. Repetitive easy tasks can inflate a test count, and selective cases can flatter an entrant. Relevant task selection, checked outcomes, version records, and independent scrutiny help establish what evidence was actually produced. A nominally cheap substitute can be costly once those safeguards are included.

Who pays for the next attempt?

Suppose a dominant AI supplier controls the only practical interface through which an available rival could be evaluated. If a restrictive term prevents a useful test, the first question is whether an access duty can restore it. A charge that leaves the rival locked out may raise money without fixing the obstruction. Where a bounded, secure arrangement can restore the route at a justified cost, opening it addresses the problem directly.

Where conduct still imposes a measurable external loss, a corrective charge should reflect that loss after feasible replacement and private responses are counted. Revenues could support independent evaluation or access. Spending the money well is a separate obligation: the public laboratory has to produce usable evidence, and affected people must be able to act on it. The firm's wealth alone supplies no estimate of the damage.

Now consider a team that could use the interface but cannot finance the work needed to reach a credible trial. Access rules alone leave it outside. Shared research and wider access to productive assets require financing, and relying entirely on existing owners lets their willingness to fund a challenge decide which challenges occur. Progressive taxation of capital income and large inheritances can supply part of that financing and spread ownership beyond its current holders. Its justification is the productive participation and independent discovery it makes possible. No verdict about each taxpayer's moral desert is needed.

Florian Scheuer and Joel Slemrod's review of wealth taxation examines the choice of tax base, valuation, avoidance, enforcement, and behavioral responses. Those choices change what a tax collects and how people respond.[174] A tax on annual wealth, a tax on capital income, and an inheritance tax reach different things. Workable capital income and inheritance taxation offer routes to financing broader participation; their investment and financing effects determine the terms. Uncertainty about contribution alone supplies neither a preferred instrument nor a tax rate.

The uses of the revenue should be concrete. A shared computing facility could let a qualified team attempt a different model. An independently governed laboratory could test failures that no supplier has an incentive to disclose. Grants could finance credible first demonstrations. Broadly held funds or employee ownership could distribute claims on productive assets beyond their existing owners. These are proposals for widening the means to attempt and benefit from economic activity; their performance has to be measured against their costs.

A payment to a household can make it easier to survive a failed attempt or to acquire training. It can also serve the straightforward purpose of protecting income. It need not generate a comparison to be worthwhile. But a policy advertised as restoring contestability has to reach the missing resource or permission: cash cannot, by itself, open a closed interface.

The public funder should publish qualification rules, allow competing methods, and provide appeal against exclusion. A grant recipient should be able to report failure without concealing it to protect the program. Continuing support should depend on what is learned and on the credible value of further work. Public finance should create alternatives to private gatekeeping while remaining answerable to independent judgment itself.

These fiscal purposes should remain visible in the accounts. A conduct-based charge seeks to change a harmful incentive. General taxation finances shared capabilities and broader participation. Pooling their revenues may be convenient, but financing a valuable laboratory does not establish that every contributor caused an equivalent informational loss. Nor should the same harm be charged twice through overlapping instruments.

Capital can finance the equipment, time, and risk that discovery requires. It can also accumulate around a route whose owner decides who else may enter. Capital income and large inheritances should help finance the shared capabilities through which other people can become productive owners. The terms must preserve reasons to

invest. Leaving accumulated returns entirely outside that responsibility allows a successful position to pass to the next generation together with the power to prevent its challengers from developing.

Choosing rules before knowing every contribution

A defensible charge needs a causal chain: identified conduct, changed comparative evidence, affected decisions and beneficiaries, and an assessed loss after private responses and feasible replacement. It must then improve the attainable outcome after productive benefits, financing, enforcement, and entry are counted.

The time horizon must match on both sides. Software changes can make some findings obsolete while leaving enduring failure modes relevant. Longer effects on learning and entry require stated assumptions and evidence. A temporary information problem should not become a permanent fiscal category after its basis has changed.

Participation can change too. Qualification costs may deter useful entrants; predictable obligations may support investment. Warnings about reduced innovation need investigation, rather than automatic acceptance or dismissal. These are responses affecting the case for a particular instrument.

Access duties, payments for verified testing, public provision, and existing private arrangements have to be compared on those terms. Where the evidence cannot support a precise corrective rate, a narrower access remedy or a limited trial can establish what works. General financing choices can still proceed through an explicit public judgment about benefits, burdens, and incentives.

We do not need agreement about the exact share of every fortune attributable to its owner before choosing a tax system. We need agreement about what that system should make possible, how its costs will be borne, and how its results can be challenged. A reward

record that cannot settle contribution should not be asked to certify the rules protecting its own distribution.

A settlement that rewards useful investment, broadens ownership, and finances independent routes to discovery gives enterprise a future beyond its current owners. A successful business can earn substantial returns while contributing to the resources through which another business might demonstrate a better way. The next task is to make those opportunities observable in the institutions economics studies.

CHAPTER 12

An economics of possible worlds

The winter wheat planted at Broadbalk in autumn 1843 was harvested in 1844. The records of that harvest were the beginning of a much longer undertaking. Fertilizers, manure, and untreated ground supplied contrasts that successive generations of researchers continued to study. The field produced grain. Keeping differences alive helped it produce evidence too.[175]

An ordinary commercial field might sensibly receive whatever treatment its owner expects to work best. Broadbalk had another purpose. Some ground continued receiving treatments whose value could be assessed against other ground. Giving every strip the apparently best treatment would have changed what the field could teach.

That does not make the experiment an argument against productive farming. It makes its second product visible. A harvest is an outcome. A maintained comparison helps investigate an explanation. The labor, land, records, and organization required to preserve that explanation have costs and possible benefits of their own.

The long record also makes a difficulty visible. An experiment cannot remain useful simply by refusing to change. Weeds, soil acidity, new varieties, and changing farming practices can alter the question a comparison answers. Rothamsted's account of its experiments describes considered modifications alongside preserved treatment differences.[176]

By the late 1940s, some plots had become so acidic that comparing two nitrogen fertilizers no longer answered the intended question. In the separate Agdell experiment, acidity favored a disease that

depressed turnip yields; the experiment required extensive modification in 1951. Researchers applied chalk to correct acidity. Broadbalk later received shorter-strawed wheat varieties and a revised cropping arrangement in 1968. Preserving a useful experiment required changing what happened on the ground.[176]

Repeated applications of manure and fertilizer altered the soils themselves, leaving different stores of nutrients and organic matter. Later investigations could study those reserves and their effects. Yesterday's treatment had become part of today's condition. A difference after many years can reflect the accumulated consequences of earlier management as well as the immediate application. Comparing this year's input, a sustained practice, or a change from the inherited state asks different questions.

A field trial compares treatments with accumulated histories. It cannot erase their earlier development. Likewise, evaluating available technologies and estimating how different early opportunities would have changed their development are different questions, with different evidence requirements.

There is no timeless instruction here to hold everything fixed. A researcher must know which difference is being maintained, what else has changed, and whether the resulting observation still bears on the intended question. A familiar treatment name cannot guarantee an unchanged experiment.

AI evaluation faces a similar continuity problem. An unchanged benchmark can become a precise answer to an obsolete question. Revising it only when a favored system performs badly can undermine comparability. Documented changes are part of the evidence.

Broadbalk's plan distinguishes cropping and management histories across treatment strips. The name alone does not identify the ground, treatment, or period producing an observation. The same discipline applies to software: a service label can survive changes in models, interfaces, and review procedures. Version history helps separate those changes from differences in assigned work.

The fields yield crops and measurements: harvests, soil conditions, and changes across treated plots. Maintaining those comparisons is part of Rothamsted's productive work. Keeping an experiment going can make questions answerable that an ordinary harvest would leave unresolved.

Economics studies how resources produce goods and services, how incentives influence behavior, and how uncertainty affects choices. The arrangements that let alternatives develop interpretable records deserve the same attention. Losing those arrangements changes what an economy can know about its own possibilities.

A second use for the harvest

Rothamsted's archived samples can be examined with methods unavailable when they were collected. The record includes physical material, not only numbers in a table. A later researcher can sometimes ask that material a new question.[177]

Sarah Bearchell and colleagues took small portions of old leaves and stems from archive containers, powdered them using liquid nitrogen, and extracted DNA. They looked for short DNA sequences to accommodate damage in the old material. Their 2005 study examined two wheat pathogens in samples spanning 1844–2003. Gaps in the archive required combining material from two plots, after checks of comparability. Changes in the pathogens' relative abundance were associated with sulfur-dioxide emissions. The researchers considered other correlated factors; they had not randomized atmospheric pollution.[178]

The useful surprise is the question made possible. Harvest material retained for one scientific purpose could contain evidence relevant to another. The archive connected a modern laboratory method with a sequence of older observations. It did not manufacture those earlier samples after discovering that they would be useful.

Someone designing a record for today's decision may not know every future question. Keeping suitable material and documentation can make later investigation possible, as Rothamsted's samples did.

That potential benefit must still justify the resources used to maintain the archive.

A preserved sample can supply another measurement of a harvest that occurred. It cannot supply the harvest from a treatment never applied to that ground. Material preserves observations from one history; maintained treatment contrasts create evidence from differences in what was done.

Digital records likewise differ in what they preserve. Requests, versions, outputs, corrections, and checked outcomes can support more questions than a success total. They still cannot invent an unobserved alternative's response to a task.

Neither statement is a reason to collect everything. Residents' records can contain sensitive information. Storage, documentation, security, and authorized access consume resources. The institution must choose what to preserve for justified purposes and how to protect it. A comparison worth maintaining may need a carefully limited record rather than unrestricted retention.

Preservation has an option value and a cost. Retention policies should specify permitted uses, provenance, exclusions, and limits on later claims while protecting the people whose data make investigation possible. The future researcher is one beneficiary among others.

The institutional task is to make those tradeoffs reviewable. A preservation decision should identify the anticipated use, likely costs, permissions, review date, and consequences of deletion. Some judgments will prove mistaken as technology changes. A documented reason for the original choice gives later decision makers a basis for revision. At Rothamsted, the connection between retained material and recorded management allows new questions to reach backward. In software procurement, a comparable ambition is a protected record that remains intelligible after an interface, model, or contractor changes. Neither ambition requires promising that every future question will be answerable.

Aged material requires suitable methods and checks on what the measurements mean. Old procurement logs present a similar problem: missing entries, changing definitions, and forgotten selection rules can obstruct interpretation. A long history helps when later investigators can understand how it was made.

A final success label can discard the original input and the correction needed to diagnose a failure. Retaining those records may enable re-evaluation, subject to reliable checks, relevance, privacy, and reproducible software versions. A useful archive preserves enough context for a justified question.

Archives, natural variation, additional measured outcomes, and designed experiments can all supply evidence about an alternative. Each route depends on a connection between the observations available and the question being asked.

The arrangement makes a difference

Ronald Fisher's 1926 paper on field experiments began with a familiar agricultural difficulty: apparently similar ground can yield differently. A treatment's success cannot be assessed solely by weighing the crop accurately. The arrangement of the comparison affects what the difference means.[179]

His discussion moved from replication to the way plots were allocated. A collection of treatment labels arranged systematically could produce a misleading estimate of error. Random arrangements, including appropriate restrictions that the analysis respects, could support a different inferential claim. The method depended on how the experiment was generated, rather than the appearance of its final diagram alone.

Broadbalk's original plots were not laid out as a modern randomized trial. Its evidence comes from maintained treatment contrasts, subsequent changes, and accumulated observations. Fisher's paper concerns experimental design and the uncertainty calculations appropriate to it. Interpreting each record depends on how its comparisons were made.

The same design issues recur in service evaluation. Different task difficulty, inconsistent judgment, or a shared faulty reference can distort apparently balanced comparisons. Repetition alone does not repair a common measurement error.

The remedy is more specific than asking for more data. Which requests entered? How were they allocated? Which outcomes were independently checked? Which service version produced them? What could the system retain or learn between requests? The answers describe the experiment that the analyst actually has.

Fisher's question about apparently similar ground remains useful for economics: what would justify attributing an observed difference to a treatment rather than the allocation process? Designed experiments, credible existing variation, and structural restrictions offer different routes. Counting an outcome accurately does not identify its cause.

Experimental design, causal inference, structural modeling, sequential learning, and partial identification already provide ways to study counterfactual questions. The comparison budget focuses attention on one part of the evidence supply: the accumulated chances that an allocation goes outside the current favorite.

Under the model's conditions explained in Chapter 3, those chances limit how precisely productive contribution can be recovered from the allocation history. A researcher can then ask whether the real market supplies other observations that overcome that limit.

An application should start with the available records. Does the analyst know the assignment probabilities? Are customer outcomes, independent tests, or other measurements available? Those observations determine whether the model describes the actual information problem.

An empirical study need not begin by claiming that an industry's lifetime budget is finite. A finite record cannot establish that conclusion merely because comparison appears to be slowing. It can investigate a defined horizon, assess uncertainty about the allocation

mechanism, and compare feasible ways of producing evidence. That would already be useful progress.

Simulation offers a narrower test. Build an artificial market whose rules are known, run the proposed estimator, and check whether it recovers what was put in. Tim Morris, Ian White, and Michael Crowther explain why a simulation study must state its aims, data-generating mechanisms, and measures of performance. A method can work in that artificial world while failing in a real market whose feedback rules are different.[180]

A credible account of an allocation process records eligibility, assignment, internal testing, rejected cases, and checked outcomes. Known probabilities should have a documented basis; estimated probabilities need uncertainty. Completed-work shares cannot substitute for them.

An experiment observing paired outcomes contains more information than an allocation-winner record. Its number of cases is not automatically a comparison budget. Applying an allocation-only limit while ignoring checked responses misstates the observation scheme. Designing additional informative outcomes can change what is learnable even when production remains concentrated.

What an empirical test would need

Kasy and Sautmann's agricultural recruitment application supplies a documented example of matching experimentation to policy choice. Six recruitment options were tested through an adaptive calling design with 10,000 numbers over seventeen waves. The outcome was completion of five recruitment questions, rather than later agricultural income.[181]

The call experiment measured recruitment. To assess the agricultural service's effects on yields or income, researchers would need to follow those outcomes as well. What gets recorded determines which decisions the evidence can support.

A useful study would distinguish assignment chances, observed outcomes, and the effect being estimated. It would ask what can be learned from allocations alone, how checked outcomes change the answer, and how sensitive the conclusion is to plausible feedback rules.

Repeated records from one event, common shocks, and shared evaluation errors can make observations dependent. Recording versions, exclusions, and failures helps later investigators judge how much independent evidence the record contains.

Feasibility includes power to detect a consequential difference, the cost of supervision, privacy protection, and what action the result can change. A spending limit controls expenditure; it does not by itself establish adequate information or net social value.

The resulting report should connect methods, uncertainty, costs, and decisions. Negative and missing results need to remain visible. A successful evaluation can favor an existing arrangement, recommend a narrower change, or show that further investigation is not worth its cost.

The permissions behind the experiment

The design still needs an institution capable of carrying it out. Requests may be controlled by the buyer, service interfaces by suppliers, and verification resources by another organization. Having each ingredient somewhere does not mean the usable experiment has been assembled.

Kenneth Arrow's 1963 study of medical care supplies an earlier institutional argument. The patient, in his account, cannot inspect the service fully before consuming it and depends on knowledge the physician possesses. Expectations that the doctor should put the patient's welfare first help organize this unequal relationship. Arrow interpreted medical institutions partly as responses to uncertainty about illness, treatment, and information. He also warned that an attempted remedy can introduce other obstacles. Institutions deserve

investigation as arrangements for managing uncertainty, with consequences that require appraisal.[182]

James Buchanan and Yong Yoon analyze an anticommons: multiple rights to exclude can leave a resource underused. Their parking example requires complementary permits from separate holders.[183] Information can face a related institutional problem when useful records, interpretation, and permission to evaluate are controlled separately.

Contracting can sometimes assemble those complements. Commitment, verifiable delivery, compensable costs, and financing determine whether a feasible agreement exists. A nominal right of access does not provide the remaining inputs for free.

This should make us wary of explaining every refusal as a plan to suppress competition. A holder may face genuine privacy, security, staff, or productive costs. The useful question is whether those costs exceed the attainable benefit, whether the beneficiary can compensate the provider, and whether the arrangement can be enforced. A refusal can be privately rational without settling the wider economic case.

It should also make us wary of assuming that a mandate supplies every missing input. A rule can require access while leaving nobody responsible for defining the outcomes or paying the evaluator. Another can fund a laboratory that lacks permission to observe the relevant tasks. A useful institutional design assembles the complements needed for the stated comparison.

Rothamsted's maintained fields and archive offer a historical reference for this less glamorous work. Treatments, measurements, material, and documentation must remain connected across time. An isolated bag of grain and a beautifully organized treatment register are different things. Their joint usefulness depends on retaining the connection between them.

Institutional alternatives include existing qualified evaluators, secure contracts, shared infrastructure, and independently verified private tests. Their costs and possible common failures should be compared before establishing a new public body.

An experiment must permit a result favoring the incumbent. An institution whose survival requires the challenger to win has compromised its evidentiary purpose.

Credibility also requires stable criteria, recorded revisions, and relevant rights of challenge. Independent review can test whether the evidence supports a conclusion; the institution acting on it remains responsible for its decision.

An existing internal test may already resolve the buyer's uncertainty. Paying for a duplicate study would then spend resources without changing the decision. A missing alternative warrants further investigation when a feasible comparison could supply useful evidence.

When the leading arrangement fails

After a leading system fails, another team needs resources to investigate, permission to test a different approach, and freedom to report what it finds. Public computing can support that work while admission rules or restrictions on publication obstruct it. What matters is whether a workable alternative can get through those steps and into use.

A failure can be isolated by independent tests or reproduced across many services sharing a common checklist and restricted evaluation route. The number of brands says little about which situation applies. Robustness concerns the practical ability to expose the failure and develop a workable alternative.

An available rival is little help if a buyer cannot transfer the records needed to test it. A dormant supplier's records can shorten recovery when they remain usable. Existing private tests or a cheaper substitute may already provide the answer. Waiting time, permissions, expense, and lost productive capacity determine whether a new access arrangement improves on those routes.

Subsidized computing that produces unusable results still consumes money, equipment, and skilled work. Official checks can

spread a common error or suppress a finding that would embarrass the sponsor. A costly application process can exclude the team that needs the resources. The value of keeping an alternative available has to exceed those costs, including displaced private experiments and slower initial development.

Faster recovery in one country could reflect better equipment, a deeper pool of engineers, or other institutions alongside the access policy. Those differences belong in the explanation. So do the applicants turned away and the findings kept from public view. A list of successful recipients leaves out the people who never reached a test.

What an economist should report

An economic account of a widely used service should establish its costs, accuracy, and reach, then explain what its operating arrangement permits participants to learn about alternatives. Productive improvement and comparative learning can move together or apart.

An informative report would state its target. Is it estimating the value of a supplier's recommendation, the effect of access on entry, the effect of a training design on workers, or the effect of an input on winning rewards? These questions can be connected, but a success on one is not an answer to all the others.

It would identify the observation process. The routing rule, the outcomes retained, internal experiments, and external tests can all matter. A conclusion drawn from allocation winners alone should say so. If more information exists, the analyst should explain what changes when it is included.

It would report the relevant uncertainty about the mechanism. Two histories with the same market shares may have different assignment processes. Two models can forecast the observed process similarly while disagreeing about how an intervention would work. Precise estimates under one maintained explanation should not conceal uncertainty about that explanation.

Blackwell and Dubins show how conditional forecasts can merge under specified assumptions. In the allocation model, that agreement can coexist with unresolved contribution. Accurate prediction of the next winner does not automatically identify the input effect behind the process.[184]

Forecasts of workload and inference about a different technology answer different questions. Each can be valuable when its evidence and assumptions fit the intended use.

It would also distinguish statistical information from decision value. A measured contrast could be precise but economically trivial. Another could remain uncertain while affecting a consequential choice. Blackwell's decision perspective supplies a foundation; collection costs, interpretation, incentives, and legal feasibility still determine whether obtaining the evidence is worthwhile.[185]

A precise but tiny difference may leave the purchase unchanged; uncertainty about a costly failure may justify a targeted test. The value of more comparison depends on the decision it can improve.

Finally, the report would state which counterfactual it is evaluating. A shadow replay holds underlying effects, initial conditions, and response rules fixed while allowing the realized history to differ. A new access rule or training policy changes the institution. Its estimated effects depend on how people would respond under the changed rules.

A useful procurement report can be brief: what question did the buyer ask, which services were compared, what happened, what remains uncertain, and what decision follows? The methods and records should be available for anyone authorized to check them.

Economics should test the rules

An economist could begin with one purchase. Record which suppliers were eligible, how the buyer selected them, what the buyer learned about their work, and whether another test changed the next decision. Establish the assignment probabilities from the actual process, with uncertainty where they must be estimated. Compare the information in the award record with the information added by checked outcomes. That would give the comparison budget a specified empirical setting instead of treating market shares as its measure.

The next step is to compare practical ways of learning more. A buyer could extend the existing contract, retain more useful records, evaluate an available rival on suitable tasks, or purchase an independent assessment. Costs, shared errors, and what each supplier has already learned belong in the design. A fresh group of tasks does not erase shared state or make observations independent by declaration.

Then examine the rules supplying access. If a rival cannot reach the trial, identify the missing permission or resource. Change a bounded part of the arrangement where feasible, observe what new work becomes possible, and assess whether the result improves a consequential choice. Count private testing that public provision displaces. A remedy that adds expense without adding usable evidence should be revised or ended.

Hayek described competition as a way to discover facts that were not known or used beforehand.[186] Economics should make the institutions performing that discovery a subject of sustained investigation. The relevant outcome includes whether a capable alternative can obtain a credible test, whether its findings reach the people who need them, and whether those people can act on what they learn.

For AI, computing capacity, power connections, skilled staff, and working systems are real constraints. An applicant with no feasible development route cannot be made competitive by a paper right to access. But a team capable of a bounded evaluation need not

reproduce an entire frontier training operation. The economist should identify the smallest workable arrangement that answers the useful question, then compare its benefits with its full costs.

These studies would give policy a means to improve. Publish the allocation rules, relevant outcomes, and grounds for changing a decision so that other investigators can check them. Retain room for a result showing that the incumbent remains best, that a private contract already supplies the necessary evidence, or that a public intervention costs more than it achieves. A commitment to discovery has to permit those findings too.

The record we choose to leave

At Rothamsted, someone had to preserve the samples, label them, and keep the treatment records intelligible. Later researchers could put new questions to that material. An archive that had been discarded could not have answered them.

An economy makes a similar choice whenever it allocates another opportunity. It can support the established supplier and also preserve an independent route through which someone else could demonstrate a better method. It can distribute a share of capital's returns more widely and give people resources with which to attempt something new. Or it can let control over the next attempt accumulate with control over the last reward.

The comparison budget gives that choice a precise statistical dimension. Under the allocation model's conditions, a finite lifetime total of chances outside the current favorite can leave contribution unresolved even as the reward record grows indefinitely. The practical concern is the opportunity to generate the missing evidence. A larger file of the incumbent's awards cannot always supply it.

At network scale, that difficulty can begin before there is a rival to count. A possible supplier may need customers, resources, and independent evaluation that the prevailing arrangement gives it no feasible way to obtain. The incumbent's returns can keep financing expansion while the challenge that might test those returns never

develops. What looks like an indefinitely repeated endorsement can partly reflect a closed route to comparison. To preserve discovery, policy has to examine the conditions for creating an alternative as well as the treatment of alternatives already established.

An economy committed to independent discovery has reason to reward useful work while preserving routes to challenge it. A substantial return can finance the next invention. Giving its recipient authority over whether anyone else can attempt one adds a different power, with different consequences. The payment and the power should be judged separately.

Start with a gatekeeper controlling an indispensable route to evaluation. If closing it threatens a lasting loss of useful comparison, protecting access preserves the possibility of discovering which supplier deserves the next contract. Documented interfaces, voluntary portability, secure evaluation access, and independently reviewed terms can make an enforceable duty concrete. The obligation should follow the actual obstruction, with privacy, security, resource costs, and investment included in its design. When useful private evidence already answers the affected decision, duplicating the experiment may buy nothing.

Act while the alternative can still develop. A team cannot be required to demonstrate years of learning before obtaining the customers through whom that learning would occur. Before closure becomes difficult to reverse, a credible attempt needs a route to its first test. Keeping that route open lets customers judge the result without promising the entrant a contract or permanent support.

Where restrictive conduct prevents valuable learning and imposes a cost on others, leaving that cost outside the decision rewards too much restriction. An access duty may prevent the loss. A corrective charge may address a remaining external harm. Its justification depends on the missing comparison, the decisions affected, and what feasible replacement would achieve. A large profit supplies neither a damage meter nor a reason to ignore a demonstrated obstruction.

Opening a gate accomplishes little for someone unable to afford the attempt. Capital income and inheritance taxation can help finance independent experimentation and broader ownership of productive assets. A shared computing facility, a credible first demonstration, or a stake in a broadly held fund can put a different means of economic participation within reach. Such financing has to preserve useful investment and account for avoidance, administration, and the benefits actually delivered. It need not wait for an impossible audit of every taxpayer's exact contribution.

Ownership and entry require attention together. A dividend can spread income while leaving the recipient dependent on one supplier's permission. Public computing can support another development route while its admission rules exclude inconvenient challengers. Broader capital ownership needs independent evaluation, published eligibility rules, and appeal alongside it. Government should be as open to a challenge to its preferred answer as a private gatekeeper is required to be.

MusicLab gave the same recordings more than one opportunity to develop a following. Lockdown finished first in one world and fortieth in another.[187] Its success was real in one history. Its neglect was real in the other. Neither chart could tell the whole story by itself.

We will rarely have both charts for an economy. We still have to choose how its rewards are taxed, which rights accompany ownership, and what opportunities remain available to people whose work has not yet acquired a record. The choice can rest on shared aims and practical evidence about institutions. It need not rest on a claim that today's distribution has finally established who deserves everything they received.

A successful economy needs reasons to keep producing and the means for other people to begin. Protecting investment, financing useful challenges, and limiting the power to close an essential gate serve those purposes together. Success can earn a reward. The next opportunity must remain open to a different answer.

The next opportunity

At Rothamsted in England, wheat grows in an experiment begun in 1843. Across its plots, researchers have compared mineral fertilizers, manure, and no fertilizer. They have kept records and saved samples of crops and soil. Generations of researchers have inherited both the harvests and the differences that help explain them.[188]

Give every plot the same treatment, and the field might produce more wheat. It would also stop producing that continuing comparison. An untreated plot can look like wasted ground when this year's crop is all that counts. Its value changes when we count what it helps people learn.

Someone has to keep making room for it. Soil must be tended, samples preserved, records understood. The experiment has changed as agricultural methods and scientific questions have changed. The knowledge available to later researchers depends on work that earlier generations chose to do.

A market leaves a different kind of field behind it. Orders, wages, investments, and download counts record where activity went. They can tell us a great deal about the businesses and people who received those opportunities. The rejected application usually leaves a thinner file.

For a beginner, an assignment can bring pay, practice, a reference, and something to show the next employer. Turn someone down for lack of experience, and the decision can help perpetuate the very lack it cites. Another year of work by the experienced applicant will not reveal what the beginner would have learned. To find that out, someone has to offer a beginning.

The same difficulty can arise when software does the work. A company might send thousands of tasks to AI agents using one model and build an impressive record of completed jobs. To discover

whether another system would perform better, it needs suitable tasks, access to that system, and a way to compare results. Another million completions by the incumbent do not automatically provide that comparison.

A rival listed on a website has permission to exist. Getting its work into an evaluation, its product in front of customers, or its findings into a decision is a further step. When a gatekeeper controls those routes, it can help decide which alternatives acquire evidence and which remain promises. That is where epistemic monopoly reaches into everyday economic life.

We can spend heavily on producing answers while spending little on discovering whether we are asking the right system. We can reward experience while closing the routes through which someone else could acquire it. A busy economy can keep repeating a choice without getting much better at explaining it.

The next AI company should have a workable route to demonstrate that it can do better. A worker rejected by one platform should have somewhere else to establish what they can do. A researcher should be able to investigate a dominant system without giving its owner authority over the findings.

If the only route to a credible test belongs to the company being challenged, who gets to decide whether the challenge is heard? Leaving that decision entirely with the owner gives it authority over the evidence of its own superiority. An enforceable access duty can keep a useful route open. The rival may need a secure interface, the worker an independent assessment, the researcher access to a credible test. Those arrangements should preserve privacy, safety, and useful investment while making the challenge possible. Their costs belong in the decision, alongside the value of finding out.

We should act while an alternative still has a route to development. Asking for a finished rival after closing the means of building one makes success a condition for obtaining the opportunity to succeed. A credible attempt should be able to reach a test before its absence becomes another reason to dismiss it.

A team still needs resources after the gate opens. Capital income and inheritance taxation can help finance independent experimentation and broaden ownership of productive assets. A shared computing facility can let that team attempt a different design. A grant can pay for a first demonstration. A share in productive capital can let more people benefit from the work machines perform. Funding should follow usable opportunities and results, with costs, investment effects, and displaced private efforts counted. An economy that relies on discovery has a reason to help pay for the means of making it happen.

When a restriction prevents valuable learning, its author should face the cost imposed on others. Opening the route may accomplish more than collecting money after closing it. Financing common capabilities need not wait for the treasury to measure the unearned portion of every fortune.

Public institutions must remain open to challenge themselves. A laboratory should be able to publish an inconvenient finding. A team refused access should have a meaningful appeal. Money for discovery should leave room for different methods, and for evidence that the established supplier remains best. The purpose is to let what happens in the test change what happens afterward.

At MusicLab, that possibility was built into the experiment. The same forty-eight recordings entered separate worlds. Listeners made choices, those choices became visible to later listeners, and the charts developed differently. Lockdown finished first in one world and fortieth in another.[189]

Imagine having only the first chart. We could play the winning recording, examine its qualities, and write a persuasive account of why it deserved to lead. With only the other chart, we could do the same for its failure. The second history makes both explanations harder to deliver with such confidence.

We can admire an achievement without treating its reward as the final word on everyone else's ability. The applicant, the business, or the idea with a thin record may have needed an opportunity before

there could be much to record. A society that wants to discover what people can do has to make some of those beginnings possible.

Most shadow futures will leave no second chart. We cannot recover every missed opportunity or run every possible experiment. We can choose rules that reward useful investment, finance credible alternatives, and prevent an essential gateway from becoming its owner's permanent power over other people's beginnings. Agreement about those rules can guide us even when agreement about each person's exact contribution remains beyond our evidence.

Lockdown's other chart existed because someone made room for another history. The recording was ready. The opportunity had to be built.

An economy should reward its winners without giving them ownership of everyone else's chance.

Notes

The other life of a song

1. Matthew J. Salganik, *Bit by Bit: Social Research in the Digital Age*, online open review edition, section 4.6.1, "Create zero variable cost data," especially figures 4.20–4.21 and the paragraph reporting Lockdown's rankings. Author's account.

2. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311, no. 5762 (2006): 854–856, doi:10.1126/science.1121066. Design, participant count, and zero initial downloads: p. 854. Inequality, variation across worlds, and the continuing role of independently assessed appeal: pp. 854–855, figures 1–3. Original article and supporting materials.

3. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *Proceedings of the National Academy of Sciences* 111, no. 19 (2014): 6934–6939, doi:10.1073/pnas.1316836111. Kickstarter sample and random assignment: pp. 6934–6935; 39 versus 70 percent receiving subsequent third-party funding: p. 6935, figure 1. Diminishing marginal returns to larger initial treatments: pp. 6936–6937, figure 3. Full paper hosted by the European University Institute.

4. Robert K. Merton, "The Matthew Effect in Science," *Science* 159, no. 3810 (1968): 56–63, especially pp. 56–57, doi:10.1126/science.159.3810.56. Original article. Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58, no. 3 (1950): 211–221, especially pp. 211–213, doi:10.1086/256940.

5. Steven Johnson, "The Genius of the Tinkerer," *The Wall Street Journal*, September 25, 2010, adapted from *Where Good Ideas Come From: The Natural History of Innovation*, archived print view, PDF p. 2, paragraphs introducing Kauffman and the expanding-house analogy.

6. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12, no. 4 (2020): 328–359, doi:10.1257/pol.20190086, abstract; June 9, 2017 author working paper, pp. 1–4 and 6–8.

7. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, "Generative AI at Work," *The Quarterly Journal of Economics* 140, no. 2 (2025): 889–942, doi:10.1093/qje/qjae044.

Abstract and Sections I, III, V.B, and VI. Published article; November 2024 working paper, pp. 1-3.

8. Joel Becker, Nate Rush, Beth Barnes, and David Rein, "Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity", 2025, abstract and pp. 1–5; Joel Becker, Nate Rush, Tom Cunningham, David Rein, and Khalid Mahamud, "We Are Changing Our Developer Productivity Experiment Design", METR, February 24, 2026, sections "Wider adoption of AI has made it more difficult to measure task-level productivity" and "Other means of measuring productivity."

9. Rothamsted Research, "Why Do We Make Changes to the Long-Term Experiments at Rothamsted?", January 17, 2024, opening paragraphs and the Broadbalk discussion. Institutional account.

The evidence success leaves behind

10. Thomas Blake, Chris Nosko, and Steven Tadelis, "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment," *Econometrica* 83, no. 1 (2015): 155–174, doi:10.3982/ECTA12423. See the authors' May 2014 NBER working paper 20171, printed pp. 1–5 and the brand-keyword experiment in Section 3. Full working paper.

11. Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58 (1950): 211–221, especially pp. 211–213, doi:10.1086/256940; Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, especially pp. 56–57, original paper; W. Brian Arthur, "Competing Technologies, Increasing Returns, and Lock-In by Historical Events," *Economic Journal* 99 (1989): 116–131, pp. 116–118 and 128, doi:10.2307/2234208.

12. Albert-László Barabási and Réka Albert, "Emergence of Scaling in Random Networks," *Science* 286, no. 5439 (1999): 509–512, especially pp. 509–511, doi:10.1126/science.286.5439.509, author-hosted article.

13. Martin Erlic, "Shadow Futures: Contribution Uncertainty and the Self-Reinforcing Market," SSRN 6003994.

14. Salganik, Dodds, and Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006): 854–856, especially pp. 854–855, original article. Lockdown's specific ranks are reported in Salganik's *Bit by Bit*, open review edition, section 4.6.1.

15. George A. Akerlof, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism," *The Quarterly Journal of Economics* 84, no. 3 (1970): 488–500, especially pp. 488–490. Original paper.

16. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, “Field Experiments of Success-Breeds-Success Dynamics,” *PNAS* 111 (2014): 6934–6939, especially 6934–6937. doi:10.1073/pnas.1316836111.

17. Line Chloé Le Goff and Philippe Soulier, “Parameter Estimation of a Two-Colored Urn Model Class,” *International Journal of Biostatistics* 13 (2017), article 20160029, doi:10.1515/ijb-2016-0029; author preprint, version 2, Sections 3.2–3.3, pp. 14–15.

18. Steven Johnson, “The Genius of the Tinkerer,” *The Wall Street Journal*, September 25, 2010, adapted from *Where Good Ideas Come From: The Natural History of Innovation* (Riverhead Books, 2010), archived print view, PDF p. 2; Stuart A. Kauffman, *At Home in the Universe: The Search for the Laws of Self-Organization and Complexity* (Oxford University Press, 1995); Kauffman, “The Adjacent Possible”, *Edge*, November 9, 2003.

19. David Blackwell and Lester Dubins, “Merging of Opinions with Increasing Information,” *Annals of Mathematical Statistics* 33 (1962): 882–886, main theorem, p. 883, doi:10.1214/aoms/1177704456; Hao Zhang, “Inconsistent Estimation and Asymptotically Equal Interpolations in Model-Based Geostatistics,” *Journal of the American Statistical Association* 99 (2004): 250–261, pp. 250–253, published article.

20. Thijs Bol, Mathijs de Vaan, and Arnout van de Rijt, “The Matthew Effect in Science Funding,” *PNAS* 115, no. 19 (2018): 4887–4890, especially pp. 4888–4889, figures 1 and 3, doi:10.1073/pnas.1719557115. Full paper.

21. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, “Generative AI at Work,” *The Quarterly Journal of Economics* 140 (2025): 889–942, abstract and Sections I, V.B, and VI, doi:10.1093/qje/qjae044.

22. Joel Becker, Nate Rush, Tom Cunningham, David Rein, and Khalid Mahamud, “We Are Changing Our Developer Productivity Experiment Design,” METR, February 24, 2026, sections on wider adoption and alternative measurement designs, research team's account.

23. Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Book I, Chapter III and Chapter X, Part II, original text; International Labour Organization, *World Employment and Social Outlook 2021*, Chapter 4, printed pp. 182 and 192, report.

24. Zahra Meghani, “Regulations Matter: Epistemic Monopoly, Domination, Patents, and the Public Interest,” *Philosophy & Technology* 34 (2021): 1449–1474, introduction and Section 6, article.

25. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, “How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree,” *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record. For the historical and foreclosure analysis, see the June 2017 author working paper, pp. 1–4 and 7–8, full text.

26. U.S. Department of Justice and Federal Trade Commission, *2023 Merger Guidelines*, overview and Guidelines 2, 3, and 10.

27. Rothamsted Research, “Why Do We Make Changes to the Long-Term Experiments at Rothamsted?”, January 17, 2024, opening and Broadbalk discussion, institutional account.

The winner and the explanation

28. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, “Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market,” *Science* 311 (2006): 854–856, especially pp. 854–855 and figures 1–3, full article. Lockdown's ranks are reported in Salganik's *Bit by Bit*, open review edition, section 4.6.1.

29. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, “Field Experiments of Success-Breeds-Success Dynamics,” *PNAS* 111 (2014): 6934–6939, doi:10.1073/pnas.1316836111. Intervention design and four settings: pp. 6934–6935; marginal returns: pp. 6936–6937 and figure 3; matching, sampling, and procedures: p. 6937. Full text.

30. Lev Muchnik, Sinan Aral, and Sean J. Taylor, “Social Influence Bias: A Randomized Experiment,” *Science* 341, no. 6146 (2013): 647–651, doi:10.1126/science.1240466. Design, site ordering, and sample: pp. 647–648; first-viewer effects and final scores: pp. 648–649, figure 1; negative-treatment correction: p. 649. Full paper.

31. Robert K. Merton, “The Matthew Effect in Science,” *Science* 159 (1968): 56–63, doi:10.1126/science.159.3810.56. Forty-first chair and named examples: p. 56; recognition as access to resources: p. 57; credit for Harriet Zuckerman's interviews: opening. Original article.

32. Thijs Bol, Mathijs de Vaan, and Arnout van de Rijt, “The Matthew Effect in Science Funding,” *PNAS* 115 (2018): 4887–4890, doi:10.1073/pnas.1719557115. Threshold design: p. 4888; participation, output measures, and further funding excluding the initial grant: p. 4889, figures 1–3; institutional discussion: p. 4890. Full paper.

33. Yang Wang, Benjamin F. Jones, and Dashun Wang, “Early-Career Setback and Future Career Impact,” *Nature Communications* 10 (2019), article 4331, doi:10.1038/s41467-019-12189-3. See “Future career impact,” analyses of screening, and the Discussion. Full article.

34. Vincent Traag, Emer Brady, Philippe Vincent-Lamarre, Flavie Bidel, Cindy Lopes-Bento, Jens Peter Andersen, and Carter Bloch, “The Matthew Effect and Early-Career Setbacks in Research Funding—a Replication Study,” *eLife*, reviewed preprint version 1, October 13, 2025, doi:10.7554/eLife.109042.1. Abstract, Sections 2-4, and Appendix A. Full reviewed preprint.

35. Armen A. Alchian, “Uncertainty, Evolution, and Economic Theory,” *Journal of Political Economy* 58 (1950): 211–221, doi:10.1086/256940, pp. 213–214.

36. George A. Akerlof, “The Market for ‘Lemons’: Quality Uncertainty and the Market Mechanism,” *The Quarterly Journal of Economics* 84 (1970): 488–500, especially pp. 488–490, full paper.

37. Thomas Blake, Chris Nosko, and Steven Tadelis, “Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment,” *Econometrica* 83 (2015): 155–174, doi:10.3982/ECTA12423. See the May 2014 NBER working paper, pp. 1–5, Section 3 and Section 4.1, printed p. 11, full working paper.

38. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, “How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree,” *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record. For the historical and foreclosure analysis, see the June 2017 author working paper, pp. 1–4 and 7–8, full working paper.

39. Matthew J. Salganik and Duncan J. Watts, “Leading the Herd Astray: An Experimental Study of Self-Fulfilling Prophecies in an Artificial Cultural Market,” *Social Psychology Quarterly* 71 (2008): 338–355, doi:10.1177/019027250807100404. Inversion design and the two named songs: pp. 341–343, table 2; appeal, recovery, and downloads: pp. 349–351, figures 8–9. Original article.

The comparison budget

40. Ashok Chandrashekar, Fernando Amat, Justin Basilico, and Tony Jebara, “Artwork Personalization at Netflix”, Netflix TechBlog, December 7, 2017, sections “Challenges,” “Contextual bandits approach,” and “Performance evaluation.”

41. Lihong Li, Wei Chu, John Langford, and Robert E. Schapire, “A Contextual-Bandit Approach to Personalized News Article Recommendation,” *Proceedings of WWW 2010*: 661–670, doi:10.1145/1772690.1772758. See arXiv version 2 (2012), Sections 2.1, 4, and 5.2.1, full paper.

42. Karthik Rajkumar, Guillaume Saint-Jacques, Iavor Bojinov, Erik Brynjolfsson, and Sinan Aral, “A Causal Test of the Strength of Weak Ties,” *Science* 377 (2022): 1304–1310, doi:10.1126/science.abl4476. See design and main results, pp. 1304–1309; tie-strength definitions, pp. 1305–1306; and interference discussion, pp. 1309–1310, full paper hosted by MIT.

43. Robin Pemantle, “A Survey of Random Processes with Reinforcement,” *Probability Surveys* 4 (2007): 1–79, doi:10.1214/07-PS094. See Section 2.1, especially pp. 3–5, for the classical urn and its variants, full survey.

44. Roberto Imbuzeiro Oliveira, “The Onset of Dominance in Balls-in-Bins Processes with Feedback,” *Random Structures & Algorithms* 34 (2009): 454–477,

doi:10.1002/rsa.20261; author preprint, introduction and Theorem 1, pp. 1–2. Thomas Gottfried and Stefan Grosskinsky, “Asymptotics of Generalized Pólya Urns with Non-Linear Feedback,” *Electronic Journal of Probability* 29 (2024): 1–56, doi:10.1214/24-EJP1157; arXiv version 3, Sections 2.1–2.3, especially Theorem 2.2.

45. Maximilian Kasy and Anja Sautmann, “Adaptive Treatment Assignment in Experiments for Policy Choice,” *Econometrica* 89 (2021): 113–132, doi:10.3982/ECTA17527, pp. 113–114 and Section 6, pp. 122–123, full published paper; authors' November 10, 2021 correction, especially corrected Theorem 1.

46. Yaneer Bar-Yam, “From Big Data to Important Information,” *Complexity* 21, supplement 2 (2016): 73–98, doi:10.1002/cplx.21785, abstract and Sections I–II; author preprint.

47. Ron Kohavi, Alex Deng, Brian Frasca, Toby Walker, Ya Xu, and Nils Pohlmann, “Online Controlled Experiments at Large Scale,” *Proceedings of KDD 2013*: 1168–1176, doi:10.1145/2487575.2488217; author version, introduction and Section 1.1, pp. 1–2.

Shadow futures

48. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, “Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market,” *Science* 311 (2006): 854–856, full paper, pp. 854–855 and figures 1–3. Lockdown's specific rankings are reported in Salganik's *Bit by Bit*, Section 4.6.1.

49. Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond, “Generative AI at Work,” *Quarterly Journal of Economics* 140 (2025): 889–942, published article, abstract and Sections I and IV.

50. Line Chloé Le Goff and Philippe Soulier, “Parameter Estimation of a Two-Colored Urn Model Class,” *International Journal of Biostatistics* 13 (2017), article 20160029, doi:10.1515/ijb-2016-0029; author preprint, version 2, October 4, 2016, Sections 3.2–3.3, pp. 14–15, and Sections 4.1–4.2, pp. 19–20.

51. Lihong Li, Wei Chu, John Langford, and Robert E. Schapire, “A Contextual-Bandit Approach to Personalized News Article Recommendation,” *WWW 2010*, 661–670, author version 2, Sections 2.1, 4, and 5.2.1.

52. Shizuo Kakutani, “On Equivalence of Infinite Product Measures,” *Annals of Mathematics* 49 (1948): 214–224, doi:10.2307/1969123. Yu. M. Kabanov, R. Sh. Liptser, and A. N. Shiryaev, “On the Question of Absolute Continuity and Singularity of Probability Measures,” *Mathematics of the USSR-Sbornik* 33 (1977): 203–221, doi. Jean Jacod and Albert N. Shiryaev, *Limit Theorems for Stochastic Processes*, second edition (2003), Chapter IV, and S. S. Gabriyelyan (2011), author version 4, Theorem 1. Lucien Le Cam and Grace Lo Yang, *Asymptotics in Statistics*, second edition (2000), Chapter 2.

53. John Snow, *On the Mode of Communication of Cholera*, second edition (London: John Churchill, 1855), Part 3, especially the discussions of Lambeth's new supply, intermingled customers, and the inquiry into household supplies; full section transcribed by UCLA.

54. David Blackwell and Lester Dubins, "Merging of Opinions with Increasing Information," *Annals of Mathematical Statistics* 33 (1962): 882–886, doi:10.1214/aoms/1177704456, main theorem, p. 883.

55. Hao Zhang, "Inconsistent Estimation and Asymptotically Equal Interpolations in Model-Based Geostatistics," *Journal of the American Statistical Association* 99 (2004): 250–261, full published paper, introduction, Section 3, Theorem 2, Corollaries 1–2, and Theorem 3, pp. 250–253.

56. Thomas Blake, Chris Nosko, and Steven Tadelis, "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment," *Econometrica* 83 (2015): 155–174, May 2014 author working paper, introduction and Section 3.

57. Line Chloé Le Goff and Philippe Soulier, "Parameter Estimation of a Two-Colored Urn Model Class," *International Journal of Biostatistics* 13(1) (2017), article 20160029, Sections 2 and 3.2, doi:10.1515/ijb-2016-0029.

58. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record; June 2017 author paper, introduction and institutional background.

The histories that became infrastructure

59. Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Book I, Chapters I and III (pins, specialization, market extent and water carriage), and Chapter X, Part II (corporation and settlement restrictions). Original text, Project Gutenberg.

60. *United States v. United States Steel Corp.*, 251 U.S. 417 (1920), opinion, pp. 438–439.

61. P. W. Anderson, "More Is Different," *Science* 177, no. 4047 (1972): 393–396, opening page and Bell Telephone Laboratories affiliation. Published article.

62. W. Brian Arthur, "Competing Technologies, Increasing Returns, and Lock-In by Historical Events," *Economic Journal* 99 (1989): 116–131, full published paper, pp. 116–118 and 128.

63. John Bardeen, "Semiconductor Research Leading to the Point Contact Transistor," Nobel lecture, December 11, 1956, full lecture, pp. 318–321: research program, earlier theory, materials, complementary contributions and the failed field-effect proposal

directing surface research. For the December 1947 device, see the Computer History Museum's collection account, with original logbook and publication references.

64. U.S. Department of Justice, *United States Attorneys' Bulletin*, February 17, 1956, vol. 4, no. 4, full issue, Antitrust Division, p. 119.

65. Computer History Museum, "1952: Bell Labs Licenses Transistor Technology", symposium and manufacturing-visit account, with proceedings and collection references.

66. *Hush-A-Phone Corp. v. United States*, 238 F.2d 266 (D.C. Cir. 1956), November 8, 1956, opinion, pp. 267–269 and notes 8–12.

67. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record. See the June 9, 2017 author paper, introduction, institutional background, and empirical design/results.

68. Paul A. David, "Clio and the Economics of QWERTY," *American Economic Review* 75 (1985): 332–337, full paper, pp. 332 and 335–336. S. J. Liebowitz and Stephen E. Margolis, "The Fable of the Keys," *Journal of Law and Economics* 33 (1990): 1–25, author text, introduction and discussion of typing studies.

69. F. A. Hayek, "Competition as a Discovery Procedure," translated by Marcellus S. Snow, *Quarterly Journal of Austrian Economics* 5 (2002): 9–23, full translation, pp. 9–11. This is a 2002 translation of the 1968 Kiel lecture.

70. Sony Group, *Sony Group History*, Volume 1, Chapter 3, "Developing the Transistor Radio", Part 2.

71. Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, especially 56–57. Article.

72. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *PNAS* 111 (2014): 6934–6939, full paper, pp. 6934–6937.

73. Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58 (1950): 211–221, pp. 213–214.

74. AT&T, *Annual Report 1955*, original report scan, printed pp. 13 and 16, PDF pp. 15 and 18. Americus equipment and solar-power trials are described on p. 13; the company's account of integration, decree terms, and expected business consequences is on p. 16.

75. FCC, *Use of the Carterfone Device in Message Toll Telephone Service*, 13 FCC 2d 420 (1968), adopted June 26, 1968, decision, pp. 420–425, especially p. 424.

Who gets to be tested

76. Amit P. Mehta, *United States v. Google*, memorandum opinion, August 5, 2024, full court opinion, printed pp. 1–4.

77. Mehta, August 2024 opinion, pp. 115–117, especially paragraphs 337–339.

78. Mehta, August 2024 opinion, pp. 34–39, especially paragraphs 86–93.

79. Mehta, August 2024 opinion, p. 49, paragraphs 136–137.

80. Zahra Meghani, "Regulations Matter: Epistemic Monopoly, Domination, Patents, and the Public Interest," *Philosophy & Technology* 34 (2021): 1449–1474, published text, introduction and Section 6.

81. *National Family Farm Coalition v. EPA*, June 3, 2020, No. 19-70115, opinion, pp. 5–7 and 27–32.

82. *United States v. Google*, final judgment, December 5, 2025, Sections III–V and IX.V, pp. 2–10 and 32–33; Mehta, remedies opinion, September 2, 2025, introduction and remedy-specific conclusions.

83. Mehta, August 2024 opinion, p. 116, paragraph 335.

84. Laura Chambers, "Defending an Open Web: What the Google Search Ruling Means for the Future", September 8, 2025.

85. Mehta, September 2025 remedies opinion, payment-ban discussion, printed pp. 119–128.

86. F. A. Hayek, "Competition as a Discovery Procedure", 2002 English translation of the 1968 lecture, pp. 9–11.

87. Mehta, September 2025 remedies opinion, user-side data discussion, especially pp. 163–164; December final judgment, Sections IV.B–C.

88. Jesper Akesson, Michael Luca, Gemma Petrie, and Kush Amlani, *Can Browser Choice Screens Be Effective?* (2023), full report, acknowledgments, executive summary, and experimental design, PDF pp. 2–5, 22–25 and 38–40, especially Table 2 and the T3/T4 comparison.

89. Plaintiffs, second compliance and enforcement status report, September 17, 2026, pp. 2–3 and 10–12; public case index, accessed October 16, 2026.

90. Lee-Anne Mulholland, Google's appeal statement, January 16, 2026; plaintiffs, response and opening cross-appeal brief, July 28, 2026, introduction, issues, and cross-appeal.

91. Matthew 2:1–18, King James Version.

92. Colleen Cunningham, Florian Ederer, and Song Ma, "Killer Acquisitions," *Journal of Political Economy* 129, no. 3 (2021): 649–702, especially pp. 649–653, published article, doi:10.1086/712506.

93. Peter Thiel, with Blake Masters, *Zero to One: Notes on Startups, or How to Build the Future* (Crown Business, 2014); Peter Thiel, "Competition Is for Losers," *The Wall Street Journal*, September 12, 2014, university-hosted article copy, pp. 5-6.

Who deserves the winnings

94. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006), 854–856, full paper, pp. 854–855 and Figures 1–3. The particular first-place and fortieth-place rankings come from Salganik's *Bit by Bit*, Section 4.6.1.

95. Shuji Nakamura, "Background Story of the Invention of Efficient Blue InGaN Light Emitting Diodes", Nobel lecture, December 8, 2014, printed pp. 70 and 72–75, especially pp. 73–74.

96. Nichia Corporation, "Nichia Settled with Former Employee over 'Reasonable Remuneration'", January 11, 2005, opening, settlement scope and payment paragraphs, and company statements 2–4.

97. DARPA, "Grand Challenge", institutional timeline, first three paragraphs; "The DARPA Grand Challenge: Ten Years Later", March 13, 2014, opening through the 2007 Urban Challenge.

98. William R. Kerr, Ramana Nanda, and Matthew Rhodes-Kropf, "Entrepreneurship as Experimentation," *Journal of Economic Perspectives* 28, no. 3 (2014): 25–48, especially 27–32 and 43–44. doi:10.1257/jep.28.3.25.

99. Amit P. Mehta, *United States v. Google*, liability opinion, August 5, 2024, p. 49, paragraphs 136–137.

100. Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva, "Optimal Taxation of Top Labor Incomes: A Tale of Three Elasticities," *American Economic Journal: Economic Policy* 6, no. 1 (2014): 230–271, especially 230–235. doi:10.1257/pol.6.1.230.

101. Casey Rothschild and Florian Scheuer, "Optimal Taxation with Rent-Seeking," *Review of Economic Studies* 83(3) (2016): 1225–1262, doi:10.1093/restud/rdw017; December 2015 author version, abstract and introduction, printed pp. 1–4.

102. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *PNAS* 111 (2014): 6934–6939, especially 6934–6935 and Figure 1. Article.

103. Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, full paper, pp. 56–57.

The intelligence factory

104. IEA, *Key Questions on Energy and AI* (2026), full report, published April 16, 2026; executive summary, pp. 9–11; physical bottlenecks, p. 21; demand outlook and accelerator definitions, pp. 24–26.

105. Matt Lakin, ORNL, "Super Construction at Super Scale: How ORNL Built a New Home for Frontier", November 14, 2023, construction account from cooling plant and piping through foundations and the pressure test.

106. ORNL, "Computer Engineers at ORNL Pioneer Approaches to Energy Efficient Supercomputing", section on David Grant's cooling-system work and the transition from Titan to Frontier.

107. LUMI consortium, "Sustainable Future", opening paragraphs on waste-heat use, accessed October 16, 2026.

108. Llama Team, AI at Meta, *The Llama 3 Herd of Models*, arXiv version 3, November 23, 2024, Section 3.3.4, pp. 13–14 and Table 5.

109. Argonne Leadership Computing Facility, "INCITE Program Awards Supercomputing Time to 75 High-Impact Projects", May 5, 2026, opening, selection paragraphs and listed awards; *INCITE Overview and Policies*, version 13, December 2024, distributed for the 2026 cycle, Section 1.2.1, printed pp. 1–2.

110. Llama Team, *The Llama 3 Herd of Models*, introduction, p. 1; Section 3, training infrastructure; Section 4, pre-training and post-training, p. 15.

111. DeepSeek-AI, *DeepSeek-V3 Technical Report*, arXiv version 2, February 18, 2025, p. 5, Table 1 and accompanying paragraphs.

112. Stanford HAI, *Artificial Intelligence Index Report 2026*, full report, Section 1.1, printed pp. 16–18, particularly p. 18 and Figure 1.1.4.

113. Jordan Hoffmann and colleagues, "Training Compute-Optimal Large Language Models," *NeurIPS* 2022, published paper, abstract, pp. 1–2, Section 4 and Section 4.1, p. 7. Authors' April 12, 2022 account.

114. William R. Kerr, Ramana Nanda, and Matthew Rhodes-Kropf, "Entrepreneurship as Experimentation," *JEP* 28(3) (2014), 25–48, published paper, pp. 27–30 and 43–44.

115. Kenneth J. Arrow, "Economic Welfare and the Allocation of Resources for Invention," in *The Rate and Direction of Inventive Activity* (1962), 609–626, NBER chapter, pp. 609 and 614–617, especially pp. 615 and 617.

116. IEA, *Powering Artificial Intelligence: Strategies for Integrating Data Centres into Electricity Systems*, October 14, 2026, full PDF, pp. 7–10 and 52–57, especially the AEP Ohio entry on p. 57.

A billion answers

117. Wei-Lin Chiang and colleagues, “Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference,” *ICML*, PMLR 235 (2024): 8359–8388, published paper, PDF pp. 1–3, Figure 1 and Section 3.1.

118. Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto, “Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators,” first released 2024, author version 2, March 10, 2025, pp. 1–4, Section 4.1 and Figure 3, p. 6, and limitations, p. 9.

119. Isaac Ong and colleagues, “RouteLLM: Learning to Route LLMs with Preference Data,” *ICLR* (2025), sections 3–5 and limitations. Version 4, February 23, 2025, 3–7 and 10.

120. Lingjiao Chen, Matei Zaharia, and James Zou, “FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance”, *TMLR*, December 2024, Section 4, pp. 4–5, and experiments and Figure 3, pp. 6–10.

121. Lihong Li, Wei Chu, John Langford, and Xuanhui Wang, “Unbiased Offline Evaluation of Contextual-bandit-based News Article Recommendation Algorithms,” *WSDM* 2011, 297–306, DOI 10.1145/1935826.1935878; author version 2, March 1, 2012, Sections 3–4, especially Theorem 1 and the Yahoo study.

122. Llama Team, AI at Meta, *The Llama 3 Herd of Models*, version 3, November 23, 2024, p. 2, paragraph on the flagship Llama 3.1 model improving smaller models during post-training.

123. Yaneer Bar-Yam, “From Big Data to Important Information,” *Complexity* 21(S2) (2016): 73–98, DOI 10.1002/cplx.21785; April 2016 author preprint, abstract and Sections I–II, pp. 1–3.

124. David Blackwell and Lester Dubins, “Merging of Opinions with Increasing Information,” *Annals of Mathematical Statistics* 33 (1962): 882–886, doi:10.1214/aoms/1177704456, pp. 882–883 and main theorem.

125. Hao Zhang, “Inconsistent Estimation and Asymptotically Equal Interpolations in Model-Based Geostatistics,” *JASA* 99 (2004), 250–261, full paper, introduction and Section 3, pp. 250–253.

126. Nicholas Carlini, “Building a C Compiler with a Team of Parallel Claudes,” February 5, 2026, sections “Make Parallelism Easy” and “Evaluation.” Anthropic, “How We Built Our Multi-Agent Research System,” June 13, 2025, section “Architecture Overview.”

127. Joon Sung Park and colleagues, “Generative Agents: Interactive Simulacra of Human Behavior,” *UIST* 2023, author version 2, August 6, 2023, Sections 3, 6.5.2, 7 and 8.2, PDF pp. 5 and 15–17.

128. Alexander Novikov and colleagues, *AlphaEvolve: A Coding Agent for Scientific and Algorithmic Discovery*, 2025 white paper, pp. 1–3 and data-center scheduling discussion pp. 13–14; Google DeepMind, May 14, 2025 account.

129. Stanford HAI, *AI Index Report 2026*, Section 1.4, inference subsection, printed p. 36.

The first rung

130. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, "Generative AI at Work", *The Quarterly Journal of Economics* 140(2) (2025): 889–942, Section VI.B and Figure VII.

131. Brynjolfsson, Li, and Raymond, abstract, Sections III and V.

132. Same paper, Section VI.B, Figure VII.

133. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, "Generative AI at Work," *The Quarterly Journal of Economics* 140(2) (2025): 889–942, Section VIII, doi:10.1093/qje/qjae044.

134. Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman, "Generative AI without Guardrails Can Harm Learning", *PNAS* 122(26), 2025, e2422633122. Author version: education_llm.pdf, pp. 4–11 for design/results and pp. 13–18, Section 4, for message/error analyses.

135. Joel Becker, Nate Rush, Beth Barnes, and David Rein, "Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity", 2025, abstract and Sections 1–2.

136. METR, "We Are Changing Our Developer Productivity Experiment Design", February 24, 2026, sections on wider adoption, recruitment, time measurement, and interpretation.

137. David H. Autor, "Why Are There Still So Many Jobs? The History and Future of Workplace Automation", *Journal of Economic Perspectives* 29(3), 2015, 3–30, especially pp. 5–7. See also James Bessen, "Toil and Technology", *Finance & Development* 52(1), March 2015, 16–19, ATM discussion.

138. Daron Acemoglu and Pascual Restrepo, "Robots and Jobs: Evidence from US Labor Markets", *Journal of Political Economy* 128(6), 2020, 2188–2244; author-hosted journal-format proof, PDF pp. 1–4.

139. International Labour Organization, *World Employment and Social Outlook 2021: The Role of Digital Labour Platforms in Transforming the World of Work* (2021), Chapter 4 conclusion, printed p. 192 (PDF p. 194), and deactivation discussion/footnote 7, printed p. 182 (PDF p. 184). Full report.

140. Directive (EU) 2024/2831, Article 9(6) and Article 29(1), official consolidated text, November 11, 2024, PDF pp. 9–10 and 19.

141. Anton Korinek and Donghyun Suh, *Scenarios for the Transition to AGI*, March 2024, sections 3 and 4.1, 22–27, and conclusion, 36–37.

Competition for future evidence

142. CMA, "Roadmap Completion Decision", January 2023, paragraphs 1.1–1.4, 2.1–2.6, 4.4–4.9.

143. Blackwell, "Equivalent Comparisons of Experiments", 1953, original pp. 265–267, decision losses and Theorems 3, 5–6; Section 4, pp. 268–269. De Oliveira, November 2017 author paper, pp. 1–5, published 2018.

144. *United States v. Google*, Final Judgment, December 5, 2025, ECF 1462, Section IV.B–C, pp. 6–7.

145. Zahra Meghani, "Regulations Matter: Epistemic Monopoly, Domination, Patents, and the Public Interest," *Philosophy & Technology* 34 (2021): 1449–1474, introduction and section 6. Article.

146. CMA, *Roadmap Completion Decision*, paragraphs 2.1–2.6 and 4.3–4.5.

147. Ipsos, "Personal Banking Service Quality – Great Britain", February 16, 2026, methodology.

148. Kasy and Sautmann, "Adaptive Treatment Assignment in Experiments for Policy Choice", 2021, pp. 113–114, 122–123, Table II. November 10, 2021 correction, corrected Theorem 1.

149. F. A. Hayek, "Competition as a Discovery Procedure", 2002 English translation by Marcellus S. Snow of the 1968 lecture, pp. 9–11.

150. Watzinger et al., "How Antitrust Enforcement Can Spur Innovation", 2020; full June 2017 author version, introduction and institutional/empirical-design sections.

151. DOJ/FTC, 2023 Merger Guidelines, overview and Guidelines 2, 3 (observability), 10.

152. Kenneth J. Arrow and Anthony C. Fisher, "Environmental Preservation, Uncertainty, and Irreversibility," 1974, pp. 316–319, doi:10.2307/1883074, article copy.

153. DMA, Regulation 2022/1925, Articles 7(1)–(3), 7(6)–(7), 6(11).

154. EuroHPC, Playground call, April 9, 2025, Description/capacity notice; access modes, Industrial Innovation; Terms of Reference, V3, March 31, 2026, Sections 1.3–1.4, 2–3.1.

155. Charter of Fundamental Rights of the European Union, Article 17(1)–(2), and accompanying explanations.

156. White House, *America's AI Action Plan*, July 2025, printed pp. 3–5, deregulation/open-model/access recommendations.

157. NSF, NAIRR operations-center announcement, September 1, 2026.

158. FTC Office of Technology staff, *Partnerships Between Cloud Service Providers and AI Developers*, January 2025, pp. 1–3, 29–30, 32–34. Company information through September 2024; public information through January 2025.

159. State Council, AI Plus opinion, dated August 21, published August 26, 2025, Sections II(2), II(4), III(9)–(13).

160. CAC and six ministries, *Interim Measures for Generative AI Services*, published July 13, effective August 15, 2023, Articles 2, 4–6.

161. David Blackwell, "Equivalent Comparisons of Experiments," *Annals of Mathematical Statistics* 24 (1953): 265–272, especially 265–267. doi:10.1214/aoms/1177729032.

162. Kenneth J. Arrow, "Economic Welfare and the Allocation of Resources for Invention," in *The Rate and Direction of Inventive Activity* (1962), 614–617. Chapter.

Taxing lost comparison

163. Swedish Environmental Protection Agency, *Reducing NOx Emissions: An Evaluation of the Nitrogen Oxide Charge*, Report 5335 (November 2003), 13–14. Report. Swedish Environmental Protection Agency, "Nitrogen Oxides," October 28, 2025, sections "Comparative Measurement of Measurement Equipment" and "Test Comparisons," accessed October 16, 2026.

164. Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva, "Optimal Taxation of Top Labor Incomes: A Tale of Three Elasticities," *American Economic Journal: Economic Policy* 6, no. 1 (2014): 230–271, especially 230–235. Article.

165. Casey Rothschild and Florian Scheuer, "Optimal Taxation with Rent-Seeking," *Review of Economic Studies* 83, no. 3 (2016): 1225–1262. doi:10.1093/restud/rdw017. See also the December 2015 author version, 1–4.

166. Kenneth J. Arrow, "Economic Welfare and the Allocation of Resources for Invention," in *The Rate and Direction of Inventive Activity: Economic and Social Factors* (Princeton University Press and NBER, 1962), 609–626, especially 609 and 614–617. Chapter.

167. Luca Colombo, Gianluca Femminis, and Alessandro Pavan, “Information Acquisition and Welfare,” *Review of Economic Studies* 81 (2014): 1438–1483, especially 1438–1440 and 1446–1448. doi:10.1093/restud/rdu015.

168. Colombo, Femminis, and Pavan, “Information Acquisition and Welfare,” Corollary 1, 1446–1447. Article.

169. California Energy Commission, “Electric Program Investment Charge—EPIC,” accessed October 16, 2026.

170. Jim Zoellick, David Carter, and Peter Lehman, *Redwood Coast Airport Microgrid*, CEC-500-2026-006 (California Energy Commission, January 2026), 1–3, 53–54, 62, 69, and 95–96. Report.

171. William R. Kerr, Ramana Nanda, and Matthew Rhodes-Kropf, “Entrepreneurship as Experimentation,” *Journal of Economic Perspectives* 28, no. 3 (2014): 25–48, especially 27–32. Article.

172. David Blackwell, “Equivalent Comparisons of Experiments,” *Annals of Mathematical Statistics* 24 (1953): 265–272, especially 265–267. doi:10.1214/aoms/1177729032.

173. Competition and Markets Authority, *Retail Banking Market Investigation Order 2017*, Part 2 and Part 3, Article 16.

174. Florian Scheuer and Joel Slemrod, "Taxing Our Wealth," *Journal of Economic Perspectives* 35, no. 1 (2021): 207–230. doi:10.1257/jep.35.1.207.

An economics of possible worlds

175. Rothamsted Research, A. J. Macdonald, editor, *Guide to the Classical and Other Long-Term Experiments, Datasets and Sample Archive*, 2018, reprinted 2019, pp. 4–9, especially p. 7.

176. Rothamsted Research, “Why Do We Make Changes to the Long-Term Experiments?”, January 17, 2024; A. J. Macdonald, ed., *Guide to the Classical and Other Long-Term Experiments, Datasets and Sample Archive*, 2018, reprinted 2019, pp. 5–7 and Broadbalk plan.

177. Electronic Rothamsted Archive, Rothamsted Sample Archive, overview and documented applications; A. J. Macdonald, ed., *Guide to the Classical and Other Long-Term Experiments, Datasets and Sample Archive*, 2018, reprinted 2019, p. 6.

178. Bearchell, Fraaije, Shaw, and Fitt, “Wheat Archive Links Long-Term Fungal Pathogen Population Dynamics to Air Pollution”, *PNAS* 102(15) (2005): 5438–5442, Materials and Methods, “Sample Selection,” “DNA Extraction,” “PCR Assay Development,” and “Validity of PCR Data,” and pathogen-abundance results.

179. R. A. Fisher, "The Arrangement of Field Experiments", 1926, original pp. 503–508, reproduced pp. 84–89.

180. Tim P. Morris, Ian R. White, and Michael J. Crowther, "Using Simulation Studies to Evaluate Statistical Methods," *Statistics in Medicine* 38 (2019): 2074–2102, Sections 3.1–3.2 and Table 1, doi:10.1002/sim.8086.

181. Maximilian Kasy and Anja Sautmann, "Adaptive Treatment Assignment in Experiments for Policy Choice," *Econometrica* 89 (2021): 113–132, especially 113–114 and 122–123. See the November 10, 2021 correction.

182. Kenneth J. Arrow, "Uncertainty and the Welfare Economics of Medical Care", *American Economic Review* 53(5) (1963): 941–973, especially pp. 941, 947–951.

183. James M. Buchanan and Yong J. Yoon, "Symmetric Tragedies: Commons and Anticommons," *Journal of Law and Economics* 43, no. 1 (2000): 1–13, especially 1–6. Article.

184. Blackwell and Dubins, "Merging of Opinions with Increasing Information", 1962, main theorem, pp. 882–883.

185. David Blackwell, "Equivalent Comparisons of Experiments", 1953, pp. 265–267, Theorems 3, 5–6, and Section 4; Henrique de Oliveira, "Blackwell's Informativeness Theorem Using Diagrams", November 2017 author paper, published 2018, pp. 1–5.

186. Hayek, "Competition as a Discovery Procedure", 2002 translation by Marcellus S. Snow of the 1968 lecture, pp. 9–11.

187. Salganik, Dodds, and Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market", 2006, pp. 854–856; Salganik, *Bit by Bit*, Section 4.6.1, open review edition, n.d.

The next opportunity

188. A. J. Macdonald, ed., *Guide to the Classical and Other Long-Term Experiments, Datasets and Sample Archive*, Rothamsted Research, 2018, reprinted 2019, pp. 5–7; Rothamsted Research, "Why Do We Make Changes to the Long-Term Experiments?", January 17, 2024.

189. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006): 854–856, doi:10.1126/science.1121066; Matthew J. Salganik, *Bit by Bit: Social Research in the Digital Age*, Section 4.6.1, open review edition, n.d.

References

Acemoglu, D., and P. Restrepo. 2020. Robots and jobs: Evidence from US labor markets. Journal of Political Economy 128:2188–2244. Source

Akerlof, G. A. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. The Quarterly Journal of Economics, 84(3), 488-500. Source

Akesson, J., M. Luca, G. Petrie, and K. Amlani. 2023. Can Browser Choice Screens Be Effective? Mozilla-commissioned research report. Source

Alchian, A. A. 1950. Uncertainty, evolution, and economic theory. Journal of Political Economy 58:211–221. Source

American Telephone and Telegraph Company. 1955. Annual Report 1955. Historical report scan. Source

Anderson, P. W. 1972. More Is Different. Science 177(4047):393–396. Source

Argonne Leadership Computing Facility. 2026. INCITE program awards supercomputing time to 75 high-impact projects. May 5. Source

Arrow, K. J. 1962. Economic welfare and the allocation of resources for invention. In The Rate and Direction of Inventive Activity: Economic and Social Factors, 609–626. Princeton University Press and NBER. Source

Arrow, K. J. 1963. Uncertainty and the welfare economics of medical care. American Economic Review 53(5):941–973. Original paper

Arrow, K. J., and A. C. Fisher. 1974. Environmental Preservation, Uncertainty, and Irreversibility. Quarterly Journal of Economics 88(2):312–319. Source

Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. The Economic Journal, 99(394), 116-131. Source

Autor, D. H. 2015. Why are there still so many jobs? The history and future of workplace automation. Journal of Economic Perspectives 29:3–30. Source

Barabási, A.-L., and R. Albert. 1999. Emergence of scaling in random networks. Science 286(5439):509–512. Article Full text

Bar-Yam, Y. (2016). From big data to important information. Complexity, 21(S2), 73–98. doi:10.1002/cplx.21785. Source

Bardeen, J. 1956. Semiconductor research leading to the point contact transistor. Nobel lecture, December 11. Source

Bastani, H., O. Bastani, A. Sungu, H. Ge, Ö. Kabakcı, and R. Mariman. 2025. Generative AI without guardrails can harm learning: Evidence from high school mathematics. PNAS 122:e2422633122. Source

Bearchell, S. J., B. A. Fraaije, M. W. Shaw, and B. D. L. Fitt. 2005. Wheat archive links long-term fungal pathogen population dynamics to air pollution. PNAS 102:5438–5442. Source

Becker, J., N. Rush, B. Barnes, and D. Rein. 2025. Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity. METR research report. Source

Bessen, J. 2015. Toil and technology. Finance & Development 52(1):16–19. Publisher record

Blackwell, D. 1953. Equivalent comparisons of experiments. Annals of Mathematical Statistics 24:265–272. Source

Blackwell, D., and Dubins, L. (1962). Merging of opinions with increasing information. The Annals of Mathematical Statistics, 33(3), 882-886. doi:10.1214/aoms/1177704456. Source

Blake, T., C. Nosko, and S. Tadelis. 2015. Consumer heterogeneity and paid search effectiveness: A large-scale field experiment. Econometrica 83:155–174. Source 2014 working paper

Bol, T., M. de Vaan, and A. van de Rijt. 2018. The Matthew effect in science funding. Proceedings of the National Academy of Sciences 115:4887–4890. Source

Brynjolfsson, E., D. Li, and L. R. Raymond. 2025. Generative AI at work. Quarterly Journal of Economics 140:889–942. Source

Buchanan, J. M., and Y. J. Yoon. 2000. Symmetric Tragedies: Commons and Anticommons. Journal of Law and Economics 43(1):1–13. Published paper

California Energy Commission. n.d. Electric Program Investment Charge program. Funding and administration account. Accessed October 16, 2026. Source

Carlini, N. 2026. Building a C compiler with a team of parallel Claudes. Anthropic Engineering, February 5. Source

Chambers, L. 2025. Defending an open web: What the Google Search ruling means for the future. Mozilla, September 8. Source

Chandrashekar, A., F. Amat, J. Basilico, and T. Jebara. 2017. Artwork personalization at Netflix. Netflix TechBlog, December 7. Source

Chen, L., M. Zaharia, and J. Zou. 2024. FrugalGPT: How to use large language models while reducing cost and improving performance. Transactions on Machine Learning Research, December. Source

Chiang, W.-L., et al. 2024. Chatbot Arena: An open platform for evaluating LLMs by human preference. ICML, Proceedings of Machine Learning Research 235:8359–8388. Source

Colombo, L., G. Femminis, and A. Pavan. 2014. Information Acquisition and Welfare. Review of Economic Studies 81:1438–1483. Published article Full text

Competition and Markets Authority. 2017. Retail Banking Market Investigation Order 2017. Implementation arrangements and service-quality survey funding. Source

Competition and Markets Authority. 2023. Retail Banking Market Investigation: Roadmap Completion Decision. January. Source

Computer History Museum. n.d. 1947: Invention of the point-contact transistor. Silicon Engine collection. Accessed October 16, 2026. Source

Computer History Museum. n.d. 1952: Bell Labs licenses transistor technology. Silicon Engine collection. Accessed October 16, 2026. Source

Cunningham, C., F. Ederer, and S. Ma. 2021. Killer Acquisitions. Journal of Political Economy 129(3):649–702. Source

Cyberspace Administration of China and six ministries. 2023. Interim measures for generative AI services. Published July 13; effective August 15. Original Chinese. Source

David, P. A. (1985). Clio and the economics of QWERTY. American Economic Review, 75(2), 332-337. Source

de Oliveira, H. 2018. Blackwell's informativeness theorem using diagrams. Games and Economic Behavior 109:126–131. Source

DeepSeek-AI. 2025. DeepSeek-V3 Technical Report. arXiv:2412.19437, version 2, February 18; first released 2024. Source

Defense Advanced Research Projects Agency. 2014. The DARPA Grand Challenge: Ten years later. March 13. Source

Defense Advanced Research Projects Agency. n.d. Grand Challenge. Institutional innovation timeline. Accessed October 16, 2026. Source

Dubois, Y., B. Galambosi, P. Liang, and T. B. Hashimoto. 2024. Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators. Author version 2, March 10, 2025. Source

Electronic Rothamsted Archive. n.d. Rothamsted Sample Archive. Institutional inventory and applications. Accessed October 16, 2026. Source

Erlic, M. Shadow Futures: Contribution Uncertainty and the Self-Reinforcing Market. SSRN 6003994. SSRN

EuroHPC Joint Undertaking. 2025. Playground access to AI factories. Call opened April 9. Source

EuroHPC Joint Undertaking. 2026. AI factories industrial innovation access: Terms of reference. V3, March 31. Source

EuroHPC Joint Undertaking. AI factories access modes. Industrial Innovation; accessed October 2026. Source

European Parliament and Council. 2022. Regulation (EU) 2022/1925, Digital Markets Act. Articles 6(11) and 7. Source

European Parliament and Council. 2024. Directive (EU) 2024/2831 on improving working conditions in platform work. Articles 9(6) and 29(1). Source

European Union. Charter of Fundamental Rights, Article 17. FRA text and explanations. Source

Federal Communications Commission. 1968. Use of the Carterfone Device in Message Toll Telephone Service. 13 FCC 2d 420, adopted June 26. Source

Fisher, R. A. 1926. The arrangement of field experiments. Journal of the Ministry of Agriculture 33:503–513. Scanned reprint, pp. 84–94. Source

FTC Office of Technology staff. 2025. Partnerships between cloud service providers and AI developers. January. Source

Gabriyelyan, S. S. (2011). Absolute continuity and singularity of two probability measures on a filtered space. Journal of Theoretical Probability, 24(3), 595-614. Source

Google DeepMind. 2025. AlphaEvolve: A Gemini-powered coding agent for designing advanced algorithms. May 14. Developer account. Source

Gottfried, T., and Grosskinsky, S. (2024). Asymptotics of generalized Pólya urns with non-linear feedback. Electronic Journal of Probability, 29, 1-56. Source

Hadfield, J., B. Zhang, K. Lien, F. Scholz, J. Fox, and D. Ford. 2025. How we built our multi-agent research system. Anthropic Engineering, June 13. Source

Hayek, F. A. (2002). Competition as a discovery procedure. The Quarterly Journal of Austrian Economics, 5(3), 9-23. Source

Hoffmann, J., et al. 2022. Training compute-optimal large language models. Advances in Neural Information Processing Systems 35. Source

Hush-A-Phone Corp. v. United States. 1956. 238 F.2d 266, D.C. Circuit, November 8. Source

INCITE. 2024. INCITE Overview and Policies. Version 13, December; distributed for the 2026 cycle. Source

International Energy Agency. 2026. Key Questions on Energy and AI. Paris: IEA. Source

International Energy Agency. 2026. Powering Artificial Intelligence: Strategies for Integrating Data Centres into Electricity Systems. October 14. Report Full PDF

International Labour Organization. 2021. World Employment and Social Outlook 2021: The Role of Digital Labour Platforms in Transforming the World of Work. Geneva: ILO. Source

Ipsos. 2026. Personal banking service quality—Great Britain. February 16; January–December 2025 fieldwork. Source

Jacod, J., and Shiryaev, A. N. (2003). Limit Theorems for Stochastic Processes (2nd ed.). Springer. Source

Johnson, S. 2010. The Genius of the Tinkerer. The Wall Street Journal, September 25. Adapted from Where Good Ideas Come From: The Natural History of Innovation (Riverhead Books). Source

Kabanov, Yu. M., Liptser, R. Sh., and Shiryaev, A. N. (1977). On the question of absolute continuity and singularity of probability measures. Mathematics of the USSR-Sbornik, 33(2), 203-221. doi:10.1070/SM1977v033n02ABEH002421. Source

Kakutani, S. (1948). On equivalence of infinite product measures. Annals of Mathematics, 49(1), 214-224. doi:10.2307/1969123. Source

Kasy, M., and A. Sautmann. 2021. Correction regarding Adaptive Treatment Assignment in Experiments for Policy Choice. Author correction, November 10. Source

Kasy, M., and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. Econometrica, 89(1), 113-132. doi:10.3982/ECTA17527. Source

Kauffman, S. A. 1995. At Home in the Universe: The Search for the Laws of Self-Organization and Complexity. Oxford University Press. Source

Kauffman, S. A. 2003. The Adjacent Possible. Edge, November 9. Source

Kerr, W. R., Nanda, R., and Rhodes-Kropf, M. (2014). Entrepreneurship as experimentation. Journal of Economic Perspectives, 28(3), 25-48. Source

Kohavi, R., A. Deng, B. Frasca, T. Walker, Y. Xu, and N. Pohlmann. 2013. Online controlled experiments at large scale. Proceedings of KDD:1168–1176. Source

Korinek, A., and D. Suh. 2024. Scenarios for the transition to AGI. March author working paper. Author paper

Lakin, M. 2023. Super construction at super scale: How ORNL built a new home for Frontier. Oak Ridge National Laboratory, November 14. Source

Le Cam, L., and Yang, G. L. (2000). Asymptotics in Statistics: Some Basic Concepts (2nd ed.). Springer. Source

Le Goff, L. C., and Soulier, P. (2017). Parameter estimation of a two-colored urn model class. The International Journal of Biostatistics, 13(1), Article 20160029. Source

Li, L., W. Chu, J. Langford, and R. E. Schapire. 2010. A contextual-bandit approach to personalized news article recommendation. Proceedings of WWW:661–670. Source

Li, L., W. Chu, J. Langford, and X. Wang. 2011. Unbiased offline evaluation of contextual-bandit-based news article recommendation algorithms. WSDM:297–306. Author version 2, March 1, 2012. Source

Liebowitz, S. J., and S. E. Margolis. 1990. The fable of the keys. Journal of Law and Economics 33:1–25. Source

Llama Team, AI at Meta. 2024. The Llama 3 Herd of Models. arXiv:2407.21783, version 3, November 23. Source

LUMI consortium. n.d. Sustainable future. Operator account of waste-heat use. Accessed October 16, 2026. Source

Macdonald, A. J., editor. 2018. Guide to the Classical and Other Long-Term Experiments, Datasets and Sample Archive. Rothamsted Research. Reprinted 2019. Source

Meghani, Z. (2021). Regulations matter: Epistemic monopoly, domination, patents, and the public interest. Philosophy & Technology, 34(4), 1449-1474. Source

Merton, R. K. (1968). The Matthew effect in science. Science, 159(3810), 56-63. Source

METR. 2026. We are changing our developer productivity experiment design. February 24. Source

Morris, T. P., I. R. White, and M. J. Crowther. 2019. Using Simulation Studies to Evaluate Statistical Methods. Statistics in Medicine 38(11):2074–2102. Published article Full text

Muchnik, L., S. Aral, and S. J. Taylor. 2013. Social influence bias: A randomized experiment. Science 341:647–651. Source

Mulholland, L.-A. 2026. Why we're appealing the DOJ search distribution case. Google, January 16. Company statement. Source

Nakamura, S. 2014. Background story of the invention of efficient blue InGaN light emitting diodes. Nobel lecture, December 8. Source

National Family Farm Coalition v. Environmental Protection Agency. 2020. Ninth Circuit opinion, No. 19-70115, June 3. Source

National Science Foundation. 2026. NSF establishes operations center for the National Artificial Intelligence Research Resource. September 1. Source

Nichia Corporation. 2005. Nichia settled with former employee over reasonable remuneration. Company statement, January 11. Source

Novikov, A., et al. 2025. AlphaEvolve: A Coding Agent for Scientific and Algorithmic Discovery. Google DeepMind white paper. Source

Oak Ridge National Laboratory. n.d. Computer engineers at ORNL pioneer approaches to energy efficient supercomputing. Accessed October 16, 2026. Source

Oliveira, R. I. (2009). The onset of dominance in balls-in-bins processes with feedback. Random Structures & Algorithms, 34(4), 454-477. Source

Ong, I., et al. 2025. RouteLLM: Learning to route LLMs with preference data. ICLR; author version 4, February 23, first released 2024. Source

Park, J. S., et al. 2023. Generative agents: Interactive simulacra of human behavior. UIST; author version 2, August 6. Source

Pemantle, R. (2007). A survey of random processes with reinforcement. Probability Surveys, 4, 1-79. Source

Piketty, T., Saez, E., and Stantcheva, S. (2014). Optimal taxation of top labor incomes: A tale of three elasticities. American Economic Journal: Economic Policy, 6(1), 230-271. Source

Proceedings of the National Academy of Sciences. 2025. Correction for Bastani et al., Generative AI without guardrails can harm learning: Evidence from high school mathematics. PNAS 122:e2518204122. Affiliation correction, August 20. Correction

Rajkumar, K., G. Saint-Jacques, I. Bojinov, E. Brynjolfsson, and S. Aral. 2022. A causal test of the strength of weak ties. Science 377:1304–1310. Source

Rothamsted Research. 2024. Why do we make changes to the long-term experiments at Rothamsted? January 17. Source

Rothschild, C., and Scheuer, F. (2016). Optimal taxation with rent-seeking. The Review of Economic Studies, 83(3), 1225-1262. Source 2015 author version

Salganik, M. J. n.d. Bit by Bit: Social Research in the Digital Age. Open review edition, Section 4.6.1. Accessed October 16, 2026. Source

Salganik, M. J., and D. J. Watts. 2008. Leading the herd astray: An experimental study of self-fulfilling prophecies in an artificial cultural market. Social Psychology Quarterly 71:338–355. Source

Salganik, M. J., Dodds, P. S., and Watts, D. J. (2006). Experimental study of inequality and unpredictability in an artificial cultural market. Science, 311(5762), 854-856. Source

Scheuer, F., and J. Slemrod. 2021. Taxing our wealth. Journal of Economic Perspectives 35(1):207-230. Article

Smith, A. 1776. An Inquiry into the Nature and Causes of the Wealth of Nations. Book I, Chapters I, III and X, Part II. Source

Snow, J. 1855. On the Mode of Communication of Cholera. Second edition. London: John Churchill. Part 3; UCLA transcription. Source

Sony Group. n.d. Developing the Transistor Radio. Sony Group History, Volume 1, Chapter 3. Part 2. Source

Stanford Institute for Human-Centered Artificial Intelligence. 2026. Artificial Intelligence Index Report 2026. Section 1.1, pp. 16-18. Source

State Council, China. 2025. Opinion on implementing the AI Plus initiative. Dated August 21; published August 26. Original Chinese. Source

Swedish Environmental Protection Agency. 2003. Reducing NOx Emissions: An Evaluation of the Nitrogen Oxide Charge. Report 5335, November; extended English summary. Source

Swedish Environmental Protection Agency. n.d. Nitrogen oxides: Charge, measurement and administration guidance. Accessed October 16, 2026. Source

The Bible. Matthew 2:1–18. King James Version. Source

Thiel, P. 2014. Competition is for losers. The Wall Street Journal, September 12. Adapted from Zero to One, with Blake Masters. Article Article copy

Thiel, P., with B. Masters. 2014. Zero to One: Notes on Startups, or How to Build the Future. Crown Business. Publisher

Traag, V., E. Brady, P. Vincent-Lamarre, F. Bidel, C. Lopes-Bento, J. P. Andersen, and C. Bloch. 2025. The Matthew effect and early-career setbacks in research funding—a replication study. eLife reviewed preprint, version 1, October 13. Source

U.S. Department of Justice and Federal Trade Commission. 2023. Merger Guidelines. December 18. Source

U.S. Department of Justice. 1956. Antitrust Division: AT&T consent decree summary. United States Attorneys' Bulletin 4(4), February 17, p. 119. Source

United States and plaintiff states. 2026. Response and opening cross-appeal brief in the Google search case, July 28. Party filing. Source

United States and plaintiff states. 2026. Second compliance and enforcement status report. United States v. Google LLC, September 17, ECF 1552. Party filing. Source

United States v. Google LLC. 2024. Memorandum opinion, Judge Amit P. Mehta, August 5, ECF 1033. Source

United States v. Google LLC. 2025. Final judgment, December 5, ECF 1462. Source

United States v. Google LLC. 2025. Remedies memorandum opinion, Judge Amit P. Mehta, September 2, ECF 1436. Source

United States v. United States Steel Corp. 1920. 251 U.S. 417, especially 438–439. Source

van de Rijt, A., Kang, S. M., Restivo, M., and Patil, A. (2014). Field experiments of success-breeds-success dynamics. Proceedings of the National Academy of Sciences, 111(19), 6934-6939. Source

Wang, Y., B. F. Jones, and D. Wang. 2019. Early-career setback and future career impact. Nature Communications 10:4331. Source

Watzinger, M., T. A. Fackler, M. Nagler, and M. Schnitzer. 2020. How antitrust enforcement can spur innovation: Bell Labs and the 1956 consent decree. American Economic Journal: Economic Policy 12:328–359. Source 2017 author version

White House. 2025. America's AI action plan. July. Source

Zhang, H. (2004). Inconsistent estimation and asymptotically equal interpolations in model-based geostatistics. Journal of the American Statistical Association, 99(465), 250-261. doi:10.1198/016214504000000241. Source

Zoellick, J., D. Carter, and P. Lehman. 2026. Redwood Coast Airport Microgrid. California Energy Commission, CEC-500-2026-006, January. Source

Acknowledgments

This book depends on researchers who made their experiments, methods, working papers and findings available for others to examine, and on the people who preserve public records and scientific archives. The notes and references identify the work behind its arguments and examples. Particular debts run to the MusicLab researchers and to the long traditions of information economics, experimental design, reinforcement, discovery and public finance discussed throughout the book.

The comparison-budget argument, its interpretation, and its proposed applications are my responsibility.